

Interventional Cardiology Boards

High-yield review and topic notes

ABIM Interventional Cardiology certification exam · October 6, 2026

Compiled for Avi · Built October 3, 2026

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Interventional Cardiology Boards — High-Yield Review (Sept 2026)

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Interventional Cardiology Boards — High-Yield Review (Sept 2026)

Condensed and updated from “Interventional Tidbits” (2021–2022 board-review lectures). Current through September 2026.

UPDATED marks a fact from the original guide that was wrong or has changed.

Board exams lag new evidence. Where a new trial conflicts with an older guideline class, both are shown. Figures, cartoons and angiographic stills from the original were left out. This is a study aid, not a substitute for the primary guidelines.

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What changed: corrections to the original guide

Topic	Original guide said	Current answer
Anaphylaxis	Methylprednisolone 12 mg IV; epinephrine described as “1:1,000 / 1:10,000”	Methylprednisolone 125 mg IV (adjunct only). Epinephrine is now labeled in mg/mL: IM 0.3–0.5 mg of 1 mg/mL; IV 0.1 mg of 0.1 mg/mL given slowly.
PDE5 inhibitors	Wait 5 half-lives (tadalafil 75 h)	Guideline: no nitrates within 24 h of sildenafil/vardenafil or 48 h of tadalafil .
Hydration (POSEIDON)	LVEDP >18 → 1 mL/kg/h; trial “excluded LV dysfunction”	LVEDP <13 → 5, 13–18 → 3, >18 → 1.5 mL/kg/h . NAC and bicarbonate give no benefit (PRESERVE).
Eye-lens dose	150 mSv/yr	US NRC limit is still 150 mSv, but ICRP recommends 20 mSv/yr (averaged over 5 yr). Know both.

Topic	Original guide said	Current answer
VKA mechanism	"Inhibits IX, XI, XII, Xa, VII, thrombin"	Blocks vitamin K–dependent synthesis of II, VII, IX, X and proteins C/S.
DOAC clearance/dosing	Apixaban 8% renal, rivaroxaban 60% renal; edoxaban 30 mg only if CrCl 15–30	Renal: dabigatran 80% , edoxaban 50% , rivaroxaban 35% unchanged, apixaban ~27%. Edoxaban 30 mg if CrCl 15–50 .
HIT	Warfarin for 6 months	Isolated HIT: 4–6 wk (up to 3 mo); HITT: 3 months . DOACs are acceptable. Start warfarin only once platelets $\geq 150k$, with ≥ 5 d overlap.
DAPT de-escalation	"Not in guidelines yet"; when stepping down, drop the P2Y12 inhibitor and keep aspirin	2025 ACS guideline: ticagrelor monotherapy ≥ 1 mo after PCI is Class 1 . HBR: stop aspirin at 1–3 mo. Step-down now favors P2Y12 monotherapy .
P2Y12 choice in ACS	Clopidogrel or ticagrelor Class 1; potent agents IIa	Prasugrel or ticagrelor is Class 1 over clopidogrel for ACS PCI.
GP IIb/IIIa	Class 1 if not preloaded with a P2Y12 inhibitor	Routine use is Class 3 . Bailout only (large thrombus, no/slow reflow): 2a.
Triple therapy	"VKA preferred over DOAC on triple therapy; INR 2–2.5"	DOAC preferred (AUGUSTUS etc.). Stop aspirin at 1–4 wk (Class 1), continue OAC + clopidogrel, then OAC alone after 12 mo (AFIRE, EPIC-CAD).
Warfarin before cath	INR <1.5 femoral / <2 radial	SCAI 2021: INR ≤ 1.8 femoral, ≤ 2.2 radial . No vitamin K.
BMS in HBR	Boards want BMS for very high bleeding risk	DES with short DAPT beats BMS in HBR (LEADERS FREE, ZEUS, SENIOR, ONYX ONE). BMS is essentially obsolete.
Intravascular imaging	2011 guideline (IIa/IIb)	Class 1 for LM/complex lesions (2025 ACS; ESC 2024 CCS). Supported by RENOVATE, OCTOBER, IVUS-ACS and 2024 meta-analysis ($\downarrow$ cardiac death, MI, ST).
FAME 3	CABG superior	1 yr: PCI missed noninferiority. 5 yr: death/stroke/MI similar (16% vs 14%); MI and repeat revascularization higher with PCI.
iFR	Identical to FFR	Noninferior at 1 yr. 5-yr DEFINE-FLAIR showed higher all-cause death with iFR (9.0% vs 6.2%), not seen in iFR-SWEDEHEART; cause unexplained.

Topic	Original guide said	Current answer
Complete revasc timing	Staged is more favorable	MULTISTARS AMI: immediate multivessel PCI is noninferior (fewer events) vs staged . Complete revasc is Class 1 in STEMI and NSTEMI-ACS. Never in shock at the index procedure.
Lytics + ticagrelor	Ticagrelor worse bleeding	TREAT: ticagrelor after lytics (<75 y) was noninferior for major bleeding . Clopidogrel remains the adjunct at lysis. After lysis, angiography at 2–24 h is Class 1 .
Transfusion in MI	Threshold Hb 8	MINT: restrictive strategy not noninferior. Hb ≥10 “may be reasonable” (2b).
VA-ECMO / MCS in shock	“ECMO has data for improved 30-day survival”	ECLS-SHOCK: no mortality benefit , more bleeding. DanGer Shock: Impella CP ↓ 180-d death in STEMI-CS (2a). Routine IABP/VA-ECMO: Class 3.
High-risk PCI with Impella	“Reasonable”	CHIP-BCIS3 (2026): no benefit, CV death higher with Impella . PROTECT IV pending. Routine use is not supported.
Post-arrest cooling	32–36 °C for 24 h	TTM2: 33 °C no better than normothermia. 2025 AHA: temperature control (32–37.5 °C) for ≥36 h , avoid fever.
Inotrope in AMI-CS	Add dopamine	Norepinephrine first. Dobutamine or milrinone for inotropy (SOAP II: dopamine → more arrhythmias).
Colchicine	COLCOT/LoDoCo2 positive	CLEAR SYNERGY (2024) negative post-MI. 2025 ACS: 2b.
LDL goal	PAD statin to LDL <100	High-intensity statin (≥50% ↓). 2026 dyslipidemia guideline: LDL <55 very-high-risk ASCVD, <70 other ASCVD . Measure Lp(a) once.
CTO PCI benefit	“No LV function data yet”; IIa	EXPLORE and REVASC: no LVEF benefit . ACC/AHA 2021: CTO PCI for refractory angina is 2b .
SVG EPD	Class 1	2a in 2021 guideline.
Coronary DCB	“When are we getting DCBs?”	AGENT DCB FDA-approved (2024) for coronary ISR.
Bivalirudin	Only harm (↑ ST)	BRIGHT-4: bivalirudin + high-dose post-PCI infusion ↓ death/bleeding vs UFH in STEMI. It is a guideline alternative to UFH.
Bilateral RAS	“Cr decreases with ACEi”	Bilateral RAS/solitary kidney: ACEi raises Cr (AKI) ; BP is volume-dependent.

Topic	Original guide said	Current answer
Acute limb ischemia	"Pain starts 2 weeks prior"; Doppler categories mis-assigned	ALI = symptoms <2 weeks . Rutherford I: arterial + venous signals audible. IIa/IIb: arterial inaudible, venous audible; IIb has weakness. III: both inaudible.
Asymptomatic carotid	"Never stent asymptomatic"; CMS pays only high surgical risk	CREST-2 (2025): CAS + intensive medical therapy ↓ stroke/death (2.8% vs 6.0%); CEA arm NS. CMS (2023) covers CAS for symptomatic ≥50% / asymptomatic ≥70% regardless of surgical risk, with shared decision-making.
TAVR	"Nothing approved for AR"; Portico; ASA only; PCI pre-TAVR only for symptoms; failure at 8–10 yr	JenaValve Trilogy approved for AR (Mar 2026) . Asymptomatic severe AS approved (EARLY TAVR; FDA 2025). Navitor replaced Portico. NOTION-3 : PCI of significant CAD ↓ MACE. NOTION 10-yr : TAVR durability ≥ SAVR. ESC 2025: TAVI ≥70 y.
Cerebral embolic protection	Unclear	PROTECTED TAVR and BHF PROTECT-TAVI: no stroke reduction . Not routine.
Mitral/tricuspid devices	ViMAC/ViR research only; no TR devices	Tendyne (2025) and SAPIEN M3 transseptal TMVR (Dec 2025) approved. TriClip and EVOQUE approved 2024. RESHAPE-HF2 and MATTERHORN expand TEER in SMR.
LAAO	ACC/AHA IIb; LAAO ≈ DOAC	2023 AF guideline: 2a if long-term OAC is contraindicated. OPTION (post-ablation) and CHAMPION-AF: noninferior to DOAC with less bleeding, but numerically more ischemic stroke. CLOSURE-AF (HBR): failed noninferiority.
HCM	ASA "Class IC"; ASA worsens MR	2024 HCM guideline: after BB/CCB, myosin inhibitor, disopyramide or SRT (Class 1) . Mavacamten and aficamten (FDA Dec 2025). ASA doesn't worsen SAM-MR, but intrinsic MV disease favors myectomy.
Platypnea-orthodeoxia	Dyspnea lying down, better upright	Reversed: dyspnea and desaturation when upright , relieved supine.
Coarctation	"75–85% of BAV patients have CoA"	Reversed: 50–75% of CoA patients have BAV (only a minority of BAV have CoA).
ASD closure / ToF PVR	2018 ACHD criteria	2025 ACHD: ASD closure if PVR ≤2 WU (consider if 2–<5 WU). ToF PVR emphasizes RVESVi >80 mL/m² .

Topic	Original guide said	Current answer
PAD antithrombotics	ASA alone for most	2024 PAD guideline: SAPT (ASA or clopidogrel), and rivaroxaban 2.5 mg BID + ASA (Class 1) for symptomatic PAD and after revascularization. CLTI: BEST-CLI favors vein bypass when good GSV.

1. Periprocedural Medications, Contrast, Sedation and CA-AKI

Metformin (ACR Manual on Contrast Media)

- Risk is lactic acidosis if AKI develops while the drug accumulates (renally cleared, $t_{1/2} \sim 6$ h).
- **eGFR ≥ 30 and no AKI:** no need to stop before or after contrast.
- **Hold at the time of the procedure and for 48 h after, then recheck renal function** if: AKI, eGFR < 30 , or an intra-arterial study that could embolize to the renal arteries.
- Board default if the stem is vague: hold the morning of the procedure, resume after 48 h with a stable Cr.

Contrast hypersensitivity

- Anaphylaxis is Type I (IgE/mast cell). Gell–Coombs II = IgG cytotoxic, III = immune complex, IV = T-cell (delayed rash).
- **Elective premedication:** prednisone 50 mg PO at 13, 7 and 1 h before, plus diphenhydramine 50 mg at 1 h. Alternative: methylprednisolone 32 mg PO at 12 and 2 h. **Urgent:** methylprednisolone 40 mg IV or hydrocortisone 200 mg IV (repeat q4h until contrast), plus diphenhydramine 50 mg IV 1 h before. Steroids given < 4 –5 h before contrast have little proven effect. H2 blockers and leukotriene antagonists: insufficient evidence.
- Isolated hives or pruritus: diphenhydramine. No epinephrine unless the airway or hemodynamics are involved.

Anaphylaxis

- **Epinephrine is the only first-line drug.** IM 0.3–0.5 mg (1 mg/mL) into the thigh, every 5–15 min. IV 0.1 mg (0.1 mg/mL) slowly, or 10 mcg boluses, then 1–4 mcg/min infusion.
- **Never give 1 mg IV push** (that is the arrest dose).
- On beta-blockers and refractory: **glucagon 1–5 mg IV over 5 min** (a rapid push causes emesis), then infusion. Adjuncts: IV fluids; albuterol for bronchospasm (not stridor); H1 blocker (skin symptoms only); methylprednisolone 125 mg IV (prevents biphasic reaction, no acute effect).

UPDATED Original listed methylprednisolone 12 mg. The dose is 125 mg. “1:1,000 / 1:10,000” labeling has been replaced by 1 mg/mL and 0.1 mg/mL.

Sedation

Minimal: normal response to voice. **Moderate** (most cath): purposeful response to voice or light touch; airway and ventilation maintained. **Deep:** purposeful response only to pain; may need airway support.

Drug	Dose	Onset / duration	Notes
Midazolam	0.5–2 mg	2–5 min / ~30 min	Anxiolytic, amnestic
Fentanyl	25–50 mcg	1–2 / 30–60 min	Minimal histamine or hypotension
Hydromorphone	0.5–2 mg	5–10 / 4–5 h	Common “fentanyl allergy” alternative; metabolites accumulate in renal failure
Morphine	1–5 mg	2–8 / 4–5 h	Histamine, hypotension; delays oral P2Y₁₂ absorption
Meperidine	12.5–50 mg	5–10 / 2–5 h	Anti-shivering; avoid in renal failure

- **Flumazenil** 0.2 mg IV over 30 s, repeat to a 1 mg max. Seizure risk in chronic benzodiazepine use.

Naloxone: spontaneously breathing 0.04–0.1 mg; apneic 0.4–2 mg; repeat q2–3 min. It is shorter-acting than most opioids, so watch for re-sedation.

Goal of reversal is adequate ventilation, not full arousal.

PDE5 inhibitors

They potentiate nitrate cGMP, which can cause profound hypotension.

No nitrates within 24 h of sildenafil or vardenafil, or 48 h of tadalafil (ACC/AHA).

- Treat hypotension with fluids and an α -agonist (phenylephrine).

Contrast-associated AKI (CA-AKI)

- **Definitions:** classic CIN = Cr $\uparrow$ ≥ 0.5 mg/dL or $\geq 25\%$ at 48–72 h. KDIGO = ≥ 0.3 mg/dL in 48 h or $\geq 1.5\times$ in 7 d. **Course:** Cr rises at 24–48 h, peaks at 3–5 d, recovers by 7–10 d.
- **Mechanism:** medullary vasoconstriction and hypoxia, reactive oxygen species, direct tubular toxicity.
Patient risk (Mehran): eGFR < 60 , DM, age > 75 , CHF/low EF, hypotension/IABP, anemia, ACS/shock.
Procedural risk: contrast volume, repeat exposure within 72 h. **Prevention:**
- Isotonic saline 1–1.5 mL/kg/h for 3–12 h before and 6–24 h after. Isotonic beats half-normal saline.
- **POSEIDON** (LVEDP-guided): 3 mL/kg/h for 1 h, then by LVEDP: $< 13 \rightarrow 5$; $13\text{--}18 \rightarrow 3$; $> 18 \rightarrow 1.5$ mL/kg/h $\times 4$ h. **N-acetylcysteine and bicarbonate: no benefit** (PRESERVE). NAC is Class 3.

Iso-osmolar vs low-osmolar contrast: no consistent difference in modern data.

- **Contrast limits:** maximum dose = 5 mL $\times$ weight (kg) / Cr, up to 300 mL. Contrast volume/CrCl > 3.7 (some use $> 2\text{--}3$) predicts AKI.
- Minimize volume: biplane, small injections, IVUS-guided minimal-contrast PCI. Stop CTO PCI at $> 3.7\times$ eGFR.

2. Radiation Physics and Safety

How the system works

- Only ~1% of tube energy becomes x-rays (the rest is heat). The small focal spot is used for fluoro, the large one for cine.
- **kVp** sets beam energy: higher kVp = more penetration, **less contrast**. **mA** sets photon quantity: dose is proportional to mA with no loss of contrast.
- **Pulse width** 4–10 ms; longer causes motion blur.
- **Frame rate**: 7.5 fps fluoro and 15 fps cine are standard. Use 30 fps only for pediatrics or LV imaging.
- **Cu spectral filters** remove low-energy photons that add dose but not image. They improve images and lower dose.
- **Automatic dose-rate control** raises output as attenuation rises (large patient, steep angle).
- **US fluoro limit**: 88 mGy/min (10 R/min); high-level mode 176 mGy/min. **Cine has no legal limit** and ~10× the dose rate of fluoro. Fluoro and cine each contribute roughly half of total dose.
- **Image quality**: contrast (iodine), spatial resolution (blur), noise (quantum mottle; more in fluoro). Anti-scatter grid and flat-panel detector.

Dose metrics

Metric	Unit	Predicts
Air kerma ($K_{a,r}$) at the interventional reference point (15 cm toward the tube from isocenter)	mGy	Deterministic (skin)
Kerma-area product / DAP ($K \times$ field area)	Gy·cm ²	Stochastic (cancer)
Effective dose (organ-weighted)	Sv (1 rem = 10 mSv)	Whole-body stochastic risk

Biologic effects

Deterministic effects have a threshold (skin, cataract). **Stochastic effects** have no threshold (cancer; linear no-threshold model).

Skin (peak skin dose):

- 2 Gy: transient erythema/epilation
- 5–10 Gy: erythema
- 10–15 Gy: moist desquamation
- 15 Gy: ulceration/necrosis (sentinel event)
- Arrange follow-up when $K_{a,r} > 5$ Gy. **Do not biopsy** a suspected radiation ulcer.
- Thresholds are lower with prior exposure, DM and collagen-vascular disease.
- Cancer risk ≈5% fatal cancer per Sv. **Younger age is the strongest risk modifier**; females have higher risk.

Occupational limits and protection

Whole body: 50 mSv/yr (NRC). Practical goal <5 mSv/yr. **Lens:** NRC 150 mSv/yr; **ICRP recommends 20 mSv/yr**. Skin/extremities: 500 mSv/yr.

- **Declared pregnancy:** fetal limit 5 mSv for the pregnancy (0.5 mSv/month); second dosimeter under the apron. A pregnant worker cannot be excluded from the lab. 0.5 mm Pb apron attenuates ~95–98% of scatter (0.35 mm ~90–95%).
- **The main source of operator dose is patient scatter**, so lowering patient dose lowers operator dose. At 1 m, scatter $\approx 1/1000$ of the patient entrance dose; inverse square law applies.

Reduce dose:

- Collimate (single best step: less dose, less scatter, better image)

Low frame rate; fluoro-save instead of cine

- Table up and detector down Avoid steep angles and magnification
- Rotate projections to spread skin dose
- Ceiling and table shields close to the patient; step back during cine
- Real-time dose monitoring

Large patients need higher kVp and mA, which means more dose and scatter with worse contrast and more blur.

3. Antiplatelet and Anticoagulant Therapy

Mechanisms and agents

Agent	Target / kinetics	PCI dosing	Key points
Aspirin	Irreversible COX-1 $\rightarrow$ $\downarrow$ TXA ₂	162–325 mg load; 75–100 mg daily	≤ 100 mg with ticagrelor (PLATO US signal). Ibuprofen blocks aspirin access to COX-1, so give aspirin first.
Clopidogrel	Thienopyridine prodrug, 2-step CYP2C19 activation; irreversible	600 mg (300 mg with lytics if ≤ 75 y; 75 mg if > 75 y)	Only P2Y ₁₂ inhibitor for elective/CCD PCI and with OAC. LOF carriers have more MACE/ST. Doubling the daily dose does not overcome LOF; switch agents.
Prasugrel	Thienopyridine prodrug (1 CYP step), irreversible; onset ~30 min–2 h	60 mg, then 10 mg (5 mg if < 60 kg)	Contraindicated after prior stroke/TIA . Generally avoid ≥ 75 y. Give only after anatomy is known (ACCOAST). Once daily.
Ticagrelor	Direct-acting, reversible , allosteric (non-ADP site)	180 mg, then 90 mg BID; 60 mg BID long-term	Dyspnea (~10–15%) and ventricular pauses (adenosine-mediated). Switch to prasugrel for dyspnea.
Cangrelor	IV reversible P2Y ₁₂ ; offset ~1 h	30 mcg/kg bolus + 4 mcg/kg/min ≥ 2 h or for PCI duration	For P2Y ₁₂ -naïve patients. CHAMPION PHOENIX $\downarrow$ periprocedural MI/ST. 2025 ACS: 2b.

Agent	Target / kinetics	PCI dosing	Key points
Eptifibatide	Peptide GP IIb/IIIa; plasma $\leq$ 2.5 h; platelet recovery ~4–8 h	180 mcg/kg $\times$ 2 boluses + 2 mcg/kg/min	Halve infusion if CrCl $<$ 50; avoid on dialysis. Not reversed by platelets.
Tirofiban	Non-peptide GP IIb/IIIa	25 mcg/kg + 0.15 mcg/kg/min	Halve infusion if CrCl $\leq$ 60.
Abciximab	Antibody fragment; long platelet-bound $\leq$; reversed by platelet transfusion	—	No longer marketed in the US.
Vorapaxar	PAR-1 (thrombin receptor) antagonist	2.08 mg daily	Prior MI or PAD ($\downarrow$ ALI). Contraindicated after stroke/TIA/ICH.

Pivotal P2Y₁₂ trials

- **TRITON-TIMI 38:** prasugrel vs clopidogrel (300 mg) in ACS PCI. $\downarrow$ CV death/MI/stroke and ST; $\uparrow$ major bleeding. Net harm with prior stroke/TIA; no net benefit $\geq$ 75 y or $<$ 60 kg.
- **PLATO:** ticagrelor vs clopidogrel in ACS. $\downarrow$ CV death/MI/stroke, $\downarrow$ **CV and all-cause death**, $\downarrow$ ST; $\uparrow$ non-CABG major bleeding.
- **ISAR-REACT 5:** prasugrel $>$ ticagrelor in ACS with an invasive strategy ($\downarrow$ death/MI/stroke, similar bleeding). **ACCOAST:** prasugrel pretreatment in NSTEMI gave no ischemic benefit and $\uparrow$ bleeding.
- **CURRENT-OASIS 7:** double-dose clopidogrel $\downarrow$ ST in the PCI subgroup; $\uparrow$ bleeding.
- **TAILOR-PCI** (genotype-guided): primary endpoint NS. **POPular Genetics:** genotype-guided de-escalation noninferior with $\downarrow$ bleeding.
- **Routine platelet-function testing is not recommended** (GRAVITAS, ARCTIC, TRIGGER-PCI).
- **Switching P2Y₁₂ inhibitors (acute/early phase)**
- **Clopidogrel** $\rightarrow$ **ticagrelor:** 180 mg regardless of the last dose.
- **Clopidogrel** $\rightarrow$ **prasugrel:** 60 mg regardless.
- **Ticagrelor** $\rightarrow$ **clopidogrel:** 600 mg 24 h after the last ticagrelor dose.
- **Ticagrelor** $\rightarrow$ **prasugrel:** 60 mg 24 h after the last dose. **Prasugrel** $\rightarrow$ **clopidogrel:** 600 mg 24 h later. In the late phase, use maintenance doses.

DAPT duration: evidence and current recommendations

Trial	Design	Result
DAPT (2014)	30 vs 12 mo DAPT after DES	$\downarrow$ MI and ST; $\uparrow$ bleeding; $\uparrow$ all-cause death (unexplained). DAPT score $\geq$ 2 favors prolonged therapy.
PEGASUS-TIMI 54	Ticagrelor 90 or 60 mg BID, 1–3 y after MI	$\downarrow$ MACE with both doses; 60 mg has better tolerability (approved dose)
SMART-DATE	6 vs 12 mo DAPT in ACS	More MI with 6 mo
TWILIGHT	High-risk PCI: 3 mo DAPT $\rightarrow$ ticagrelor alone	$\downarrow$ bleeding; no ischemic penalty

Trial	Design	Result
TICO	ACS: 3 mo DAPT → ticagrelor alone	↓ net events, driven by ↓ bleeding
ULTIMATE-DAPT / T-PASS	ACS: ticagrelor monotherapy after 1 mo (or <1 mo)	↓ bleeding; no ↑ ischemic events
STOPDAPT-2	1 mo DAPT → clopidogrel (mostly CCD)	Superior net outcome (↓ bleeding)
STOPDAPT-2 ACS	Same strategy in ACS	Not noninferior (↑ MI): avoid early clopidogrel monotherapy in ACS
MASTER DAPT	HBR: 1 mo vs ≥3 mo DAPT	Noninferior for ischemia; ↓ bleeding
HOST-EXAM; SMART-CHOICE 3	Chronic clopidogrel vs aspirin monotherapy after PCI	Clopidogrel ↓ thrombotic events (and bleeding in HOST-EXAM)
LEADERS FREE, ZEUS, SENIOR, ONYX ONE	HBR: DES + 1 mo DAPT vs BMS	DES superior; no role for BMS

ACS (2025 ACC/AHA)

- **Default DAPT is 12 mo** in non-HBR patients (Class 1).
- **Prasugrel or ticagrelor over clopidogrel** for PCI (Class 1). Prasugrel may be preferred in NSTEMI-ACS (ISAR-REACT 5). **Transition to ticagrelor monotherapy ≥1 mo after PCI** if DAPT is tolerated (Class 1).
- HBR: stop aspirin at 1–3 mo and continue the P2Y12 inhibitor (2a).
- De-escalation from prasugrel/ticagrelor to clopidogrel after 1 mo: 2b.
- Upstream P2Y12 inhibitor only if angiography is delayed >24 h (2b). No routine pretreatment before early cath. GP IIb/IIIa: **routine use Class 3**; bailout (thrombus, no/slow reflow) 2a. Cangrelor: 2b.

Chronic coronary disease (2023 CCD guideline)

- Clopidogrel-based DAPT for **6 mo** after PCI.
- HBR: 1–3 mo of DAPT, then P2Y12 inhibitor monotherapy (2a).
- Extended DAPT beyond 12 mo only for high ischemic and low bleeding risk (2b).
- Rivaroxaban 2.5 mg BID + aspirin (COMPASS) for high ischemic / low bleeding risk.

UPDATED The original said short-DAPT and aspirin-free strategies were “not in guidelines yet” and to keep aspirin when de-escalating. Both are outdated. P2Y12 monotherapy is now guideline-endorsed.

Antithrombotics with OAC (AF + PCI)

- **Trials:** WOEST, PIONEER AF-PCI, RE-DUAL, AUGUSTUS, ENTRUST-AF PCI. DOAC + P2Y12 inhibitor bleeds less than VKA triple therapy.
- **Recommendations:**
- DOAC over VKA

- **Clopidogrel** is the P2Y₁₂ inhibitor of choice Aspirin ≤100 mg, **stopped 1–4 wk after PCI** (Class 1, 2025 ACS)
- Continue OAC + clopidogrel to 12 mo (6 mo in CCD/HBR)
- Then **OAC monotherapy** (AFIRE: rivaroxaban; EPICCAD: edoxaban)
- Extend aspirin toward 30 d only for very high thrombotic, low bleeding risk.
- Add a PPI with DAPT/OAC combinations in GI-bleed-risk patients.
- Bleeding scores: PRECISE-DAPT ≥25 or ARC-HBR (1 major or 2 minor) = HBR.

UPDATED The original said “VKA preferred over DOAC while on triple therapy, INR 2–2.5.” The current answer is DOAC, clopidogrel and early aspirin discontinuation.

Parenteral anticoagulation

UFH and LMWH

UFH: 70–100 U/kg without GPI; 50–70 U/kg with GPI. ACT 250–300 s (HemoTec) or 300–350 s (Hemochron); 200–250 s with GPI.

- **Enoxaparin in PCI:**
- Last SC dose <8 h: nothing more
- 8–12 h: 0.3 mg/kg IV
- 12 h or none: 0.5–0.75 mg/kg IV
- **Avoid switching UFH ↔ LMWH** (↑ bleeding; SYNERGY). Anti-Xa goal 0.5–1.0 IU/mL; check it in CKD or at extremes of weight.
- **Fondaparinux** alone causes catheter thrombosis (OASIS-5), so add UFH at PCI.
- **Sheath pull:** ACT <150–180 s; 4–8 h after LMWH; ≥2 h after bivalirudin; fibrinogen >150 mg/dL after lytics. OK to pull on GPI (more oozing).

Bivalirudin

- **Dosing:** 0.75 mg/kg bolus + 1.75 mg/kg/h. CrCl <30 → 1 mg/kg/h; dialysis → 0.25 mg/kg/h. Not reversed by protamine.
- **HORIZONS-AMI:** ↓ bleeding and mortality vs UFH + GPI, but ↑ acute ST.
- **HEAT-PPCI, MATRIX, VALIDATE-SWEDEHEART:** no advantage over UFH with contemporary radial practice and selective GPI.
- **BRIGHT-4 (2022):** bivalirudin + **high-dose post-PCI infusion (1.75 mg/kg/h for 2–4 h)** ↓ 30-d death/BARC 3–5 bleeding vs UFH in STEMI, without excess ST. Bivalirudin is a guideline alternative to UFH in STEMI PCI and the agent of choice in HIT.

Heparin-induced thrombocytopenia

- Anti-PF₄/heparin IgG. Platelets fall >50% at 5–10 d; rapid onset if heparin exposure within the prior 30–100 d; delayed onset up to 3 wk.

Risk is higher with UFH than LMWH, surgical than medical, and female than male. Untreated thrombosis risk is ~50%. Use the 4T score.

- **Stop all heparin** (including flushes and heparin-coated lines).
- Start a direct thrombin inhibitor: **argatroban** (hepatic; avoid in liver failure; raises INR) or **bivalirudin** (preferred for PCI). Fondaparinux or DOACs are alternatives. Send PF4 ELISA; confirm with serotonin release assay. Warfarin only after platelets $\geq 150k$, overlapping ≥ 5 d until INR is therapeutic.
- **Duration:** HIT 3 mo; isolated HIT 4–6 wk (up to 3 mo).

Oral anticoagulants

- **Renal clearance:** dabigatran ~80%, edoxaban ~50%, rivaroxaban ~35% (unchanged), apixaban ~27%.
- **AF dose reductions:**
 - Apixaban 2.5 mg BID if ≥ 2 of: age ≥ 80 , weight ≤ 60 kg, Cr ≥ 1.5
 - Rivaroxaban 15 mg if CrCl ≤ 50
 - Edoxaban 30 mg if CrCl 15–50 (avoid if CrCl > 95) Dabigatran 75 mg BID if CrCl 15–30 (US)
- **VKA still required:** mechanical valves; rheumatic moderate–severe MS with AF (INVICTUS: rivaroxaban inferior); high-risk (triple-positive) APS.
- DOACs are acceptable in obesity, with active cancer (apixaban/edoxaban or LMWH), and increasingly for LV thrombus.
- **VKA pharmacology:** blocks synthesis of factors II, VII, IX, X and proteins C/S. Overlap with a parenteral agent ≥ 5 d and until INR is therapeutic ≥ 24 h.
- **Before cath:**
 - Warfarin: **INR ≤ 2.2 radial, ≤ 1.8 femoral** (SCAI 2021). No vitamin K.
 - Apixaban/rivaroxaban: hold 24–48 h
 - Dabigatran: 48 h, longer (72–96 h) if CrCl < 50
 - Emergency: go radial
- **Bridging:** DOACs need none. VKA bridging only for very high thrombotic risk (mechanical mitral valve, stroke or embolism within 3 mo). BRIDGE trial: no bridging for AF. **Resume OAC** once hemostasis is secure (~4–6 h after radial; next day after femoral). **Before CABG:**
 - Clopidogrel 5 d; prasugrel 7 d; ticagrelor 3–5 d Eptifibatide/tirofiban 2–4 h; cangrelor ~1 h
 - Urgent CABG: hold clopidogrel/ticagrelor ≥ 24 h if possible

4. Bleeding, Vascular Access and Access Complications

Bleeding

Bleeding is the most common PCI complication and independently predicts mortality.

Access-site bleeding has declined with radial access; nonaccess bleeding (mostly GI) now predominates, especially in ACS.

BARC:

- 1: not actionable
- 2: overt, actionable
- 3a: Hb ↓3–<5 g/dL or any transfusion
- 3b: Hb ↓≥5, tamponade, surgery, or vasoactive drugs
- 3c: ICH or intraocular
- 4: CABG-related
- 5: fatal

Risk factors: female, **low body weight** (not obesity), older age, CKD, anemia, ACS, prior bleed, thrombocytopenia, OAC/NSAID/steroids, femoral access, high heparin dose, GPI, lytics.

Avoidance: radial access; weight-based dosing; avoid UFH ↔ LMWH crossover; avoid routine GPI; smaller sheaths; ultrasound/fluoro-guided CFA access; early sheath removal; PPI.

Radial vs femoral

RIVAL (2011): no difference in the primary endpoint; benefit in the STEMI subgroup and high-volume centers.

RIFLE-STEACS: ↓ mortality in STEMI. **MATRIX (2015):** radial ↓ BARC 3/5 bleeding, NACE and all-cause death in ACS. **SAFARI-STEMI (2020):** stopped for futility; neutral mortality.

Radial is the default, Class 1 in ACS (↓ bleeding, vascular complications, mortality).

SIHD: radial ↓ access bleeding but no mortality benefit. **Radial downsides:** learning curve; slightly more radiation/contrast early; longer door-to-balloon (minutes); crossover ~5–10%.

Distal radial (DISCO RADIAL, 2022): RAO not significantly lower (0.3% vs 0.9%); higher crossover.

Allen/Barbeau testing is not required before radial access (RADAR: no hand ischemia). Ultrasound improves first-pass success (RAUST).

- **Radial artery occlusion (RAO) and radial troubleshooting**
- **RAO** occurs in ~1–5% (up to 10%). Risk: female, small artery, large sheath, spasm, low heparin, occlusive hemostasis.

Prevention: heparin ≥50 U/kg or 5,000 U (give even if INR is therapeutic); **patent hemostasis** (PROPHET); prophylactic ulnar compression (PROPHET-II); smaller/hydrophilic sheaths.

RAO treatment: ipsilateral ulnar compression 1 h or anticoagulation (LMWH ~4 wk). About half recanalize spontaneously.

- **Anatomic variants:** radial loop (almost always with an accessory recurrent radial; cross with 0.014" wire or hydrophilic wire, then straighten); high-bifurcating radial (small vessel, spasm); subclavian/innominate tortuosity (age ≥75, short stature, HTN; deep inspiration or stiff wire; consider left radial).
- **Forearm perforation:** keep going. Cross with a 0.014" wire, balloon-assisted tracking, and seal with a long sheath or catheter; compressive wrap afterward.
- **Catheter kink/knot** (no torque transmission, damped trace): **do not pull**. Fix the catheter proximally (BP cuff, manual pressure) and untwist; straighten with a wire; snare from femoral access.

Severe spasm/entrapment: sedation, IA verapamil/NTG, SC nitroglycerin, warm compress; **inflate a brachial BP cuff 5 min then release** (reactive hyperemia); deep sedation or anesthesia if needed.

- **Radial cocktail:** verapamil 2.5–5 mg ± NTG 100–200 mcg + heparin.
- **Radial pseudoaneurysm:** compression band; thrombin or surgery if it fails.

Femoral access

- **Target:** CFA over the middle of the femoral head, below the inferior epigastric artery (IEA) and above the bifurcation.
- **High stick** (above IEA) → retroperitoneal bleed. **Low stick** (SFA/profunda) → pseudoaneurysm or AV fistula. **Ultrasound guidance** (FAUST): ↓ attempts, venipuncture and time to access. Improves CFA cannulation mainly with high bifurcations. Standard of care with fluoroscopic landmarking.
- **Emergency blind stick:** ~2 cm below the midpoint of the ASIS–pubic tubercle line.
- **Access-complication risk:** female, age ≥65, low weight or BMI ≥40, HTN, CKD, CHF, PAD, AMI, sheath size and dwell time, anticoagulation.

Specific complications

Retroperitoneal hematoma (0.15–0.75%)

- Hypotension/tachycardia, flank/back/lower abdominal pain; bradycardia may occur. ~75% present within 3 h.
- Treat before imaging if unstable: manual compression, stop/reverse anticoagulation, fluids/blood. CT if stable.
- Persistent bleeding: contralateral access, balloon tamponade, covered stent or coil; surgery rarely.

Pseudoaneurysm

Pulsatile tender mass with bruit.

<2 cm: often closes spontaneously; rescan in ~2 wk.

- ≥2–3 cm, symptomatic, expanding or on anticoagulation: **US-guided thrombin injection** (~97% success; best for narrow neck).
- US compression: 75–98% success, painful, lower success on anticoagulation.
- Surgery: infected, rapidly expanding, skin necrosis, or neuropathy.

AV fistula

- Continuous bruit/thrill; low stick; left-sided; simultaneous artery and vein access.
- ~⅓ close by 1 yr; observe small ones.
- Large (high-output HF, steal, limb edema): covered stent or surgery.
- **Dissection:** retrograde flaps usually seal with antegrade flow. Flow-limiting → PTA/stent.

Vascular closure devices

- Shorten time to ambulation; **do not** ↓ **bleeding or vascular complications** in RCTs. Device failure raises complications ~3×.
- Avoid with: puncture above IEA or at/below bifurcation, CFA <5 mm, heavy calcification/PAD, infection risk, collagen or bovine allergy (Angio-Seal). Infection is usually a **mycotic pseudoaneurysm**.
- **Brachial access**: higher complication rate (thrombosis, median nerve compression from hematoma). Avoid closure devices.

5. Coronary Pathobiology, MI Definitions and Coronary Anomalies

Atherosclerosis

Sequence:

- Endothelial dysfunction (↓ NO; earliest step, independently prognostic)
- LDL entry and oxidation
- Monocyte adhesion → macrophage → foam cell (fatty streak)
- VSMC migration and collagen synthesis (fibrous cap) Necrotic core and vasa vasorum neovascularization (intraplaque hemorrhage)
- **Vulnerable plaque (TCFA)**: cap <65 μm, large lipid core, macrophages/T cells, few VSMCs, MMP activity, positive remodeling, spotty calcium. **Rupture occurs at the cap shoulder**.
- **Remodeling index**: lesion EEM area / reference EEM area. Positive >1.05 (ACS, vulnerable); negative <0.95 (elderly, DM, stable). Early plaque grows outward, so the angiogram looks normal.

ACS substrates:

- **Rupture** (~60–75%): TCFA; more often STEMI.
- **Erosion** (25–40%): intact cap, SMC and proteoglycan-rich, few inflammatory cells; young women and smokers; more often NSTEMI-ACS. OCT diagnosis; some managed without stenting (**EROSION**). **Calcified nodule** (5%): elderly, CKD, DM; tortuous RCA and LM.

Most ACS arises from non-obstructive (<50%) plaques because they vastly outnumber severe lesions, although each severe lesion carries higher individual risk. 20–75% of ACS patients have multiple ruptured plaques (the “vulnerable patient”).

Vasomotor testing: acetylcholine is endothelium- *dependent* (dilates normal arteries, constricts dysfunctional ones). Adenosine, NTG and nitroprusside are endothelium- *independent*.

- **Neoatherosclerosis** develops earlier and more often in DES (months to 1–2 yr) than BMS (>2–5 yr).

Inflammation trials

CANTOS: canakinumab (IL-1β) ↓ MACE 15%, ↑ fatal infection; LDL unchanged.

CIRT: low-dose methotrexate neutral (IL-6/CRP not lowered).

- **COLCOT** (post-MI) and **LoDoCo2** (CCD): colchicine ↓ MACE; more GI effects and a non-CV death signal. **CLEAR SYNERGY (2024)**: colchicine after MI had no MACE benefit.

- Colchicine 0.5 mg is FDA-approved; guidelines give it 2b.
- **Universal definition of MI (4th)**
- **Type 1:** atherothrombotic.
- **Type 2:** supply/demand mismatch. Worse prognosis, driven by non-cardiac death.
- **Type 3:** cardiac death before biomarkers.
- **Type 4a** (PCI-related, ≤ 48 h):
 - Normal baseline: cTn $>5\times$ the 99th percentile URL. Elevated but stable/falling baseline: $>20\%$ rise to $>5\times$ URL.
 - Plus new ischemic ECG changes, angiographic complication, or new imaging loss of viable myocardium.
- **Type 4b:** stent thrombosis. **Type 4c:** restenosis. **Type 5** (CABG): $>10\times$ URL plus new Q waves, graft occlusion, or imaging evidence.
- **SCAI definition** (prognostically significant periprocedural MI): CK-MB $\geq 10\times$ ULN ($\geq 5\times$ with new Q waves) or cTn $\geq 70\times$.

Pretreatment with antiplatelet therapy and a high-intensity statin load $\downarrow$ periprocedural MI.

SCAD, vasospasm and INOCA

SCAD: $\sim 90\%$ women; peripartum, FMD, extreme exertion/emotion. Look for an abrupt caliber change or long smooth narrowing (type 2).

- **Conservative management preferred** (most heal). PCI only for ongoing ischemia, instability or LM/proximal involvement (high failure rate).
- Prefer IVUS over OCT (contrast flush can propagate). Screen for FMD. Beta-blockers $\downarrow$ recurrence.
- **Vasospasm:** VSMC hyperreactivity $\pm$ endothelial dysfunction; East Asian ancestry, smoking, 5-FU, cocaine. Treat with CCB $\pm$ nitrates; avoid non-selective BB. **INOCA/MVD thresholds:** CFR < 2.0 – 2.5 , IMR ≥ 25 , abnormal ACh test.
- Epicardial spasm: $\geq 90\%$ constriction with symptoms and ECG changes.
- Microvascular spasm: symptoms and ECG changes without epicardial spasm.
- **CorMicA:** stratified therapy guided by invasive testing improved angina.

Non-invasive assessment: PET or CMR myocardial blood flow reserve (2a).

Coronary anomalies

Seen in 0.6–1.5% of angiograms; $\sim 80\%$ benign. **Second leading cause of SCD in young athletes** (after HCM). SCD is often the first event, typically during or after exertion. Stress testing is unreliable because obstruction is dynamic.

Most common: separate LAD/LCx ostia, then **LCx from the right sinus or RCA** (retroaortic, benign). **Clues:** long LM without LCx filling; an unperfused territory.

High-risk: anomalous LEFT from the right sinus with an interarterial course. Other high-risk features are slit-like ostium, acute take-off angle, intramural segment (length) and lateral compression on IVUS. Anomalous RCA from the left sinus is more common but lower risk.

- **Anomalous LM courses from the right sinus:**
- **Interarterial:** normal-length LAD; “anterior dot” in RAO; dangerous.
- **Retroaortic:** “posterior dot”; benign.
- **Septal (intramyocardial):** gives septal branches; short LAD.
- **Prepulmonic:** short LAD.
- **Work-up:** CCTA (or CMR in young patients to avoid radiation), ischemia testing, IVUS.
- **Treatment:**
- **Surgery (unroofing preferred)** for anomalous left or right coronary with ischemia or symptoms (Class 1). Reasonable for asymptomatic anomalous LEFT with interarterial/intramural course, especially young or athletic patients.
- Anomalous RCA without symptoms or ischemia: observation, shared decision-making.
- Bypass grafts fail from competitive flow; reimplantation risks ostial stenosis. PCI reserved for poor surgical candidates.
- **Engagement:** RCA from left sinus → AL1/AL2, JL; LCx from right sinus → JR4, AR, MP; LM from right sinus → JL/AL. **Coronary fistula:** congenital or acquired (PCI, CABG, biopsy, ablation, myectomy). Origin RCA ≈ LAD > LCx; >90% drain to the right heart/PA.
- Close (catheter or surgery) if large, or small–moderate with ischemia, arrhythmia, ventricular enlargement/dysfunction, or endocarditis. Small asymptomatic: no closure; follow every 3–5 yr.
- **ALCAPA** (Bland–White–Garland): infantile form (~85– 90%, HF with feeding) vs adult form (RCA collaterals → steal, MR, VT, SCD). Surgical reimplantation.
- **Myocardial bridge:** BB ± non-DHP CCB (avoid nitrates), ± ivabradine. Unroofing/myotomy if refractory. **Avoid stenting** (perforation, fracture, ISR).
- **Ectasia:** ≥1.5× reference diameter. ACS with ectasia and thrombus: antithrombotic medical therapy.

6. Intracoronary Imaging and Physiology

IVUS and OCT

	IVUS	OCT
Energy	Ultrasound, 20–60 MHz (higher frequency = better resolution, less depth)	Near-infrared light; blood must be cleared (contrast flush)
Axial resolution	~100–200 μm (HD ~40 μm)	10–20 μm
Penetration	4–8 mm ; sees EEM through plaque	1–2 mm; lipid attenuates signal (EEM often hidden)

	IVUS	OCT
Best for	Ostial/LM, large vessels, CTO/subintimal tracking, CKD (little contrast) , SCAD, plaque burden, sizing	Cap thickness/TCFA, calcium thickness and arc , thrombus (red = attenuating, white = low attenuation), erosion, edge dissection, malapposition, tissue protrusion, ISR mechanism
Limits	Calcium shadowing, blood speckle	Cannot image ostial LM; contrast load
Artifacts	NURD (mechanical rotational), ring-down (solid-state/phased array), obliquity (overestimates size), guidewire shadow	Incomplete blood clearance, sew-up (motion), sunflower/merry-go-round (eccentric catheter), saturation, fold-over, air bubbles

Outcome evidence

IVUS-XPL and **ULTIMATE**: IVUS-guided DES ↓ TVF. Meeting ULTIMATE optimization criteria lowered TVF further.

- **ADAPT-DES** (registry): IVUS benefit greatest in ACS and complex disease.

RENOVATE-COMPLEX-PCI (2023): imaging-guided complex PCI ↓ TVF (7.7% vs 12.3%).

- **OCTOBER (2023)**: OCT-guided complex bifurcation PCI ↓ MACE.
- **ILUMIEN IV (2023)**: OCT ↑ MSA and ↓ ST, but TVF was not significantly lower.
- **OCTIVUS**: OCT noninferior to IVUS.
- **IVUS-ACS (2024)**: ↓ TVF in ACS. **Network meta-analysis (2024)**: imaging guidance ↓ cardiac death, MI, ST and TVR.
- **Guidelines**: Class 1 for LM and complex lesions (2025 ACS; ESC 2024 CCS). ACC/AHA 2021: 2a for PCI guidance, 2a for mechanism of stent failure.

UPDATED The original cited 2011 IIa/IIb recommendations and said “OCT has less data.” Imaging-guided PCI now carries a Class 1 recommendation for complex disease.

Optimization targets

ULTIMATE: MSA >5.0 mm² (IVUS) or >90% of distal reference lumen; edge plaque burden <50% within 5 mm; no edge dissection involving media >3 mm.

OCT: MSA >4.5 mm²; expansion >80–90%. **LM “5-6-7-8” rule (IVUS MSA)**: ostial LCx 5, ostial LAD 6, polygon of confluence 7, proximal LM 8 mm².

- **ISR/ST predictors**: MSA <5.5 mm² (IVUS), stent length >40 mm, underexpansion, geographic miss, edge dissection, tissue protrusion. Acute malapposition alone is not predictive.

OCT calcium score (arc >180° = 2, thickness >0.5 mm = 1, length >5 mm = 1): score 4 predicts underexpansion, so modify the calcium first.

Imaging for lesion severity

LM: IVUS MLA ≥6 mm² → safe to defer (LITRO; ~FFR 0.80 in Western patients; ~4.5–4.8 mm² in Asian series). MLA <6 → revascularize or confirm with physiology.

- **Non-LM**: MLA ≥4 mm² identifies non-ischemic lesions, but **a small MLA does not prove ischemia**. Use physiology.

- **FLAVOUR (2022)**: FFR-guided PCI noninferior to IVUS-guided for intermediate (40–70%) lesions, with fewer PCIs.
- Vulnerable-plaque features (PROSPECT): plaque burden >70%, MLA <4 mm², TCFA. **PREVENT (2024)**: preventive PCI of vulnerable non-flow-limiting plaque ↓ target vessel events; not yet guideline practice.

Coronary physiology

Indices and cutoffs

FFR = Pd/Pa at maximal hyperemia (full form (Pd–Pv)/(Pa–Pv); subtract RA pressure if RA is markedly elevated).

≤0.80 = ischemic (older 0.75 validation).

- Lesion-specific and epicardial; independent of HR, BP and contractility.
- Lower FFR (<0.60) = higher event risk and larger benefit.

Non-hyperemic pressure ratios (iFR, RFR, dPR): **≤0.89**. No adenosine; wave-free diastole. iFR pullback with coregistration predicts post-PCI result.

Angiography-derived FFR (QFR): FAVOR III China beat angiography, but **FAVOR III Europe (2024)**: **QFR inferior to FFR**.

CFR = hyperemic/resting flow (Doppler or thermodilution). Integrates epicardial and microvascular function. <2.0 abnormal (<2.5 for MVD); normal 3–5.

- **IMR** ≥25 = microvascular dysfunction.

FFR/iFR discordance: FFR+/iFR– → focal lesion with preserved hyperemic flow. FFR–/iFR+ → diffuse disease or impaired microvascular reserve. Deferred discordant lesions do well.

Key trials

DEFER: PCI of FFR ≥0.75 lesions gave no benefit at 5 and 15 yr; deferral is safe. **FAME**: FFR-guided multivessel PCI ↓ death/MI/revascularization with fewer stents. Intermediate (50–70%) angiographic lesions were most often misclassified.

FAME 2: FFR ≤0.80 → PCI + OMT ↓ urgent revascularization (not death/MI) vs OMT. Deferred FFR-negative lesions had low event rates.

- **FAME 3 (3VD)**: 1 yr, FFR-PCI failed noninferiority to CABG (MACCE; repeat revascularization). **5 yr: death/stroke/MI 16% vs 14% (NS)**; MI and repeat revascularization higher with PCI.

DEFINE-FLAIR and **iFR-SWEDEHEART (2017)**: iFR noninferior at 1 yr with fewer revascularizations and less discomfort. **5-yr DEFINE-FLAIR**: MACE NS, but all-cause death 9.0% vs 6.2% (HR 1.56) with iFR; iFR-SWEDEHEART showed no difference.

- **Guidelines (2021 ACC/AHA)**: FFR or iFR is **Class 1** for intermediate stenosis in SIHD without prior ischemia testing.

UPDATED Add the 5-yr FAME 3 convergence and the unexplained 5-yr iFR mortality signal. Both are now commonly tested.

STEMI non-culprit physiology

DANAMI-3-PRIMULTI and **COMPARE-ACUTE**: FFR-guided complete revascularization ↓ repeat revascularization vs culprit-only. Non-culprit FFR is safe acutely. **FLOWER-MI (2021)**: FFR- vs angiography-guided complete revascularization: no difference; fewer stents with FFR.

- **FRAME-AMI (2023)**: FFR-guided better than angiography-guided.

FULL REVASC (2024): FFR-guided complete revascularization did not ↓ death/MI/unplanned revascularization vs culprit-only.

In acute STEMI, microvascular stunning can cause **false-negative FFR**. Non-culprit assessment is reasonable; culprit FFR is not valid.

Pitfalls

False-negative (FFR too high):

- Inadequate hyperemia (caffeine, theophylline, low IC dose)
- Guide damping/deep engagement (disengage during measurement)

Guide side holes with IC adenosine

- Large guide

Severe microvascular dysfunction (acute STEMI, LVH >13 mm, LVEF <30%)

Small territory or non-viable myocardium

Pre-TAVR severe AS (FFR may fall after TAVR; iFR less affected)

False-positive (FFR too low):

- **Drift**: check pressure waveforms. A true distal gradient loses the dicrotic notch; drift keeps a similar shape. Pull the sensor back to the guide tip and re-equalize (target 1.00 ± 0.02).
- Wire-induced pseudostenosis (accordion)
- Spasm (always give IC NTG)
- Contrast in the guide (flush with saline)
- Sensor placed too distal in a diffusely diseased vessel
- **Serial lesions**: hyperemic pullback, treat the largest stepup first, then re-measure (lesions interact).

Adenosine: IV 140 mcg/kg/min or IC boluses (RCA ~50–100 mcg, LCA ~100–200 mcg). Watch for AV block in the RCA. Dipyridamole potentiates adenosine.

7. Chronic Coronary Disease, Revascularization, Left Main and Bifurcation

Testing (2021 chest pain / 2023 CCD guidelines)

No known CAD

- **Low risk**: defer testing (Class 1), or CAC / exercise ECG (2a).

- **Intermediate–high risk:** CCTA (favored <65 y, not on OMT) or stress imaging (favored ≥65 y or known CAD burden). PET > SPECT (2a).
- Exercise ECG is reasonable if the ECG is interpretable and the patient can do ≥5 METs.
- CT-FFR is reasonable for 40–90% proximal/mid stenosis.
- Add CAC to stress imaging (2a). CAC or CCTA in asymptomatic screening without indication: Class 3.

Known CAD

- Stress imaging for new or worsening symptoms (Class 1).
- CCTA reasonable for stents ≥3 mm, grafts, or nonobstructive CAD progression.
- **No routine periodic stress testing** when stable.
- **High-risk non-invasive findings:** ≥10% ischemic myocardium (SPECT/PET), ≥3 segments on stress echo, CMR ≥12% or ≥3 segments, T1D, exercise hypotension. **Go to invasive angiography:** CCTA showing ≥50% LM or ≥70% 3-vessel disease with symptoms (Class 1); or persistent limiting angina / high-risk ischemia on GDMT. Physiology if no prior testing.
- **Viability:** LGE extent has best sensitivity for recovery; dobutamine CMR best specificity. **STICH:** viability did not predict CABG benefit. Stunning = transient dysfunction with normal flow; hibernation = chronic hypoperfusion.

Medical therapy

- BB or CCB first-line for angina, then long-acting nitrate or ranolazine.
- **Ranolazine:** late Na⁺ current inhibitor; no HR/BP effect; QT prolongation; CYP3A interactions.
- **Long-term BB without a specific indication** (MI within ~1 yr, LVEF ≤50%, arrhythmia, angina) does not ↓ events. BB benefit after MI with preserved EF is contested: REDUCE-AMI and REBOOT neutral, BETAMI-DANBLOCK favorable.

Vasospastic angina: CCB + nitrates.

- High-intensity statin. **2026 dyslipidemia guideline:** LDL <55 mg/dL for very-high-risk ASCVD, <70 for others; add ezetimibe, then PCSK9i / inclisiran / bempedoic acid.

SGLT2i and GLP-1 RA in DM or obesity; BP <130/80; icosapent ethyl if TG 135–499 on statin (REDUCE-IT). Colchicine 2b. EECP 2b.

Revascularization trials

- **COURAGE (2007):** PCI + OMT did not ↓ death/MI in SIHD.
- **ISCHEMIA (2020)** (moderate–severe ischemia; excluded LM ≥50%, LVEF <35%, recent ACS, unacceptable angina): no ↓ death/MI; **better angina/QoL**. ISCHEMIA-EXTEND: ↓ CV death, ↑ non-CV death, all-cause similar.
- **REVIVED-BCIS2 (2022):** PCI in LVEF ≤35% with viability did not ↓ death/HF hospitalization.
- **STICH/STICHES:** CABG ↓ all-cause death at 10 yr in ischemic cardiomyopathy (LVEF ≤35%).
- **FREEDOM:** DM with multivessel disease, CABG ↓ death/MI; ↑ stroke.

- **SYNTAX:** PCI failed noninferiority. CABG better for SYNTAX ≥ 33 ; LM subgroup similar (more revascularization with PCI, more stroke with CABG).
- **EXCEL** (LM, SYNTAX ≤ 32): 3-yr noninferior. 5-yr primary NS (22.0% vs 19.2%), but **all-cause death higher with PCI (13.0% vs 9.9%)**, plus more ischemia-driven revascularization.
- **NOBLE** (LM): 5-yr MACCE worse with PCI (28% vs 19%), driven by non-procedural MI and repeat revascularization; no mortality difference.
- **Individual-patient meta-analysis (11 RCTs):** CABG ↓ mortality in multivessel disease with DM or high SYNTAX; similar in LM with low–intermediate SYNTAX.

Guideline recommendations (ACC/AHA 2021 revascularization, 2023 CCD)

- **Heart team** for complex multivessel/LM disease, DM or LV dysfunction (Class 1).

To improve survival:

- Significant LM → CABG (Class 1); PCI reasonable for low–intermediate complexity LM (2a). Multivessel CAD with LVEF $\leq 35\%$ → CABG (Class 1). DM + multivessel CAD including LAD → CABG with LIMA (Class 1); PCI if poor surgical candidate (2b). 3VD with normal EF → CABG 2b; PCI benefit uncertain. Complex multivessel disease (SYNTAX > 33) → CABG over PCI (2a).
- **For symptoms:** revascularize refractory angina despite GDMT (Class 1). Patients should be on adequate antianginal therapy first.

DES over BMS in all (Class 1), including HBR.

- **Risk scores:** SYNTAX low ≤ 22 , intermediate 23–32, high ≥ 33 (anatomic complexity, irrelevant to surgeons). STS PROM low $< 4\%$, intermediate 4–8%, high $> 8\%$. If on the fence and the patient has AF with high HASBLED, favor CABG (avoids triple therapy).

Appropriate use in SIHD (condensed)

- **Rarely appropriate:** 1–2-vessel disease without proximal LAD, asymptomatic or not on antianginals, with low-risk findings.
- **Appropriate:** symptoms on ≥ 1 –2 antianginals, or highrisk ischemia.
- **3VD / LM:** CABG appropriate across scenarios. LM PCI rarely appropriate with high SYNTAX in asymptomatic patients.
- **“Severe” for AUC:** $\geq 70\%$ non-LM ($\geq 50\%$ LM), or 40–69% with FFR ≤ 0.80 / iFR ≤ 0.89 .

NCDR CathPCI metrics: risk-adjusted mortality, bleeding, AKI, 30-d readmission, discharge meds (aspirin/P2Y12/statin), radiation dose, cardiac rehab referral, STEMI door-to-device times, AUC. Outliers often reflect documentation or abstraction errors.

Left main assessment

- Angiography is unreliable (short segment, no reference, ostial catheter effects, overlap).
- **IVUS MLA $< 6 \text{ mm}^2$ or FFR ≤ 0.80 / iFR ≤ 0.89** is significant. Downstream LAD/LCx disease can falsely raise LM FFR. Measure in the non-diseased daughter or treat the distal lesion first.

LM STEMI or shock → emergency PCI. Elective LM is never ad hoc; heart team.

Bifurcation PCI

Plaque forms on the lateral wall (low shear) and spares the carina. **Carina shift**, more than plaque shift, pinches the side branch after main-vessel stenting; most pinches are not FFR-significant.

- **Medina (MB prox, MB distal, SB):** 1 = $\geq 50\%$ stenosis. Descriptive, not prognostic.
- **Provisional stenting is the default** (Class 1):
 - Wire both branches (jailed SB wire).
 - Stent the MV sized to distal MV.
 - **POT** with a short NC balloon sized to the proximal MV. Treat SB only for TIMI < 3 , severe dissection, or severe ostial compromise with ischemia (T/TAP or culotte).
- **Planned 2-stent for complex bifurcations (DEFINITION criteria):**
 - **Major:** SB lesion length ≥ 10 mm AND SB DS $\geq 70\%$ (distal LM) or $\geq 90\%$ (non-LM).
 - **Minor:** moderate–severe calcification, multiple lesions, angle $< 45^\circ$ or $> 70^\circ$, MV RVD < 2.5 mm, thrombus, MV lesion ≥ 25 mm.
 - **Complex = 1 major + ≥ 2 minor.**
- **Trials:**
 - **DKCRUSH-V:** DK crush $\downarrow$ TLF vs provisional in true distal LM bifurcations.
 - **DEFINITION II:** systematic 2-stent (mostly DK crush) $\downarrow$ TLF in complex bifurcations.
 - **EBC MAIN (2021):** provisional $\approx$ planned 2-stent in distal LM at 1 and 3 yr; simpler and cheaper.
 - **EXCEL/NOBLE:** most distal LM treated provisionally.

DK crush steps:

- Wire both; SB stent with 2–3 mm protrusion; balloon crush.
- Rewire through a proximal cell; 1st kissing.
- MV stent; POT.

Rewire; 2nd kissing (NC balloons); final POT.

- In 2-stent techniques, **final kissing with NC balloons is mandatory**. Imaging is essentially mandatory in LM.

8. Acute Coronary Syndromes, Cardiogenic Shock and MCS

NSTE-ACS: risk and timing (2025 ACC/AHA)

ECG within 10 min; hs-troponin 0/1–2 h pathways. Troponin level correlates with risk even below the MI cutoff.

- **TIMI score (1 point each):** age ≥ 65 ; ≥ 3 risk factors; known CAD $\geq 50\%$; aspirin in the past 7 d; ≥ 2 angina episodes in 24 h; ST deviation ≥ 0.5 mm; positive biomarker.

GRACE > 140 = high risk.

- **Timing of invasive strategy:**

Immediate (<2 h): refractory angina, hemodynamic or electrical instability, acute HF, mechanical complication.

- ****Early (<24 h):**** high risk (GRACE >140, dynamic ST changes, rising troponin) (2a). TIMACS: benefit of early cath limited to GRACE >140.
- **Invasive during hospitalization** for intermediate/high risk (Class 1).
- **Selective invasive** for low risk (negative troponin, no ST change); Class 3 for early invasive with low likelihood of ACS or prohibitive comorbidity.
- **SENIOR-RITA (2024):** invasive strategy in NSTEMI ≥75 y ↓ non-fatal MI, no mortality difference.
- **Avoid:** IV BB if shock risk (COMMIT); O₂ unless SpO₂ <90% (DETO2X); NSAIDs; excess morphine (delays P2Y₁₂ absorption).
- **Transfusion (MINT 2023):** restrictive strategy not noninferior. Hb ≥10 may be reasonable (2b).
- **Complete revascularization** is Class 1 in NSTEMI-ACS (angiography- or physiology-guided).

UPDATED The original used 2014 timing tiers (72 h), “Hb threshold 8,” and Class 1 for GPI if not preloaded. See the 2025 recommendations above.

STEMI reperfusion

- **Primary PCI targets:** FMC-to-device ≤90 min at a PCI-capable center; ≤120 min with transfer (door-in-door-out ≤30 min).
- Each 30-min delay ↑ 1-yr mortality ~7.5% (relative).
- Primary PCI beats lytics (↓ death, reinfarction, stroke).
- **Fibrinolysis** if PCI is not achievable within 120 min and symptoms ≤12 h (Class 1). Give within 30 min of arrival (≤10 min of diagnosis prehospital).
- 12–24 h with ongoing ischemia, large territory or instability: lysis 2a if no PCI available.
- **Lysis for ST depression: harm (Class 3)**, except posterior MI.

Absolute contraindications to lysis:

- Any prior ICH
- Structural cerebrovascular lesion (AVM, aneurysm) or malignant intracranial neoplasm
- Ischemic stroke within 3 mo (except acute stroke within 4.5 h)
- Suspected aortic dissection
- Active bleeding or bleeding diathesis (not menses) Significant closed-head/facial trauma within 3 mo
- Intracranial/intraspinal surgery within 2 mo
- Severe uncontrolled HTN unresponsive to therapy
- **Lytic facts:** ICH ~1%. ~50–60% achieve TIMI 3 flow by 90 min. Reocclusion/reinfarction 5–10%.
- **Adjuncts with lysis:**
- Aspirin 162–325 mg.

- **Clopidogrel 300 mg (≤ 75 y) or 75 mg (> 75 y)**, continued ≥ 14 d up to 12 mo. **TREAT:** switching to ticagrelor after lysis (< 75 y) is noninferior for bleeding. Anticoagulation ≥ 48 h, ideally for the hospitalization (≤ 8 d) or until revascularization.
- **Enoxaparin preferred** (ExTRACT-TIMI 25: $\downarrow$ death/MI vs UFH). < 75 y: 30 mg IV + 1 mg/kg SC q12h. ≥ 75 y: no bolus, 0.75 mg/kg q12h. CrCl < 30 : daily.
- UFH: aPTT 1.5–2 $\times$ control for 48 h. Fondaparinux acceptable, but not alone for PCI.
- STREAM: half-dose TNK in ≥ 75 y (ICH).
- **After lysis:**
- **Failed reperfusion** ($< 50\%$ ST resolution at 60–90 min, ongoing pain, instability) $\rightarrow$ **immediate rescue PCI** (Class 1).
- Successful $\rightarrow$ **angiography within 2–24 h** (Class 1, pharmacoinvasive).
- STREAM: prehospital TNK within 3 h + planned cath $\approx$ primary PCI outcomes ($\sim 1/3$ needed urgent cath). **Facilitated PCI** (immediate PCI right after full-dose lytic) is harmful.

Primary PCI details:

- **Routine aspiration thrombectomy: Class 3** (TASTE, TOTAL: $\uparrow$ stroke). Bailout only.
- Deferred stenting has no benefit (DANAMI-3-DEFER). UFH, or bivalirudin (BRIGHT-4 regimen). GPII bailout only.
- **OAT:** stable patients > 24 h (days 3–28) with an occluded IRA and no ischemia: no benefit to PCI (Class 3).

Complete revascularization in STEMI

PRAMI, CvLPRIT, DANAMI-3-PRIMULTI, COMPAREACUTE, COMPLETE (2019): non-culprit PCI $\downarrow$ CV death/MI (COMPLETE) vs culprit-only.

Class 1 for hemodynamically stable STEMI.

Timing:

- **MULTISTARS AMI (2023):** immediate multivessel PCI noninferior (fewer MI/unplanned revascularizations) vs staged at 19–45 d.

BIOVASC: immediate $\approx$ staged in ACS.

- 2025 guideline has a slight preference for single procedure; staging within 45 d is acceptable.
- **FIRE (2023):** physiology-guided complete revascularization in ≥ 75 y $\downarrow$ events.
- Complex residual disease (high SYNTAX) $\rightarrow$ heart team (staged CABG).
- **Cardiogenic shock: culprit-only at index PCI** (CULPRITSHOCK: $\downarrow$ 30-d death/RRT). Routine non-culprit PCI at index is Class 3 (harm).

Cardiogenic shock

Epidemiology: complicates ~ 5 – 10% of STEMI (less NSTEMI). Leading cause of in-hospital AMI death; mortality 40–50%. Often develops after admission. 70–80% have multivessel disease.

- **NSTEMI-shock:** older, more comorbidity, often an occluded LCx.

- **Hemodynamics:**
- SBP <90 ≥30 min or pressors needed; CI <1.8 unsupported (<2.2 supported); PCWP >15; RAP >10.
- **CPO** = MAP × CO / 451; **<0.6 W** is the strongest hemodynamic mortality predictor (SHOCK).
- **PAPi** = (PASP – PADP)/RAP; <0.9–1.0 = RV failure.
- RAP/PCWP >0.8 = RV-predominant.
- **SCAI SHOCK stages:**
- **A** at risk
- **B** beginning: hypotension/tachycardia without hypoperfusion
- **C** classic: hypoperfusion needing a pressor, inotrope or MCS; lactate ≥2
- **D** deteriorating despite initial therapy
- **E** extremis: CPR/ECMO, lactate ≥5, pH ≤7.2

2025 guideline:

- Emergency culprit revascularization regardless of age (Class 1).
- Microaxial flow pump in selected STEMI-CS (2a).
- **Routine IABP or VA-ECMO: Class 3 (no benefit).** Transfer to a shock center; MCS as bridge for mechanical complications (2a).
- **Pressors: norepinephrine first.** Inotropes: dobutamine or milrinone (DOREMI: similar). Dopamine ↑ arrhythmias (SOAP II). Epinephrine ↑ refractory shock (OptimaCC). PA catheter for unclear or deteriorating cases.
- Red flags: lactate ≥4, Cr ≥2, CPO <0.6, age ≥70, OHCA.

UPDATED The original said ECMO has “sufficient data demonstrating improved survival” and recommended dopamine as the add-on inotrope. Both are incorrect now.

- **Trials:**
- **SHOCK (1999):** emergency revascularization — 30-d NS, ↓ **6-mo and long-term mortality**.
- **IABP-SHOCK II:** no benefit at 30 d, 1 yr or 6 yr.
- **ECLS-SHOCK (2023): VA-ECMO did not ↓ 30-d mortality;** ↑ bleeding and vascular complications.
- **DanGer Shock (2024):** Impella CP in STEMI-CS (excluded comatose OHCA and RV failure) ↓ **180-d death (45.8% vs 58.5%);** ↑ bleeding, limb ischemia, RRT.

Cardiac arrest

- **TTM2 (2021):** 33 °C no better than normothermia with fever prevention.
- **2025 AHA:** deliberate temperature control (32–37.5 °C) for **≥36 h** in comatose patients.
- **COACT/TOMAHAWK:** no benefit of immediate angiography after OHCA without ST elevation. Immediate cath is for STEMI or ongoing instability.

Mechanical circulatory support

Device	Configuration / flow	Key physiology / contraindications
IABP	Descending aorta; +0.5–1 L/min	Diastolic augmentation (↑ coronary perfusion), ↓ afterload and MVO ₂ ; needs native function/rhythm. Contraindicated: ≥ moderate AR, aortic dissection/aneurysm, severe PAD. Useful bridge in VSR or acute MR.
Impella 2.5 / CP / 5.5	LV → ascending aorta; ~2.5 / 3.5–4 / 5.5 L/min	Unloads LV (↓ LVEDP, wall stress). Hemolysis, bleeding, limb ischemia. Contraindicated: LV thrombus, mechanical AVR, severe AS/AR (relative), severe PAD, VSD (right-to-left shunt). Migration into the LV shows a ventricular placement signal.
Impella RP / RP Flex	IVC (femoral) or IJ → PA; ~4 L/min	RV failure; avoid with PA/IVC thrombus.
TandemHeart	LA (transseptal) → femoral artery; up to 4–5 L/min	Needs transseptal puncture. Cannula migration into RA → desaturated return (hypoxemia). Contraindicated with LA thrombus or VSD.
ProtekDuo	RA → PA via IJ dual-lumen (± oxygenator)	Ambulatory RV support.
VA-ECMO	Venous drainage → femoral arterial return; biventricular + respiratory support	Only full cardiopulmonary support. ↑ LV afterload (LV distension, pulmonary edema, AR/MR worse, LV thrombus): vent with Impella (ECPELLA) or IABP. Differential (North–South) hypoxemia. Limb ischemia (distal perfusion cannula).

MCS for high-risk PCI

- **BCIS-1:** elective IABP gave no ↓ MACCE (later long-term mortality signal).
- **PROTECT II:** Impella 2.5 vs IABP — 30-d NS; 90-d per-protocol trend favoring Impella.
- **CHIP-BCIS3 (ACC 2026):** Impella CP in LVEF ≤35% complex PCI gave **no benefit and higher CV death** (26.7% vs 14.5%).
- **PROTECT IV:** results pending.

Routine MCS for high-risk PCI is not supported.

Mechanical complications of MI

- **Timing:** bimodal (<24 h and 3–5 d). Suspect with any new murmur or shock ≥3 d after MI, especially late presentation or first MI without reperfusion.

Ventricular septal rupture

- Risk: older, female, anterior MI, no prior smoking, late presentation, higher Killip/HR.
- Anterior MI → apical VSR; inferior MI → basal VSR (worse, harder to repair).
- Harsh holosystolic murmur; **O₂ step-up RA → RV.**

- Highest shock mortality (~87% in SHOCK registry). Afterload reduction and IABP/Impella bridge (Impella risks shunt reversal/embolization); surgery. Device closure for residual defects or as a bridge.

Papillary muscle rupture

- **Posteromedial muscle** (single supply from PDA; inferior MI).
- Abrupt pulmonary edema/shock; murmur may be soft or absent.
- **Giant v waves, no step-up.**
- Nitroprusside (if BP allows), IABP/Impella, emergent surgery.
- Papillary muscle *dysfunction* gives slower, milder MR.

Free-wall rupture

- Older women, HTN, first MI, late lysis.
- Sudden tamponade, PEA, equalized diastolic pressures. Pericardiocentesis only as a temporizing bridge to surgery. LV pseudoaneurysm (narrow neck) → surgery.

RV infarct

- Hypotension, clear lungs, ↑ JVP; RAP/PCWP >0.8. Maintain preload (avoid nitrates/diuretics), AV synchrony, inotropes, RV MCS.
- **Bezold–Jarisch reflex:** bradycardia/hypotension with inferior reperfusion; atropine and fluids.
- **Takotsubo with LVOTO:** avoid inotropes/nitrates. Fluids, BB, phenylephrine.

9. Complex PCI: Calcium, CTO, Bypass Grafts, ISR and Stent Thrombosis

Calcium and plaque modification

- ~1/3 of PCI lesions have moderate–severe calcium (underestimated by angiography). Risk: CKD/ESRD, DM, age, smoking.
- Consequences: device-delivery failure, stent underexpansion/malapposition, polymer damage, dislodgement, perforation, ISR/ST.
- Goal of atherectomy is **modification** for balloon/stent expansion, not debulking to a final result.
- **Balloons**
- Semi-compliant: deliverable, but “dog-bone” edge dissection.
- Non-compliant: focused force; used for predilation of fibrocalcific lesions and post-dilation.
- Cutting/scoring: ostial lesions, ISR, prevent slippage. Inflate slowly; do not exceed rated pressure (blade fracture/perforation).
- Super-high-pressure balloon (OPN).
- **If pre-dilation fully expands the balloon**, you can stent without atherectomy.

Rotational atherectomy

- Diamond-coated front of an elliptical burr. **Differential cutting** ablates inelastic tissue and deflects elastic wall. **Orthogonal displacement of friction** at >60k rpm.
- Debris <10 µm (smaller than capillaries), but a large burden still causes slow flow.
- **RotaWire**: 0.009" shaft with 0.014" radiopaque tip. Floppy (less bias) vs Extra Support (more bias, straightens vessel). Burr must not reach the radiopaque tip (fracture).
- **Burr sizing**: 1.25/1.5 mm (6F), 1.75 (7F), 2.0 (8F), up to 2.5 mm (9–10F). **Burr:artery ratio 0.5–0.6**; >0.7 → perforation/complications (STRATAS, CARAT). Most cases use a 1.25–1.5 mm burr.
- **Technique**:
 - 135–180k rpm; favor lower (~140–160k).
 - **Avoid decelerations >5,000 rpm.**
 - Pecking motion; runs ≤15–20 s.
 - **Never stop the burr distal to the lesion.**
 - Rotaglide flush (egg/olive-oil allergy) ± verapamil/NTG in flush.

Complications

- Slow/no-reflow: small burr, low speed, short runs, vasodilators; treat downstream lesions that trap debris.
- **Bradycardia/AV block** with dominant RCA/LCx: TVP, or atropine/aminophylline.
- Perforation (oversized burr, angulation, wire bias). **Burr entrapment** (“Kokeshi,” usually small burr through a tight lesion):
 - Lock the advancer and pull the whole system while controlling the guide.
 - Second guide/wire with balloon dilation beside the burr. Cut the drive shaft, advance a guide extension over it for coaxial traction.
 - Surgery as last resort.
- **Trials**: ERBAC/COBRA/DART showed ↑ procedural success but no restenosis benefit. **ROTAXUS**: routine rota ↑ success, no MACE benefit (and more late loss). **PREPARE-CALC**: rota ≈ modified balloons.

Orbital atherectomy

- Eccentric 1.25 mm diamond-coated crown, **coated on both sides** (sands forward and backward).
- Orbit diameter ↑ with speed (80k “low” → 120k “high”), slower advancement and more passes.
- **ViperWire**: 0.012" shaft, 0.014" tip. **ViperSlide**: soybean oil, egg-yolk phospholipids.
- Runs ≤30 s; start low speed; ~1 mm/s advance.
- **ORBIT II**: safe and effective (single arm). **ECLIPSE (2024)**: routine OA vs balloon prep in severe calcium — no ↓ 1-yr TVF, no ↑ MSA.
- **Laser (ELCA) and intravascular lithotripsy**
- **ELCA** is a 308 nm excimer laser over a standard 0.014" wire.

- Mechanisms: **photochemical** (bond breakage), **photothermal** (vapor bubble), **photomechanical** (shock wave).
- **Flush saline continuously; no contrast or blood** in the field (both absorb energy → dissection).
- Advance ~0.5–1 mm/s; titrate fluence and rate. Uses: uncrossable lesions, **underexpanded stents** (especially with contrast), thrombus, SVG, CTO cap. Relative contraindication in unprotected LM. Laser (not rota) is approved for SVG and ISR.
- **IVL (Shockwave C2+)**
- 2.5–4.0 × 12 mm; 1:1 sizing; 4 atm → pulses → 6 atm. Circumferential and deep calcium fracture. May trigger ventricular capture (“shocktopics”).

Low perforation rate (DISRUPT CAD III).

Crossing profile is bulky and may need pre-dilation.

Wire and device crossing toolbox

- **Wires:** tapered hydrophilic/polymer (microchannels), buddy wire, stiffer support wire, knuckle. Pull back the stiff segment if you see an accordion pseudolesion.
- **Support:** microcatheter (Corsair, Turnpike/Gold, Tornus), guide extension, anchor balloon, larger or more supportive guide.
- **Modification:** smaller balloon, “grenadoplasty” (rupture balloon in lesion), rota/laser/IVL, subintimal strategies.

Chronic total occlusion

- **Definition:** TIMI 0 for ≥3 mo (probable if duration unknown).
- Present in ~15–20% of angiograms; PAD is the strongest predictor. Non-IRA CTO in STEMI ↑ mortality (worst with shock).
- **Operator/center volume** matters for success and complications. Current success ~85–90%. MACE ~2–3%; death ~0.5–1% (PROGRESS-CTO, OPEN-CTO).
- **J-CTO** (1 point each: blunt cap, calcification, bend >45°, length ≥20 mm, prior failed attempt): 0 easy; 1 intermediate; 2 difficult; ≥3 very difficult Higher scores → dissection/re-entry or retrograde approach
- **Hybrid algorithm** (dual injection is essential). Four questions: proximal cap ambiguity, lesion length (≥20 mm), distal target quality, interventional collaterals. Clear cap, short lesion → **antegrade wire escalation**. If subintimal: parallel wire, or move to ADR/retrograde. Long lesion, good distal target → **antegrade dissection re-entry** (knuckle wire, CrossBoss, Stingray LP; STRAW to decompress hematoma).
- Ambiguous cap, poor distal vessel, or prior failure with good collaterals → **retrograde** (septal > epicardial; reverse CART).
- **Stop the procedure if:** air kerma >5 Gy (strongly consider at >7 Gy); contrast >3.7× eGFR; >3 h; or remaining options are too risky. Consider an “investment procedure” (subintimal plaque modification, re-attempt ~2–3 mo).
- **Complications:** perforation (epicardial collaterals especially dangerous; **post-CABG loculated tamponade**), donor-vessel injury/ischemia, septal hematoma (usually benign, rarely serious).

- **Evidence:**

EuroCTO (2018): ↑ angina and QoL vs OMT.

DECISION-CTO (2019): no MACE difference; high crossover.

EXPLORE (non-IRA CTO after STEMI) and **REVASC:** no LVEF improvement. No RCT survival benefit.

ACC/AHA 2021: CTO PCI for refractory angina after treating non-CTO lesions: **2b**.

Saphenous vein grafts

Failure modes:

- Early (<1 mo): thrombosis/technical, ~10–15%
- 1 mo–1 yr: intimal hyperplasia
- 1 yr: accelerated atherosclerosis with friable lipid-rich, thin-capped plaque
- **Patency:** ~25% of SVGs failed at 12–18 mo (PREVENT IV); ~50% occluded by 10 yr. LIMA-LAD patency >90% at 10 yr; late LIMA failure from competitive flow, subclavian stenosis or anastomotic issues.

VEST (external stent): no patency improvement.

Strategy:

- Prefer **native-vessel PCI** over SVG PCI when feasible (2a).
- Redo CABG if failed or unused LIMA, multiple graft failures, heart team.
- Radial access is reasonable for graft angiography (more contrast/crossover, fewer vascular complications).
- **SVG PCI risks:** distal embolization and no-reflow (10–20%); periprocedural MI (predicts late death). Risk ↑ with plaque volume, degeneration score, female sex, CKD.
- **Embolic protection** (FilterWire EZ, Spider): needs a ~2.5–3 cm distal landing zone; can clog; not possible for distal anastomosis lesions. **2021 guideline: 2a** (was Class 1). **GPI: no benefit (Class 3).**
Covered stents: no benefit.
- No PCI of chronically occluded SVGs (Class 3).
- No-reflow prevention without EPD: IC nicardipine 100–200 mcg, verapamil 100–250 mcg, adenosine, nitroprusside; direct stenting; undersized stents.
- **DES vs BMS in SVG:** ISAR-CABG ↓ TLR with DES; DIVA and BASKET-SAVAGE mixed. DES is generally used; any benefit wanes over time.
- **LIMA PCI:** rare (2–3%/yr failure). Challenges: ischemia, visualization, damping, wire pseudolesions, guide dissection. Engage with IMA or JR4 catheter.

In-stent restenosis

BMS ISR is diffuse neointimal hyperplasia (SMC-rich, collagen III), peaking at 6–8 mo. BMS ISR rate ~20–30%. **DES ISR** is focal/edge, proteoglycan-rich, with late catchup, frequent peri-strut inflammation and **neoatherosclerosis**. Endothelialization can take years. Current DES ISR ~1–2%/yr.

~1/3 of ISR presents as ACS.

Definition: $\geq 50\%$ DS within the stent or 5 mm edges (symptoms/ischemia), or $\geq 70\%$ DS; IVUS MLA $< 4 \text{ mm}^2$ ($< 6 \text{ LM}$).

Risk factors:

Patient: DM (insulin resistance blunts mTOR-inhibitor effect), CKD/HD, smoking, female sex, ACS. Lesion: small vessel, long lesion, bifurcation, ostial, calcified, CTO, SVG.

Procedure: underexpansion (strongest), geographic miss, residual edge plaque, stent gap, fracture. Device: thick struts, polymer/drug hypersensitivity.

Management:

Image first to define the mechanism.

- Underexpansion → high-pressure NC, IVL, or laser for calcium behind struts.
- Neointima/neoatherosclerosis → **DES** (Class 1) or **DCB**. The **AGENT paclitaxel DCB is FDA-approved (2024)**; AGENT IDE showed ↓ TLF vs POBA.
- Cutting/scoring balloon to prevent slippage (2b). Recurrent multi-layer DES-ISR → **intravascular brachytherapy**, DCB, or CABG. Avoid a third layer. Laser > POBA for debulking ISR. Rota for ISR is mixed (ROSTER favorable, ARTIST unfavorable).

Stent thrombosis

ARC definitions:

Definite: angiographic or pathologic confirmation.

- **Probable:** unexplained death $\leq 30 \text{ d}$, or target-territory MI without angiography.

Possible: unexplained death $> 30 \text{ d}$.

- **Timing:** acute $< 24 \text{ h}$; subacute 1–30 d; late 30 d–1 yr; very late $> 1 \text{ yr}$.
- **Rates:** $< 1\%$ at 30 d, then $\sim 0.2\text{--}0.6\%/ \text{yr}$. Mostly early. Presents as STEMI/SCD; 30-d mortality $\sim 10\text{--}25\%$.
- **Early ST** is procedural: underexpansion/small MSA, edge dissection, geographic miss, tissue protrusion, inflow/outflow disease, inadequate antithrombotic therapy (e.g., after stopping bivalirudin).
- **Late/very late ST:** premature DAPT discontinuation (strongest early predictor), late-acquired malapposition, neoatherosclerosis, fracture, hypersensitivity. Patient factors: ACS, DM, CKD, low EF, long total stent length.
- After 6 mo on current DES, the ST risk from DAPT interruption is low. **Treatment:**
- Primary PCI; aspiration only if large thrombus.
- **Intravascular imaging** to find the mechanism; stent again only for dissection, underexpansion not fixed by balloon, or new disease.
- Switch or reload to ticagrelor/prasugrel; GPI; check adherence.
- Consider genotyping (CYP2C19 LOF) and prolonged DAPT.
- Ticagrelor, prasugrel and cangrelor have lower ST rates than clopidogrel.

- **DAPT score:** age ≥ 75 -2, 65–74 -1, < 65 0. +1 each for smoking, DM, MI at presentation, prior PCI/MI, stent < 3 mm, paclitaxel stent. +2 for CHF/LVEF $< 30\%$ or SVG PCI. **≥ 2 favors > 12 mo DAPT.**

10. Management of PCI Complications

Coronary perforation

- **Epidemiology:** ~0.3–0.5% overall (higher in CTO and atherectomy). Associated with tamponade, MI, emergency surgery and death. **Ellis classification:**
- **I:** extraluminal crater without extravasation
- **II:** pericardial/myocardial blush without a jet **III:** jet ≥ 1 mm
- **III-CS:** cavity spilling into a chamber (usually benign)
- **Immediate steps (all types):**
- **Proximal balloon tamponade** first; keep wire position. Support hemodynamics.
- Stat echo (IC agitated contrast/Definity can localize an ongoing leak). Pericardiocentesis if tamponade (echo or fluoro guided; autotransfusion). Notify surgery early. No further anticoagulant/GPI.
- **Large-vessel perforation** → **covered stent** (PK Papyrus, 5–6F; Graftmaster). Use a **dual-guide “block-and-deliver”** (ping-pong): the second guide delivers the covered stent while the first keeps the balloon up, minimizing un-tamponaded time. Residual leak → look for geographic miss.
- **Distal wire perforation** (polymer-jacketed or stiff wires): often recognized late, leading to delayed tamponade. Balloon occlusion.
- **Embolize via microcatheter:** coils (detachable allow repositioning), autologous fat or subcutaneous tissue (mix with contrast), thrombin, microspheres. Aspirate through the microcatheter to collapse the branch.
- Covered stent across the branch origin as a last resort.
- **Collateral perforation** (CTO): embolize **from both sides**. Epicardial collateral perforation is more dangerous than septal.
- **Protamine** is not routine. Use it (partial) only for a persistent leak after tamponade, after all equipment is out. Avoid after a covered stent if sealed (thrombosis). It does not reverse bivalirudin.
- **Prior CABG:** loculated, often posterior effusions that cannot be tapped percutaneously → CT-guided drainage or surgery. Mortality is high. SVG/LIMA perforation is catastrophic.
- **After a covered stent:** higher ST and restenosis; longer DAPT.
- **Myocardial staining** (end-hole injection or pigtail entrapment) is usually benign; large stains can cause VT. **RV perforation/free-wall rupture** is the dreaded RVventriculography complication.
- **Intramycardial hematoma or deep-wall dissection** (e.g., bridged LAD, post-dilation) is usually stable. Diagnosis of exclusion; echo.

No-reflow / slow flow

- **Definition:** TIMI < 3 (or TIMI 3 with myocardial blush grade 0–1) without epicardial obstruction.

- **Mechanisms:** distal embolization; ischemia-reperfusion (neutrophils, ROS, endothelial swelling, edema); vasospasm; individual susceptibility.
- **Settings:** STEMI (high thrombus, >6 h ischemia), SVG, atherectomy, long lesions.
- **Risk factors:** older age, DM, CKD, HF/Killip >2, large infarct, long ischemic time, lesion >15 mm.
- **Prognosis:** ~2× long-term mortality.
- **Algorithm:**
 - Exclude dissection, thrombus, air and spasm (IC NTG). Aspirate thrombus if present.
 - **Deliver vasodilators distally** through a microcatheter or aspiration catheter, not the guide.
 - **IC doses (distal):**
 - Adenosine 30–60 mcg boluses (up to 1–2 mg); AV block in RCA; endothelium-independent.
 - Nitroprusside 50–200 mcg (hypotension).
 - Verapamil 100–250 mcg (up to 1 mg; bradycardia). Nicardipine 50–200 mcg.
 - Epinephrine 50–200 mcg (useful with hypotension/bradycardia; causes tachycardia).
 - Choose the agent by HR and BP: bradycardic/hypotensive → epinephrine; tachycardic/hypertensive → nitroprusside or CCB.

NTG does not treat microvascular obstruction.

- GPI can be a bailout for thrombus; no proof it treats noreflow.

Prevention: direct stenting; avoid aggressive postdilation in thrombotic lesions; EPD in SVG; small burrs, low speed and short runs; pretreatment with nicardipine/verapamil (not adenosine) in small studies; high-intensity statin.

- **“Epicardial no-reflow”** during atherectomy: debris trapped at a tight downstream lesion. Flush; if no improvement, balloon the distal lesion (do not stent the proximal segment yet).

Dissection and abrupt closure

- **NHLBI types:** A–B minor; C (extraluminal cap); D (spiral); E (filling defect); F (occlusion). Types C–F need stenting. Abrupt closure is mostly a POBA-era problem (dissection + thrombus). It is often transient, but stent it.
- **Catheter-induced ostial dissection** (classically Amplatz in the RCA; guide extensions): presents with ST elevation, VF or collapse.
 - Stop contrast injections.
 - Keep or obtain distal wire position in the true lumen (confirm with IVUS).
 - Stent the ostium/entry point promptly to seal and stop antegrade and retrograde propagation, then cover distal extension.
 - Check the aortic root for retrograde extension. Limited extension (<40 mm) usually seals once the ostium is sealed; extensive → CT/TEE, surgery.
 - Coronary dissection with a damped guide pressure trace: never inject; pull back.

Air embolism

- **Presentation:** abrupt cutoff or slow flow in one or more vessels without a dissection plane; chest pain, ST elevation, VF, hypotension.
- **Causes:** inadequate aspiration/flushing during exchanges, loose connections, balloon rupture, large devices in small guides.

Treatment:

100% O₂.

- Forceful saline injection or aspiration through a catheter; disrupt bubbles with a wire or balloon. Vasopressors or CPR if needed.

Most resolve.

Prevention: meticulous flushing and back-bleeding; wet connections.

Equipment loss / stent dislodgement

Risk: calcium, proximal tortuosity, direct stenting without predilation, non-coaxial guide on withdrawal (most common), previously deployed damaged stents catching new ones.

- 70–90% of lost devices are retrievable.
- **If the wire is still through the stent:** advance a small (1.5 mm) balloon beyond or into the stent, inflate at low pressure, and withdraw the balloon-stent-guide as a unit. **Other options:** twist two wires to entangle; snare (gooseneck/multi-loop, may need a larger guide or 6F catheter as a “mother” sheath); biopsy forceps or basket. **If unretrievable: crush against the wall** with another stent, or deploy in place if feasible. Leave small wire fragments in distal branches.
- A 300 cm wire doubled in a 4–5F catheter works as a makeshift snare.

Other intraprocedural events

- **Vagal reaction vs RP bleed:** both hypotensive. Vagal = bradycardia, responds to atropine/fluids. RP = tachycardia (bradycardia possible), transient response to fluids only.
- **Acute stent thrombosis** after bivalirudin stops or with no P2Y₁₂ load: re-anticoagulate, reload with ticagrelor/prasugrel, GPII/cangrelor.
- **Atrial septal occluder embolization** (0.5–1%; usually immediate):
 - Risk: deficient rims, undersizing, floppy septum, excessive tension.
 - Retrieve with snare/forceps into a sheath ≥2F larger than delivery (bevel the tip).
 - Move the device to the PA, aorta or IVC before manipulating; ventricular location often needs surgery. CXR/TTE before discharge.

11. Peripheral Artery, Renal, Subclavian and Carotid Disease

PAD epidemiology and presentation

- **Prevalence:** ~7% at age 40, ~20% at >80. Most are asymptomatic; ~1/3 have typical claudication; <5% have CLTI.
- **Polyvascular disease** (CAD/CVD) in ~~40%~~ **5-yr mortality 15–30% (75% CV)**.
- **Risk factors:** smoking > DM > dyslipidemia > HTN.
- Smoking favors aorto-iliac disease.
- DM favors femoropopliteal/tibial disease.
- Young smoker with distal disease → Buerger's.
- **At-risk groups:** ≥65 y; 50–64 y with risk factors; <50 with DM + 1 other risk factor; known atherosclerosis elsewhere.
- **Natural history** (5 yr):
 - Claudication: 70–80% stable, 10–20% worsen, 1–2% progress to CLTI.
 - **CLTI at 1 yr:** ~25% amputation, ~25% dead.
- Major amputation: BKA 5–8% perioperative mortality (AKA higher); ~50% of BKA patients regain ambulation.
- **Classification:** Rutherford 0–6 (4 = rest pain, 5 = minor tissue loss, 6 = major tissue loss). CLTI = rest pain, nonhealing wound or gangrene ≥2 wk; stage with Wifl.

Claudication differential:

- Neurogenic (spinal stenosis): worse standing, relieved by sitting or flexion (“shopping-cart sign”). Venous: heaviness, relieved by elevation. Chronic compartment syndrome (athletes); hip/knee arthritis; Baker cyst.

Ulcers:

- **Arterial:** distal/toes, between digits, punched-out, dry pale base, painful.
- **Venous:** medial malleolus, shallow, exudative, hemosiderin, lipodermatosclerosis.
- **Neuropathic:** plantar pressure points, callus, insensate.

CLTI rest pain is nocturnal and relieved by dependency (sleeping in a chair).

Testing

Resting ABI is first-line (higher ankle pressure of that leg / higher arm pressure):

- **≤0.90 abnormal**
- 0.91–0.99 borderline
- 1.00–1.40 normal
- **>1.40 non-compressible** → TBI (≤0.70 abnormal)

- **Exercise ABI** if resting ABI is normal/borderline with symptoms. Abnormal = ankle pressure $\downarrow$ >30 mmHg or ABI $\downarrow$ $>20\%$. $\sim\frac{1}{3}$ with normal resting ABI become abnormal.
- **Segmental pressures:** ≥ 20 mmHg drop between levels localizes stenosis.
- **Doppler waveforms:** triphasic (normal) $\rightarrow$ biphasic $\rightarrow$ monophasic (severe). PVR loses the dicrotic notch, then amplitude.
- **Wound healing:** toe pressure ≥ 30 mmHg and TcPO₂ $\geq 30-40$ mmHg favor healing; $<20-30$ predicts poor healing. ABI can be falsely normal in CLTI (calcified tibials). **Anatomic imaging** (duplex, CTA, MRA) only when revascularization is being considered. Invasive angiography only with intent to treat.

Medical therapy (2024 ACC/AHA PAD guideline)

- **Smoking cessation** (Class 1).
- **Supervised exercise therapy** (Class 1): $\geq 30-45$ min, $\geq 3\times/\text{wk}$, ≥ 12 wk; structured community programs also OK.
- **High-intensity statin** for $\geq 50\%$ LDL $\downarrow$; add ezetimibe/PCSK9i if LDL ≥ 70 .
- ACEi/ARB; BP control.

Antithrombotics:

- SAPT (aspirin or clopidogrel; CAPRIE favored clopidogrel) for symptomatic PAD.
- **Rivaroxaban 2.5 mg BID + aspirin** (COMPASS; VOYAGER PAD after revascularization) $\downarrow$ MACE and MALE (Class 1).

Short DAPT after endovascular therapy is reasonable. No full-dose OAC without another indication (WAVE). Ticagrelor = clopidogrel (EUCLID). Vorapaxar $\downarrow$ ALI (not after stroke).

- **DM:** SGLT2i and GLP-1 RA $\downarrow$ MACE. **STRIDE (2025):** semaglutide $\uparrow$ walking distance in PAD + T2DM.
- **Cilostazol** 100 mg BID (PDE3): $\uparrow$ walking distance; **contraindicated in HF.**
- **Pentoxifylline and chelation: not recommended.**

UPDATED The original said “statin to LDL <100 ” and “ASA alone for most.” Current targets are $\geq 50\%$ LDL reduction with add-on therapy if LDL ≥ 70 , and low-dose rivaroxaban + ASA for symptomatic PAD.

Revascularization

- **Asymptomatic PAD:** never revascularize to prevent progression.
- **Claudication:** only if lifestyle-limiting after GDMT and supervised exercise. Exercise first, even for Rutherford 3. **CLTI:** revascularize (Class 1) with a multidisciplinary team. Aim for **in-line flow to the wound angiosome**, revascularize early (heel wounds), close follow-up. Wet gangrene $\rightarrow$ debridement first.
- **BEST-CLI (2022):** surgical bypass with adequate single-segment GSV $\downarrow$ MALE/death vs endovascular-first. **BASIL-2 (2023):** for infrapopliteal disease, vein bypass first $\uparrow$ amputation/death vs endovascular-first.

Anatomy:

- Endovascular is best for iliac and focal femoropopliteal lesions ($<10-15$ cm).

- Long, complex aorto-iliac, CFA/profunda disease → surgery (CFA endarterectomy).
- Treat inflow before outflow.
- Infrapopliteal intervention for claudication is not durable; avoid.
- **Paclitaxel devices:** FDA (2023) found no excess mortality signal on updated data.
- **Popliteal aneurysm ≥ 2 cm → repair.**

Acute limb ischemia (symptoms < 2 weeks)

- **6 P's:** pain, pallor, pulselessness, poikilothermia, paresthesia, paralysis.
- **Etiology:** embolic (AF, LV thrombus, aneurysm, paradoxical; lodges at CFA or popliteal bifurcation) vs thrombotic (native PAD, stent or graft thrombosis; APS, JAK2, HIT, malignancy).
- **Immediate IV heparin** (Class 1).

Rutherford	Sensory	Motor	Arterial Doppler	Venous Doppler	Action
I viable	None	None	Audible	Audible	Urgent evaluation
IIa marginally threatened	Minimal (toes)	None	Often inaudible	Audible	Prompt revascularization
IIb immediately threatened	Beyond toes, rest pain	Mild–moderate	Inaudible	Audible	Emergent revascularization (≤ 6 h)
III irreversible	Profound anesthesia	Paralysis/rigor	Inaudible	Inaudible	Primary amputation

- **Revascularization:** catheter-directed thrombolysis or mechanical thrombectomy for I–IIa; surgery or mechanical options for IIb.
- Watch for **compartment syndrome** after reperfusion (fasciotomy).
- **Subacute ischemia** (weeks): long, easily crossed thrombotic occlusions. Use thrombolysis/thrombectomy and consider EPD (embolization risk).

Subclavian stenosis

- Left $\sim 4\times$ more common than right; present in $\sim 11\%$ of PAD patients. Usually asymptomatic; **inter-arm SBP difference > 15 mmHg.**
- **Screen before CABG with IMA:** arm BP difference, chest radiation, vasculitis, known PAD.
- **Treat for:** arm claudication, symptomatic vertebral steal, **coronary-subclavian steal** (LIMA angina), planned LIMA use, dialysis access. Endovascular first ($> 90\%$ 3-yr patency).

Renal artery disease

- **ASTRAL and CORAL:** no benefit of stenting atherosclerotic RAS for BP, renal function or events. **Routine stenting is not recommended.**
- **Consider stenting for:** recurrent flash pulmonary edema; rapidly declining renal function with bilateral/solitary kidney stenosis; truly resistant HTN on optimal meds. **Significance:** systolic gradient ≥ 20 mmHg, mean ≥ 10 mmHg, Pd/Pa ≤ 0.90 , or renal FFR ≤ 0.80 . **Clues:**

- HTN onset <30 or >55 y; resistant, malignant or accelerating HTN.
- Kidney size difference >1.5 cm or unexplained atrophy. **Flash pulmonary edema**; Cr ↑ >30% after ACEi/ARB. Unexplained CKD; multivessel CAD/PAD.
- **Physiology:**
- **Unilateral RAS:** renin-dependent HTN. Contralateral kidney has pressure natriuresis (no volume overload). **ACEi lowers BP.**
- **Bilateral RAS or solitary kidney:** volume-dependent; renin suppressed. **ACEi causes AKI (Cr rises)** with little BP effect.
- **Fibromuscular dysplasia:** young/middle-aged women, “string of beads” in mid/distal renal and ICA/vertebral arteries.
- **Balloon angioplasty without stent** for HTN (high cure rate).

One-time head-to-pelvis CTA/MRA (aneurysms, dissection); strong SCAD association.

UPDATED The original said Cr decreases with ACEi in bilateral RAS. Creatinine rises.

Carotid disease

Definitions and natural history

- **Symptomatic** = ipsilateral TIA, stroke or amaurosis fugax **within 6 months** (not dizziness or syncope). Stroke risk is highest in the first 2 wk.
- **Measurement:** NASCET % = $(1 - \text{residual lumen} / \text{normal distal ICA}) \times 100$.
- **Duplex ≥70%:** ICA PSV >230 cm/s and ICA/CCA ratio >4. CTA/MRA confirm.
- **Carotid disease causes 10–20% of ischemic strokes**, mostly **embolic**. Revascularization prevents embolism, not hypoperfusion.
- Asymptomatic stenosis also marks systemic risk (strokes in other territories). Total occlusion: no revascularization.
- **Arch anatomy:**
- Common origin of innominate and L CCA (“bovine,” ~15–25%).
- L vertebral from the arch (~3–5%).
- **Aberrant right subclavian** (arteria lusoria, ~1%): arises distal to L SCA, retroesophageal; makes right radial access hard.

Type III arch ↑ CAS embolic risk.

Symptomatic stenosis

- **NASCET/ECST:** CEA benefit is greatest for 70–99% (large ARR), modest for 50–69%, harmful <30%.
- Revascularization for 70–99% is Class 1 (CEA if perioperative risk <6%). **Operating within 2 wk** of the event is reasonable (2a).

CAS is an alternative, especially for high surgical risk or anatomy. **CREST**: no difference in the composite; CAS had more minor strokes, CEA more MI. Age >70 favors CEA.

Asymptomatic stenosis

- **ACAS/ACST** (pre-modern statins): modest CEA benefit, mostly in men <75.
- **SPACE-2**: no benefit of either revascularization strategy vs best medical therapy (underpowered). **ACST-2 (2021)**: CAS ≈ CEA.
- **CREST-2 (NEJM 2025)**, ≥70% asymptomatic stenosis, credentialed operators:

CAS + intensive medical management ↓ periprocedural stroke/death + 4-yr ipsilateral stroke (2.8% vs 6.0%, P=0.02; NNT ~31).

- **CEA + intensive medical management: 3.7% vs 5.3% (P=0.24, NS).**

CMS NCD (Oct 2023): CAS covered for symptomatic ≥50% and asymptomatic ≥70% **regardless of surgical risk**. Requires shared decision-making, pre/post neurologic assessment and imaging confirmation. Facility certification requirement removed.

UPDATED The original said “no board scenario where we stent an asymptomatic patient” and “CMS only reimburses high surgical risk.” Both are outdated after CREST-2 and the 2023 NCD. Individual patient selection and life expectancy still matter.

High surgical-risk features (favor CAS/TCAR)

- **Anatomic**: prior CEA with restenosis, neck radiation or radical dissection, lesion at/above C2 or below clavicle, contralateral occlusion, contralateral laryngeal nerve palsy, immobile neck, tracheostomy.
- **Physiologic**: age ≥80, NYHA III–IV, LVEF <30%, unstable angina or recent MI, LM/multivessel CAD needing surgery within 30 d, severe lung disease. Young patients generally do better with CEA. **TCAR** (flow reversal) is an alternative.

CAS technique and complications

Plan with arch imaging (CTA). Embolic risk comes from arch manipulation, crossing, and **post-dilation (highest risk step)**. Accept 10–20% residual stenosis.

EPD required. Distal filters fail with poor landing zone or ECA collaterals. Proximal occlusion (MoMA) works before crossing but needs collateral tolerance.

- **Visible thrombus → do not stent.**
- **Baroreceptor response** (bradycardia/hypotension): hold antihypertensives the morning of the procedure; atropine before post-dilation; phenylephrine/norepinephrine; midodrine or pseudoephedrine if persistent. Less common after prior CEA (denervated). **CEA-specific risks**: MI, cranial nerve injury, neck hematoma, early thrombosis.

Hyperperfusion syndrome (days after CAS/CEA): ipsilateral headache, seizures, focal deficits, ICH. Highest risk with tight stenosis, poor collaterals/contralateral occlusion, and HTN. Treat with **strict BP control** (labetalol, clonidine), ICU and imaging.

12. Structural Heart Disease

Transcatheter aortic valve replacement

Indications

- **Severe AS:** $V_{max} \geq 4$ m/s, $MG \geq 40$ mmHg, $AVA \leq 1.0$ cm². Symptom onset is the “cliff” for survival. **2020 ACC/AHA (current US VHD guideline)**, symptomatic severe AS:
- Age <65 or life expectancy >20 yr → SAVR. 65–80 → shared decision, SAVR or TF-TAVR. >80 or life expectancy <10 yr → TF-TAVR. High or prohibitive surgical risk → TAVR.

Asymptomatic severe AS

- **Class 1 AVR:** LVEF <50%; abnormal exercise test (↓ BP or symptoms); undergoing other cardiac surgery.
- **2a:** very severe ($V_{max} \geq 5$, $MG \geq 60$), rapid progression (≥ 0.3 m/s/yr), BNP >3× normal.

Newer evidence:

- **EARLY TAVR (2024):** TF-TAVR in asymptomatic severe AS ↓ death/stroke/unplanned CV hospitalization vs surveillance. **FDA approved TAVR for asymptomatic**

severe AS (May 2025).

AVATAR/EVOLVED: early SAVR benefit.

- **ESC/EACTS 2025:** TAVI favored for **≥ 70 y** (was 75); intervention for asymptomatic severe high-gradient AS at low procedural risk.

Low-risk RCTs:

- **PARTNER 3:** TAVR ↓ 1-yr death/stroke/rehospitalization; composite similar at 5 yr.

Evolut Low Risk: noninferior.

- **NOTION 10-yr:** similar mortality; less SVD with TAVR. **DEDICATE (2024):** TAVR noninferior/superior to SAVR at low–intermediate risk (≥ 65 y).
- **Bicuspid** anatomy was excluded from most RCTs; increasingly treated in selected anatomy.
- **STS risk:** low <4%, intermediate 4–8%, high >8%. **Frailty, porcelain aorta, hostile chest** (radiation, patent grafts) favor TAVR.

UPDATED The original said “if you can’t do TF, palliative care.” Alternative access (transcarotid, subclavian/axillary, transcaval) is used before palliation. TF remains best (~95% of cases).

Devices (US)

- **SAPIEN 3/Ultra RESILIA** (balloon-expandable, intraannular): approved for native AS at all risk levels, aortic/mitral/pulmonic ViV, and asymptomatic severe AS. **Evolut FX+** (self-expanding, supra-annular): better hemodynamics in small annuli. **SMART (2024):** less bioprosthetic valve dysfunction than SAPIEN in small annuli; clinical outcomes similar.
- **Navitor** (Abbott, self-expanding intra-annular; successor to Portico): high/extreme risk.

- **JenaValve Trilogy: FDA-approved March 2026** for symptomatic severe **aortic regurgitation** at high surgical risk (ALIGN-AR). It is the first TAVR approved for AR; offlabel TAVR in AR otherwise embolizes (no calcium anchor).

Pre-procedure CT

- Annulus sizing, LVOT calcium, iliofemoral access (minimal lumen diameter vs sheath), coronary heights, sinus dimensions.
- **Coronary obstruction risk (native):** coronary height <10–12 mm, sinus of Valsalva <30 mm, bulky leaflet calcium, low sinotubular junction. Left main is involved more often than RCA.
- **ViV: valve-to-coronary distance (VTC) <4 mm high risk** (4–6 gray zone). Externally mounted leaflets (Mitroflow, Trifecta) are highest risk. Also check valve-to-STJ distance for sinus sequestration in TAV-in-TAV.

CAD and TAVR

- 50% of TAVR patients have CAD. Coronary evaluation (CCTA or angiography) before TAVR is Class 1.
- **ACTIVATION (2021):** routine pre-TAVR PCI did not meet noninferiority; ↑ bleeding.
- **NOTION-3 (2024):** PCI of ≥1 lesion with FFR ≤0.80 or DS ≥90% (proximal, ≥2.5 mm) ↓ death/MI/urgent revascularization at 2 yr vs conservative care; ↑ bleeding.
- 2020 guideline: PCI reasonable for ≥70% proximal disease (2a). Complex multivessel disease (SYNTAX >32) favors SAVR + CABG.
- **Coronary access after TAVR** is harder with supra-annular SE frames (use JL3.5/JR4 through a frame cell ± guide extension) than with short BE frames (JL/EBU). Commissural alignment helps.

UPDATED The original said “PCI before TAVR only for symptomatic proximal lesions since PCI doesn’t improve outcomes.” NOTION-3 supports treating significant CAD.

Conduction disturbances

- **PPM rates:** ~5–10% with BE valves, ~15–25% with SE valves (higher in real-world practice than in trials). New LBBB ~20%.
- **Predictors: baseline RBBB** (strongest), deep implant, short membranous septum, LVOT/device-landing-zone calcium, oversizing, SE valve, intraprocedural transient CHB.
- **Cusp-overlap view** (overlapping RCC and LCC isolates the NCC, elongates the LVOT, removes parallax) allows a higher implant and ↓ PPM.

New LBBB or PR/QRS prolongation: monitor 24–48 h, ambulatory ECG, or EP study (HV interval). Progressive changes or transient CHB → PPM.

Stroke, antithrombotics, leaflet thrombosis, durability

- **Stroke:** ~2–3% at 30 d, mostly periprocedural. New MRI lesions in >80%.
- **Cerebral embolic protection** (Sentinel): captures debris in ~99% but **PROTECTED TAVR (2022)** and **BHF PROTECTTAVI (2025)** showed no stroke reduction. Not routine. **Antithrombotics:**

- **POPular TAVI:** aspirin alone bleeds less than aspirin + clopidogrel; OAC alone bleeds less than OAC + clopidogrel.
- **GALILEO:** rivaroxaban without an OAC indication ↑ death/bleeding. **ATLANTIS:** apixaban not superior. **ENVISAGE:** edoxaban noninferior to VKA in AF, with more GI bleeding.
- **Default is lifelong low-dose aspirin;** OAC only if otherwise indicated. No routine DAPT.
- **HALT/subclinical leaflet thrombosis** (CT: HALT/HAM): common, reversible with OAC, weakly linked to stroke. Not the same as clinical valve thrombosis (↑ gradients, which is treated with OAC). No routine CT screening or empiric OAC.

Durability: NOTION 10-yr, PARTNER 2/3, Evolut data show TAVR SVD ≤ SAVR through 5–10 yr. Younger patients have earlier failure; plan lifetime management (TAV-inSAV favorable; TAV-in-TAV needs leaflet-overhang and commissural planning).

Valve-in-valve

Good option for failed surgical valves at high redo risk. **Does not fix prosthesis–patient mismatch** (the small label size persists), so consider bioprosthetic valve fracture.

- **BVF:** high-pressure NC balloon fractures fracturable rings (e.g., Magna/Magna Ease, Mitroflow; not Hancock II or Trifecta) and ↓ gradient.
- Higher implant ↓ gradient but ↑ coronary risk. **BASILICA** (electrosurgical leaflet laceration) or chimney/snorkel stenting for coronary obstruction risk.

TAVR vs SAVR complication profile

SAVR: more AF, AKI, bleeding, longer length of stay. **TAVR:** more PVL (≥moderate ↑ mortality), PPM/LBBB, vascular complications.

TAVR complications

- **Coronary obstruction** (~0.7% native, higher in ViV) Presents as hypotension, ST changes or VF right after deployment; can be delayed.
- Mortality ~40–50%.
- Prevention in high-risk anatomy: wire and undeployed stent in the coronary (“chimney”), BASILICA, recapturable valve.
- Treatment: PCI (chimney), hemodynamic support, surgery.

After chimney stenting, a flapping lucency or slow flow means deploy the parked stent.

Annular rupture

Risk: BE > SE; oversizing >20% area; LVOT/annular calcium; small annulus; aggressive post-dilation. Presents as sudden hypotension/tamponade.

- Pericardiocentesis, reverse heparin, surgery; small contained ruptures may be managed conservatively.

Embolization/malposition

Risk: loss of pacing capture during BE deployment, undersizing, high implant, bulging septum, low calcium (AR), large annulus, male/large BSA.

SE valves: recapture or snare.

- **Keep the wire through the valve.** Pull an embolized BE valve with a partially inflated balloon into the aorta and deploy in a safe zone (descending aorta or distal to the great vessels), then implant a second valve. Ventricular embolization usually needs surgery. Associated with 30-d mortality.

Tamponade: LV wire perforation (stiff wire, delivery tip), RV pacing lead, aortic root rupture. Diagnose with echo and aortic root/LV angiography.

Vascular injury, aortic dissection, severe PVL (post-dilate, second valve, plug), acute AR, MR from wire entanglement.

Mitral valve

Mitral regurgitation

- **Primary (degenerative) vs secondary (ventricular or atrial functional).**

Surgery for severe primary MR:

- Symptomatic (Class 1).
- Asymptomatic with LVEF $\leq 60\%$ or LVESD ≥ 40 mm (Class 1).
- Asymptomatic repair reasonable with durable-repair likelihood $>95\%$ and mortality $<1\%$, or with progressive LV dilation, new AF or PASP >50 .

TEER for primary MR:

- **EVEREST II:** surgery more effective (more residual MR and later surgery with TEER) but TEER safer; symptoms similar at 5 yr.
- Indicated for severe symptomatic primary MR at high or prohibitive surgical risk with favorable anatomy (2a).
- **CLASP IID:** PASCAL noninferior to MitraClip.
- **TEER for secondary MR:**
 - **COAPT:** MitraClip + GDMT $\downarrow$ HF hospitalization and all-cause death (NNT ~ 6 for HFH at 2 yr) vs GDMT. Criteria: LVEF 20–50%, LVESD ≤ 70 mm, PASP ≤ 70 , persistent severe MR on maximal GDMT/CRT.
 - **MITRA-FR:** neutral. Explained by “disproportionate” vs “proportionate” MR relative to LV size, lower MR severity, larger LVs and more residual MR.
 - **RESHAPE-HF2 (2024):** TEER $\downarrow$ HFH/CV death in moderate-to-severe SMR.
 - **MATTERHORN (2024):** TEER noninferior to surgery for SMR.

ESC 2025: Class I A in eligible SMR. ACC/AHA 2020: 2a. **Unfavorable TEER anatomy:** MVA <4 cm² or baseline MG >5 mmHg (stenosis), heavily calcified grasping zone, short posterior leaflet (<7 mm), Barlow disease with multiple segments, cleft, wide flail gap/width, rheumatic disease. **TMVR:**

Tendyne (transapical; FDA 2025).

SAPIEN M3 (transseptal; FDA Dec 2025, ENCIRCLE): symptomatic MR not suitable for surgery or TEER, and MAC-related disease.

Main risk is **LVOT obstruction** (neo-LVOT $<1.7\text{--}2.0\text{ cm}^2$); LAMPOON or alcohol septal ablation to mitigate. SAPIEN 3 approved for mitral ViV. Valve-in-ring and valve-in-MAC carry higher risk (MITRAL trial).

UPDATED The original said “LVEF $>50\%$ = surgery, $<50\%$ = TEER in the US” and that TMVR is research-only. Both are outdated.

Mitral stenosis

- **Rheumatic**: commissural fusion, leaflet and subvalvular thickening; annulus spared; younger patients. AF markedly worsens prognosis.
- **Degenerative (MAC)**: annular calcium, elderly, high surgical risk; **not amenable to balloon valvotomy**. **PMBC for rheumatic MS** (Class 1): symptomatic, MVA $\leq 1.5\text{ cm}^2$, favorable morphology, no LA thrombus, less than moderate MR.
- **Wilkins score** (mobility, thickening, calcification, subvalvular; 1–4 each): **≤ 8 favorable**. Commissural calcium is unfavorable.
- Success: MVA $\geq 1.5\text{ cm}^2$ (or $\geq 50\%$ increase) with $\leq 2+$ MR.
- Complications: severe MR (new giant v wave), tamponade, embolism, residual iatrogenic ASD.
- **Anticoagulation**: VKA for rheumatic MS with AF, prior embolism or LA thrombus (INVICTUS: rivaroxaban inferior).
- In rheumatic MS, thrombus is often in the LA body (not just the appendage), so LAAO is inappropriate.
- **Hemodynamics**: PCWP–LVEDP gradient (PCWP overestimates; transseptal LA pressure is best). Hypotension with low LV filling despite high right-sided pressures.

Tricuspid, left atrial appendage, HCM

Tricuspid regurgitation

Mechanisms:

- Ventricular functional (left heart disease/PH, RV dilation, leaflet tethering).
- Atrial functional (AF, annular dilation with preserved RV).

CIED lead-related.

Primary (endocarditis, carcinoid, Ebstein).

2020 ACC/AHA surgery:

- Severe TR at time of left-sided surgery (Class 1). Progressive TR with annulus $>40\text{ mm}$ (21 mm/m^2) or prior RHF at left-sided surgery (2a). Isolated severe TR with RHF, before severe RV dysfunction (2a).

Transcatheter:

- **TriClip T-TEER** (FDA Apr 2024; TRILUMINATE: $\uparrow$ QoL, no $\downarrow$ death/HFH at 1 yr).

- **EVOQUE TTVR** (FDA Feb 2024; TRISCEND II: ↑ QoL/function; PPM ~15–25%; bleeding). **ESC 2025 (IIa)**: symptomatic high-risk patients without severe RV dysfunction or pre-capillary PH.

Left atrial appendage occlusion

- ~90% of non-valvular AF thrombi form in the LAA. Only ~50% of eligible patients receive OAC, often underdosed. **Trials:**
- **PROTECT AF/PREVALE** (Watchman vs warfarin), 5-yr pooled: noninferior composite; ↓ hemorrhagic stroke, disabling stroke and CV death; numerically more ischemic stroke.
- **PRAGUE-17**: LAAO noninferior to DOAC (net composite; ↓ non-procedural bleeding at 4 yr). **Amulet IDE**: Amulet noninferior to Watchman for safety and efficacy, superior closure; more early procedural complications (effusion, embolization). **OPTION (2024)**: after AF ablation, LAAO vs OAC ↓ non-procedural bleeding; noninferior for death/stroke/embolism.
- **CHAMPION-AF (2026)** (Watchman FLX vs DOAC, OACeligible): noninferior for CV death/stroke/SE (5.7% vs 4.8%); ↓ non-procedural bleeding (10.9% vs 19.0%); **ischemic stroke/SE numerically higher** (3.2% vs 2.2%). **CLOSURE-AF (2026)** (HBR or OAC-contraindicated patients vs physician-directed best medical care, mostly DOAC): **LAAO failed noninferiority** for the net composite.
- **Guideline (2023 ACC/AHA/HRS AF)**: LAAO **2a** for moderate–high stroke risk with a contraindication to long-term OAC from a non-reversible cause; **2b** for high bleeding risk or patient preference.
- **FDA label**: CHA₂DS₂-VASc high enough for OAC with an appropriate rationale to seek a non-drug alternative. **Post-Watchman FLX regimen**: OAC + aspirin for 45 d (or DAPT from the start), imaging (TEE/CT), DAPT to 6 mo, then aspirin. Amulet: DAPT.
- **Complications**: pericardial effusion needing intervention ~0.5–1%, device embolization <0.5%, procedural stroke. ICE-guided procedures avoid TEE and general anesthesia. - **Device-related thrombus (2–4%):**** ↑ **stroke 3–5×**. **Can occur anytime (mostly early)**. **Risk: hypercoagulable state, pericardial effusion, CKD, **deep implant (>10 mm below the pulmonary ridge)**, permanent AF, low EF, large LAA.

Treat with OAC; recurrence common.

- **Peri-device leak: any leak, even <5 mm**, ↑ **ischemic events** (NCDR). Options: plugs, coils, second device, RF ablation.

HCM and alcohol septal ablation

Hemodynamics:

- **Brockenbrough–Braunwald–Morrow**: post-PVC ↑ LV–Ao gradient with ↓ **aortic pulse pressure** (in AS the pulse pressure rises).

Spike-and-dome aortic waveform.

- Gradient ↑ with Valsalva, standing, amyl nitrite, isoproterenol; ↓ with phenylephrine, squatting, handgrip.

SAM-related MR (posteriorly directed) improves with phenylephrine. Anterior or central MR suggests intrinsic MV disease; get TEE.

- **Obstruction thresholds**: gradient ≥30 mmHg; **≥50 mmHg** (rest or provoked) with drug-refractory symptoms is the SRT threshold.

2024 AHA/ACC HCM guideline:

- Non-vasodilating BB first, then non-DHP CCB (avoid verapamil with severe gradients or hypotension). If still symptomatic: **myosin inhibitor** (mavacamten; **aficamten FDA Dec 2025**; REMS/echo monitoring), **disopyramide**, or **SRT** at an experienced center (Class 1).
- VALOR-HCM: mavacamten ↓ SRT eligibility. MAPLEHCM: aficamten > metoprolol.
- Stop vasodilators and high-dose diuretics.

Myectomy vs ASA:

- **Myectomy preferred** with concomitant surgical disease (intrinsic MV or papillary muscle anomalies, CAD, other valve disease), **septum >30 mm**, and in young patients.
- **ASA** if surgery is contraindicated or high risk, or by informed patient preference.
- Center mortality <1% for both; similar long-term survival. Myectomy gives more complete gradient relief and fewer repeat procedures. SRT is for symptoms, not mortality.

ASA technique:

- TVP; wire and OTW balloon in the target septal (1st septal may arise from ramus/diagonal).
- Selective angiography to exclude leak into LAD/collaterals.
- **Myocardial contrast echo** confirms the basal septum at SAM contact and excludes papillary muscles/RV. 1–3 mL ethanol (~1 mL per 10 mm septal thickness); LAO-cranial view.
- Gradient falls over weeks to months with remodeling (acute drop from stunning).

Conduction:

ASA → RBBB; myectomy → LBBB.

- CHB needing PPM ~10–15% after ASA, mostly within 24–48 h (monitor ≥48 h).

Baseline LBBB or 1st-degree AVB markedly ↑ CHB risk; favor myectomy or plan PPM/ICD. Best results with septum 15–30 mm.

Congenital and shunt interventions

Atrial septal defect

Device closure only for secundum defects. Primum, sinus venosus (associated partial anomalous pulmonary venous return) and coronary sinus defects need surgery.

- **Device requirements:** rims ≥5 mm (deficient aortic rim is common, OK, but ↑ erosion); defect ≤38 mm. **Measure Qp:Qs** and PVR.
- **Closure indications:** RA/RV enlargement or Qp:Qs ≥1.5 with net left-to-right shunt and no cyanosis.
- **2025 ACHD:** PVR ≤2 WU → closure; 2–<5 WU → consider in expert centers; high PVR/Eisenmenger → no closure.
- Small ASD (<1 cm, Qp:Qs <1.5, no RV enlargement) → no closure.
- **LV diastolic dysfunction** (older, HFpEF): test balloon occlusion. Marked LAP rise → fenestrated device or no closure (the ASD is a pop-off).

- **Complications:**
- AF (early ~3%, late more)
- Air embolism (keep hub low; hydrate)
- Embolization
- **Erosion** (Amplatzer ~0.1–0.3%; oversizing, deficient aortic rim)
- Tamponade, device thrombus
- **After closure:** aspirin ± clopidogrel and endocarditis prophylaxis for 6 mo.
- Unrepaired ASD long term: RHF, atrial arrhythmias, PH, paradoxical embolism.

Patent foramen ovale

- **RESPECT, CLOSE, REDUCE, DEFENSE-PFO:** closure ↓ recurrent stroke in **age 18–60 with embolic-appearing (cryptogenic) stroke** after full work-up (MRI, vessel imaging, TEE, ≥30-d rhythm monitoring, hypercoagulable/VTE testing as indicated).
- Greatest benefit with **atrial septal aneurysm or large shunt**, high RoPE (≥7), and PASCAL “probable/highly likely.”
- SCAI 2022 guideline: case-by-case above 60.
- Device-related AF is mostly early and transient. Poor Valsalva under sedation can miss a shunt on TEE.
Non-stroke indications: platypnea-orthodeoxia (dyspnea/desaturation **upright**, relieved supine); decompression illness in divers; paradoxical systemic embolism. Migraine is not an indication (PREMIUM, PRIMA).

Ventricular septal defect

- **Closure:** Qp:Qs ≥1.5 with LV volume overload and PASP <50% systemic / PVR <1/3 systemic; prior endocarditis; residual post-surgical defect.
- Perimembranous device closure risks **CHB**. Muscular defects are device-friendly.
- Post-MI VSR → surgery (device for residual or bridging). Long-standing VSD → RV overload/PH; step-up at the RV level.

Pulmonary stenosis and RVOT

- **Valvular PS** is the most common isolated obstructive CHD (~7–10%); post-stenotic PA dilation.
- **Severity:** moderate peak gradient 36–64 mmHg; severe >64 mmHg (mean >35).

Balloon valvuloplasty for symptomatic moderate–severe or asymptomatic severe **domed** PS. Dysplastic valve (Noonan), sub/supravulvar stenosis, hypoplastic annulus or severe PR → surgery. **Complications:**

- PR
- Ectopy, RBBB/AVB
- Perforation
- **“Suicide RV”** (dynamic infundibular obstruction after relief): IV fluids + BB; avoid inotropes

- **Repaired ToF** (PR → RV dilation → dysfunction/VT): **PVR (surgical or transcatheter)** for symptomatic $\geq$ moderate PR.
- Asymptomatic: RV dilation (**RVESVi >80 mL/m²** emphasized in 2025 ACHD; RVEDVi $\geq$ 160, RV:LV EDV $\geq$ 2), RV/LV dysfunction, RVOT obstruction (RVSP $\geq$ 2/3 systemic), progressive exercise intolerance, sustained arrhythmia.
- **TPVR**: Melody and SAPIEN (conduits/bioprostheses); Harmony and Alterra (native/patched RVOT). **Before TPVR: coronary compression testing** (balloon in RVOT with simultaneous coronary angiography; anomalous coronaries are common in ToF). Stent the conduit first to prevent fracture. Watch for aortic root compression.

Coarctation of the aorta

- **Associations**: **50–75% of CoA patients have BAV**; Turner syndrome (~~12–15%~~); ~~Shone complex (supramitral ring, parachute MV, subaortic stenosis, CoA)~~; intracranial aneurysm (~~10%~~, screen).
- **Findings**: upper-extremity HTN; arm–leg BP gradient; radio-femoral delay; interscapular murmur; rib notching and “3 sign”; diastolic runoff (continuous forward flow) in the descending/abdominal aorta.
- **Imaging**: CMR or CTA for all repaired and unrepaired patients (Class 1).
- **Intervention** (with HTN): peak-to-peak gradient $\geq$ 20 mmHg, or <20 mmHg with significant anatomic narrowing plus collaterals, LV dysfunction or AR. **Stenting (covered stent preferred in adults**: COAST/COAST II) beats balloon alone (aneurysm, restenosis).
- Balloon should not exceed the diameter of the adjacent normal aorta (at the diaphragm). Stent should be expandable to adult size ($\geq$ 20 mm). Staged re-dilation for severe lesions.
- **Complications**: dissection/rupture (have covered stents ready), aneurysm, stent migration, femoral access injury.

13. Hemodynamics and Cath-Lab Pearls

Calculations

Mixed venous O₂ (Flamm) = $(3 \times \text{SVC} + \text{IVC}) / 4$.

- **Qp:Qs** = $(\text{SaO}_2 - \text{MvO}_2) / (\text{PvO}_2 - \text{PaO}_2)$. Use SaO₂ for PvO₂ if not sampled (unless there is a right-to-left shunt). **Significant step-up**: $\geq$ 7% at atrial level, $\geq$ 5% at ventricle or PA.
- **SVR** = $80 \times (\text{MAP} - \text{RAP}) / \text{CO}$. **PVR (WU)** = $(\text{mPAP} - \text{PCWP}) / \text{CO}$.
- **CPO** = $\text{MAP} \times \text{CO} / 451$. **PAPi** = $(\text{PASP} - \text{PADP}) / \text{RAP}$. **Before closing a left-to-right shunt**: Qp:Qs $\geq$ 1.5, PASP <50% systemic, PVR <1/3 SVR (or low WU per lesion). **Gorlin**: $\text{AVA} = \text{CO} / (\text{HR} \times \text{SEP}) / (44.3 \times \sqrt{\text{MG}})$. **Hakki**: $\text{AVA} \approx \text{CO} / \sqrt{(\text{peak-to-peak gradient})}$.

Pressure-waveform diagnoses

- **Tamponade**: equalized diastolic pressures (RA $\approx$ RVEDP $\approx$ PCWP), pulsus paradoxus >10 mmHg, blunted y descent. **Constriction vs restriction**
- Both: dip-and-plateau, equalization within 5 mmHg, Kussmaul sign.

- **Constriction-specific: ventricular interdependence**, meaning discordant RV/LV systolic pressures with respiration (inspiration: RV ↑, LV ↓); **systolic area index >1.1**; RVEDP/RVSP >1/3. **Restriction**: concordant pressures; PASP >55–60; LVEDP – RVEDP >5.
- **Severe MR**: giant v wave on PCWP/LA. After PMBC, a new tall v wave = iatrogenic MR.
- **HCM**: Brockenbrough sign. **AS**: post-PVC pulse pressure ↑; Carabello sign (↑ arterial pressure on crossing the valve).
- **Coronary damping or ventricularization**: ostial lesion, spasm, non-coaxial or deep catheter. Don't inject; pull back.

Ventriculography and injections

- **RAO 30°**: anterior, apical, inferior walls and MR. **LAO 45– 60° ± cranial**: septum, lateral wall, VSD.
- **Rate**: ~30–45 mL at 10–15 mL/s via pigtail. Use <10 mL/s via end-hole catheters.
- **Avoid or reduce with**: LVEDP/PCWP ≥25 (pulmonary edema; pretreat NTG/nitroprusside), critical LM, severe AS, LV thrombus.
- **Pigtail position**: avoid entrapment in trabeculae (staining, VT). Warning signs: close to inferior wall, loop distorts in systole, pigtail unfurls on withdrawal. RV angiography always causes ectopy.
- **Catheter selection**
- **Left-sided SVGs**: JR4, LCB, AL1 (upward take-off: LCB, hockey stick).
- **Right-sided SVGs** (downward): MP, JR4, RCB.
- **LIMA**: IMA or JR4 from the vertical subclavian (Bernstein/vertebral to enter the subclavian).
- **RCA**: anterior or high take-off → AR, hockey stick, AL. Inferior/vertical → MP. Superior course → IM, hockey stick.
- **LCA**: dilated root → JL5/JL6; small root → JL3–3.5.
- **Post-TAVR**: Evolut → JL3.5 / JR4 through frame cells ± guide extension. SAPIEN → JL4 or EBU 3.5–3.75, JR4. **Guide-extension or deep-engagement dissection**: Amplatz + RCA is the classic culprit.
- **Anatomy and venous pearls**
- **CFA bifurcation** is below the femoral head in ~65% and below its center in ~95%.
- **Persistent left SVC** drains to a dilated coronary sinus behind the LA. Coronary sinus diverticula are linked to accessory pathways and SCD.
- **Azygos** runs posteriorly along the spine. **Scimitar**: anomalous right PV to IVC. A Swan catheter outside the cardiac border may be in an anomalous vein.
- **After transseptal puncture**, park the wire in the left upper PV. Sudden collapse after PFO/ASD/LAAO → suspect LAA/LA perforation.
- **PV stenosis after ablation** (<1%): dyspnea, hemoptysis; balloon ± stent.
- **SVC syndrome** (malignant): stent, with radiation as indicated.
- **Periprocedural**:

- NPO ~2 h clears / 6 h solids for sedation (not in emergencies).
- Pregnancy test in women of child-bearing potential. Half-dose basal insulin the night before; hold oral agents the morning of; early case slot.
- **Repaired CHD** (root or coronary manipulation, e.g., arterial switch) with angina → coronary evaluation (CCTA or angiography).
- **Repaired CHD with QRS widening or syncope** → EP referral.
- Endomyocardial biopsy tamponade ~0.5–1%.
- **Acute aortic dissection:** SBP <120 mmHg and HR <60 (IV BB first, then vasodilator). Acute mesenteric ischemia: pain out of proportion to exam.

14. Statistics

Test performance

Term	Formula
Sensitivity	$TP / (TP + FN)$, among diseased
Specificity	$TN / (TN + FP)$, among non-diseased
PPV	$TP / (TP + FP)$
NPV	$TN / (TN + FN)$
LR+ / LR-	$Sens/(1-Spec)$; $(1-Sens)/Spec$

- **Sensitivity and specificity are fixed properties of the test. PPV and NPV depend on prevalence (Bayes).**
- **Example** (sensitivity 90%, specificity 90%, n = 1000):
- Prevalence 10%: PPV 50%, NPV 99% (a positive test is unhelpful; a negative rules out). Prevalence 90%: PPV 99%, NPV 50%.
- Prevalence 50%: PPV = NPV = 90%.
- Testing is most useful at **intermediate pretest probability**. Pretest probability (history, exam, ECG) is the most important input.
- **Verification bias:** short follow-up misclassifies false negatives as true negatives and inflates apparent NPV.

Effect measures

- **RR** = risk in treated / risk in control. **RRR** = $1 - RR$. **ARR** = risk(control) – risk(treated). **NNT** = $1/ARR$. **Lytics example:** mortality 11.6% → 8.7%. ARR 2.9%, RRR 25%, NNT ≈ 34 (29 lives per 1000).
- **GLOBAL LEADERS example:** 3.8% vs 4.4%. RR 0.87, RRR 13%, ARR 0.6%, NNT ≈ 167 (not significant).
- **Odds ratio** = odds(treated) / odds(control). Approximates RR only when events are rare. Used in case-control designs.

- **Hazard ratio:** time-to-event (Cox). **Win ratio:** hierarchical composite ranking of pairs, prioritizing death (e.g., CHIPBCIS3).

Inference and design

- **p-value:** probability of data this extreme if the null is true. **95% CI:** precision; excluding 1.0 (ratio) or 0 (difference) = significant. More events give narrower CIs. **Type I error (α)** = false positive. **Type II (β)** = false negative. **Power** = $1 - \beta$ ($\uparrow$ with sample size, event rate, effect size).
- **Superiority vs noninferiority:** NI tests whether the upper CI bound stays below a prespecified margin. A trial can show NI and then superiority if the CI lies entirely on the benefit side.
- **Crossover, non-adherence and low event rates bias toward the null.** In superiority trials that is conservative; in NI trials it is anti-conservative, so review per-protocol analyses too.
- **ITT** analyzes patients as randomized (preserves randomization). **Per-protocol/as-treated** analyzes what patients received (biased). Example: CABANA ITT neutral, as-treated favored ablation.
- **Designs:** RCT (investigator assigns; randomization removes selection bias). Cohort (defined by exposure; prospective/retrospective). Case-control (defined by outcome).
- **Propensity scores** balance measured, not unmeasured, confounders.
- **Surrogate endpoint:** biomarker substituting for a clinical outcome (smaller trials, possibly misleading). **Composite endpoint:** $\uparrow$ power, but components may differ in importance and direction.
- **Equipoise:** genuine uncertainty that justifies randomization.
- **Epidemiology:** incidence = new cases over time; prevalence = existing cases at a point in time; case-fatality = deaths among those with the disease.

Key sources for updates: 2025 ACC/AHA/ACEP/NAEMSP/SCAI ACS guideline; 2023 AHA/ACC chronic coronary disease guideline; 2021 ACC/AHA/SCAI revascularization guideline; 2024 ACC/AHA PAD guideline; 2024 AHA/ACC HCM guideline; 2025 ACC/AHA ACHD guideline; 2026 ACC/AHA dyslipidemia guideline; 2020 ACC/AHA and 2025 ESC/EACTS valvular guidelines; 2023 ACC/AHA/HRS AF guideline; 2025 AHA CPR/ECC guideline; CMS carotid stenting NCD (Oct 2023); FDA device and drug approvals through March 2026; primary trial publications cited in text.

01 – Coronary Physiology and Intravascular Imaging

Pressure-derived indices

Index	Hyperemia?	Ischemic cutoff	Notes
FFR (Pd/Pa)	Yes	≤ 0.80	Validated vs stress testing; DEFER used 0.75. Gray zone 0.75–0.80
iFR	No (wave-free period)	≤ 0.89	DEFINE-FLAIR, iFR-SWEDEHEART: non-inferior to FFR at 1 y; fewer symptoms/less adenosine
RFR, dPR, DFR, Pd/Pa (resting)	No	≤ 0.89 (Pd/Pa ≤ 0.91 -0.92)	Equivalent whole-cycle resting indices
Post-PCI FFR	Yes	Target ≥ 0.90	TARGET-FFR: low post-PCI FFR predicts events
CFR	Yes	< 2.0 abnormal	Ratio of hyperemic to resting flow; combines epicardial + microvascular
IMR	Yes	≥ 25 abnormal	IMR = Pd $\times$ Tmn(hyperemic). Post-STEMI IMR > 40 predicts poor recovery

Hyperemic agents. IV adenosine 140 $\mu\text{g/kg/min}$ (central line preferred, 2–3 min; go to 180–200 if inadequate); IC adenosine bolus LCA 100–200 μg , RCA 50–100 μg (short-lived, AV block with RCA); regadenoson 0.4 mg IV bolus (no titration, long effect, useful with peripheral IV); IC nicardipine 200 μg or IC nitroprusside 50–100 μg (hypotension); papaverine (QT prolongation, torsades – rarely used). Adenosine is blocked by caffeine/theophylline (hold caffeine 12–24 h); reversal with aminophylline. Avoid adenosine in severe bronchospastic disease; sinus/AV block usually self-limited ($t_{1/2} < 10$ s).

FFR technique pearls that get tested.

- Equalize at the guide tip, measure with guide disengaged for ostial lesions and always without side-hole catheters (side holes falsely raise Pa and give false-normal FFR).
- Pullback under steady hyperemia to localize the pressure drop; for serial lesions the distal lesion's FFR is underestimated until the proximal lesion is treated (re-measure after each stent). Apparent FFR of each lesion in tandem is not simply additive.
- Drift: pull the sensor back to the guide tip; acceptable Pd/Pa 1.00 ± 0.02 .
- Guide damping/ventricularization (deep-seated or small ostium) lowers Pa and can falsely normalize FFR.
- Whip artifact, wire in a false lumen, and pressure recovery are other artifact sources.

When FFR misleads.

- Culprit lesion in acute MI (microvascular stunning $\rightarrow$ falsely high FFR). Non-culprit FFR is acceptable in the acute phase but slightly overestimates (FLOWER-MI/FRAME-AMI context).
- Severe LVH, microvascular disease, severe AS (limited hyperemia $\rightarrow$ FFR falsely high).
- Left main with downstream LAD/LCx disease: measure into the non-diseased branch; a stenosed daughter vessel raises the measured LM FFR.
- Bridging collaterals and coronary steal.
- Exercise/very high resting flow (e.g., iFR more affected).

Evidence. DEFER (FFR ≥ 0.75 , deferral safe), FAME (FFR-guided vs angio in MVD: fewer stents, lower MACE), FAME 2 (FFR ≤ 0.80 PCI + OMT vs OMT: fewer urgent revascularizations, no death/MI benefit), FAME 3 (FFR-guided PCI failed non-inferiority to CABG in 3VD at 1 y; **5-y death/stroke/MI similar, 16% vs 14%, with more MI and repeat revascularization after PCI**), DEFINE-FLAIR / iFR-SWEDEHEART (iFR $\approx$ FFR at 1 y; **5-year DEFINE-FLAIR showed higher all-cause death with iFR, 9.0% vs 6.2%, not seen in iFR-SWEDEHEART, cause unexplained**), FUTURE (trend to harm with FFR in all-comers, stopped early), FLOWER-MI (FFR-guided non-culprit not better than angio-guided in STEMI), FIRE (physiology-guided complete revascularization in ≥ 75 y MI reduced events), FULL REVASC (2024: FFR-guided complete revascularization did not reduce death/MI/unplanned revascularization vs culprit-only), FRAME-AMI (FFR-guided beat angiography-guided), FLAVOUR (FFR-guided non-inferior to IVUS-guided for intermediate lesions with fewer PCIs), FAVOR III Europe (2024: angiography-derived QFR inferior to FFR).

Anatomic-physiologic correlations (frequently asked)

- Angiographic 50–70% lesions: $\sim 35\%$ are FFR-positive; 70–90%: $\sim 20\%$ are FFR-negative.
- IVUS MLA for a non-LM lesion: $< 4.0 \text{ mm}^2$ was the classic cutoff; best FFR correlation is $\sim 2.4\text{--}3.0 \text{ mm}^2$ (poor specificity, so physiology is preferred for non-LM).
- IVUS MLA in LM: **$< 6 \text{ mm}^2$** (Western; LITRO study) or **$< 4.5 \text{ mm}^2$** (Asian) $\rightarrow$ revascularize; $\text{MLA} > 6 \text{ mm}^2 \rightarrow$ defer safely.
- Coronary CT-FFR (FFR_{ct}): PLATFORM/ADVANCE reduced unnecessary cath; ≤ 0.80 positive.

IVUS vs OCT

	IVUS	OCT
Energy	Ultrasound 20–60 MHz	Near-infrared light ($\sim 1300 \text{ nm}$)
Axial resolution	$\sim 100\text{--}150 \mu\text{m}$	$\sim 10\text{--}20 \mu\text{m}$
Penetration	4–8 mm (sees whole vessel, EEM)	1–2 mm (no EEM behind lipid)
Blood clearance	Not needed	Needed (contrast 10–14 mL; saline/dextran options)
Best for	Vessel size, plaque burden, remodeling, ostial LM, CKD (no contrast), calcium arc/thickness (cannot see behind calcium—shadowing)	Thrombus, dissection, stent apposition/expansion, TCFA, calcium thickness (penetrates calcium), ISR mechanism
Weak for	Thin caps, small dissections, thrombus	Ostial LM, large vessels $> 4.5 \text{ mm}$, poor flow/occlusion, renal failure

OCT tissue signatures. Fibrous = signal-rich homogeneous. Calcium = signal-poor with **sharp** borders, low attenuation. Lipid/necrotic core = signal-poor with **diffuse** borders, high attenuation. Red thrombus = high backscatter, high attenuation (shadow). White (platelet) thrombus = less attenuating. TCFA cap $< 65 \mu\text{m}$. Macrophages = punctate bright spots. Cholesterol crystals = thin linear bright lines. Neoatherosclerosis in ISR = lipid/calcium within neointima (late ST substrate).

IVUS tissue signatures. Calcium = bright echo with acoustic shadowing (arc, length). Attenuated plaque (signal drop-out without calcium) = necrotic core, predicts no-reflow. Echolucent plaque. Positive remodeling (EEM at lesion / reference > 1.05) = vulnerable. NIRS-IVUS: $\text{maxLCBI}_{4\text{mm}} > 400$ predicts plaque-level and

patient-level events (LRP, PROSPECT II).

Stent optimization targets (imaging-guided PCI).

- MSA ≥ 5.5 mm² by IVUS (≥ 4.5 mm² by OCT) for non-LM; or MSA $> 80\%$ of distal reference lumen.
- LM (Kang criteria, IVUS): ostial LCx ≥ 5.0 , ostial LAD ≥ 6.3 , polygon of confluence ≥ 7.2 , proximal LM ≥ 8.2 mm² (the "5-6-7-8" rule).
- Malapposition < 0.4 mm and < 1 mm length is benign acutely; large late-acquired malapposition associates with very late ST.
- Edge dissection needing treatment: $> 60^\circ$ arc, > 3 mm length, involves media, or lumen < 4 mm² (OCT).
- Plaque burden $> 50\%$ at stent edge and MSA < 5.5 mm² predict edge restenosis/ST.
- Calcium: IVUS arc $> 270^\circ$, length > 5 mm; OCT calcium score (arc $> 180^\circ$, thickness > 0.5 mm, length > 5 mm = score 4) predicts under-expansion $\rightarrow$ atherectomy/IVL before stenting.

Evidence for imaging guidance. ULTIMATE (IVUS-guided lower TVF at 1–3 y), RENOVATE-COMPLEX-PCI (IVUS/OCT-guided complex PCI reduced TVF), ILUMIEN IV (OCT: larger MSA, fewer ST, no difference in TVF at 2 y), OCTOBER (OCT-guided bifurcation PCI reduced MACE), OCTIVUS (OCT non-inferior to IVUS), IVUS-XPL (long lesions). ADAPT-DES: IVUS reduced ST/MI. 2021 ACC/AHA: intravascular imaging 2a; **2025 ACC/AHA ACS and ESC 2024 CCS: Class 1 for LM and complex lesions** (RENOVATE, OCTOBER, IVUS-ACS 2024, 2024 network meta-analysis showing lower cardiac death, MI, ST).

Coronary anatomy and angiographic views

- LM: AP caudal, LAO caudal ("spider"), RAO caudal for LM–LCx. Ostial LM: AP or shallow LAO with slight cranial/caudal, using a shallow cusp injection to avoid pressure damping.
- LAD/diagonals: RAO cranial, AP cranial, LAO cranial (proximal LAD/diagonal bifurcation).
- LCx/OMs: RAO caudal, LAO caudal.
- RCA: LAO (proximal/mid), RAO (mid), AP or LAO cranial (PDA/PL bifurcation), lateral for mid.
- Dominance = which artery gives the PDA (right $\sim 85\%$, left $\sim 8\%$, co-dominant $\sim 7\%$).
- Foreshortening: lesion looks shorter and denser; find orthogonal views before stent sizing.
- TIMI flow 0–3; TIMI myocardial perfusion grade (blush) 0–3; corrected TIMI frame count (normal ~ 21 frames at 30 fps; LAD divided by 1.7).
- Rentrop collateral grade 0–3 (3 = complete epicardial filling); Werner CC0–2 for CTO collateral size (CC2 = side-branch-like, ≥ 0.4 mm).

Anomalous coronaries (case-selection favorites)

- Anomalous LM/LAD from right sinus with **interarterial** course (between aorta and PA): associated with sudden death, especially < 35 y with symptoms/ischemia $\rightarrow$ surgical unroofing. Intramural segment on CT is the culprit. Anomalous RCA from left sinus with interarterial course: surgery if symptoms/ischemia; otherwise less clear.
- Retroaortic LCx from right sinus: benign, most common anomaly ("dot" sign behind aorta).
- ALCAPA (Bland-White-Garland): steal, surgical reimplantation.
- Coronary fistula: close if symptomatic/large shunt/aneurysmal; coil/plug.

- Myocardial bridging: systolic compression, "milking"; beta-blockers; avoid nitrates and stents (fracture, perforation); surgery/unroofing if refractory.

02 – Pharmacology

Procedural anticoagulation

Drug	Dose in PCI	Monitoring / target	Key facts
Unfractionated heparin	70–100 U/kg (50–70 U/kg with GPI)	ACT 250–300 s HemoTec; 300–350 s Hemochron; 200–250 s with GPI	Binds AT-III; $t_{1/2}$ ~60–90 min; reversal protamine 1 mg/100 U given in last 2–3 h (max 50 mg, slow). Protamine reactions: prior NPH insulin, fish allergy, vasectomy, prior protamine
Bivalirudin	0.75 mg/kg bolus, 1.75 mg/kg/h; CrCl <30: 1.0 mg/kg/h; dialysis 0.25 mg/kg/h	ACT rises (not needed to titrate)	Direct thrombin inhibitor; $t_{1/2}$ 25 min; no reversal; less bleeding vs UFH+GPI (ACUITY, HORIZONS-AMI) but ↑acute stent thrombosis in STEMI unless post-PCI infusion continued 2–4 h at PCI dose (BRIGHT, MATRIX). Agent of choice in HIT. VALIDATE-SWEDEHEART: no difference vs UFH with radial + potent P2Y12
Enoxaparin	0.5 mg/kg IV (STEEPLE, ATOLL); if last SC dose <8 h nothing; 8–12 h add 0.3 mg/kg IV; >12 h full IV dose	Anti-Xa (not ACT)	Never "cross" with UFH (↑bleeding, SYNERGY). Renal accumulation
Fondaparinux	Not for PCI alone	—	Catheter thrombosis (OASIS-5/6) → give UFH during PCI
Argatroban	350 µg/kg bolus, 25 µg/kg/min	ACT 300–450	HIT alternative; hepatic clearance

HIT: 4Ts score; stop all heparin (incl. flushes, coated catheters); bivalirudin/argatroban; no platelet transfusions; no warfarin until platelets recovered.

Oral P2Y12 inhibitors

	Clopidogrel	Prasugrel	Ticagrelor
Load / maintenance	600 mg / 75 mg	60 mg / 10 mg (5 mg if <60 kg or ≥75 y)	180 mg / 90 mg BID (60 mg BID after 1 y, PEGASUS)
Type	Thienopyridine prodrug, 2-step CYP2C19 activation; irreversible	Thienopyridine prodrug, 1-step (esterases + CYP3A4/2B6); irreversible	Cyclopentyl-triazolo-pyrimidine, direct-acting, reversible; active metabolite via CYP3A4
Onset (load)	2–6 h (600 mg)	30 min–2 h	30 min–2 h (delayed with opioids, STEMI, hypothermia; crushing helps)
Hold before CABG	5 days	7 days	5 days (ESC: 3 days)

	Clopidogrel	Prasugrel	Ticagrelor
Key trial	CURE, CREDO, CURRENT-OASIS 7 (600 mg load better in PCI)	TRITON-TIMI 38: ↓MI/ST, ↑major bleeding; net harm in prior stroke/TIA (contraindicated), no net benefit ≥75 y or <60 kg	PLATO: ↓CV death, ↓ST, ↑non-CABG bleeding; interaction with aspirin >100 mg (use ≤100 mg)
Special	CYP2C19 *2/*3 loss-of-function → poor response (Asian prevalence higher); PPI interaction (omeprazole/esomeprazole) pharmacodynamic, COGENT found no clinical harm; pantoprazole preferred	Do NOT give upstream in NSTEMI-ACS before anatomy (ACCOAST: bleeding, no benefit); no benefit in medically managed ACS (TRILOGY)	Side effects: dyspnea (~14%, adenosine reuptake inhibition, benign), ventricular pauses, ↑uric acid, ↑creatinine. Avoid with strong CYP3A4 inhibitors/inducers; ↑simvastatin/lovastatin levels (cap 40 mg)

- ISAR-REACT 5: in ACS with planned invasive strategy, **prasugrel** (given after anatomy in NSTEMI-ACS) beat ticagrelor for death/MI/stroke (6.9% vs 9.3%) with no excess bleeding.
- Pretreatment: STEMI – load at first medical contact. NSTEMI-ACS – ESC 2020 says no routine pretreatment if going to early angiography; clopidogrel pretreatment reasonable if delay.
- Switching: escalation (clopidogrel → ticagrelor/prasugrel) load anytime; de-escalation from ticagrelor to clopidogrel: 600 mg clopidogrel load 24 h after last ticagrelor dose.
- Genotype-/platelet-function-guided: TAILOR-PCI negative; POPular Genetics (STEMI) safe de-escalation; TROPICAL-ACS; ALPHEUS.

IV antiplatelets

- **Cangrelor**: IV ATP analog, reversible, $t_{1/2}$ 3–6 min, full recovery 60 min. 30 µg/kg bolus then 4 µg/kg/min for ≥2 h or duration of PCI. CHAMPION PHOENIX: ↓periprocedural MI/ST vs clopidogrel. Transition: ticagrelor any time (during or after); clopidogrel/prasugrel **only after** infusion stops (cangrelor blocks their binding). Bridging for surgery (BRIDGE trial, 0.75 µg/kg/min).
- **GP IIb/IIIa inhibitors** (bail-out for thrombus/no-reflow, not routine):
 - Eptifibatide: 180 µg/kg bolus ×2 (10 min apart), 2 µg/kg/min; CrCl <50: 1 µg/kg/min; contraindicated on dialysis. Renal clearance; $t_{1/2}$ 2.5 h; platelet recovery 4–8 h.
 - Tirofiban: high-dose bolus 25 µg/kg, 0.15 µg/kg/min (halve infusion if CrCl ≤60).
 - Abciximab (no longer marketed): Fab fragment, long platelet effect, thrombocytopenia (1–5%), reversible only with platelet transfusion; eptifibatide/tirofiban: wait 4 h.
 - Thrombocytopenia within hours of GPI: stop drug, check pseudothrombocytopenia (citrate tube), transfuse if bleeding.
 - Intracoronary vs IV: no clear benefit (AIDA STEMI, INFUSE-AMI ↓infarct size only).

Aspirin

- 162–325 mg load (non-enteric, chewed); 81 mg maintenance (ADAPTABLE: 81 = 325). Irreversible COX-1 acetylation. NSAIDs (ibuprofen) block the site if given before aspirin. Aspirin allergy → rapid desensitization protocol; clopidogrel monotherapy if not feasible.

- **2025 ACC/AHA ACS: transition to ticagrelor monotherapy ≥ 1 month after PCI is Class 1** if DAPT is tolerated (TWILIGHT, TICO, ULTIMATE-DAPT, T-PASS). HBR: stop aspirin at 1–3 months, keep the P2Y12 inhibitor (2a). De-escalation to clopidogrel after 1 month: 2b. STOPDAPT-2 ACS: early clopidogrel monotherapy was NOT non-inferior in ACS (more MI). HOST-EXAM / SMART-CHOICE 3: chronic clopidogrel beats aspirin monotherapy. Cangrelor: 2b. Routine GPI: Class 3, bailout 2a.

DAPT duration – the decision map

- **ACS:** 12 months default (ticagrelor or prasugrel preferred). Extend beyond 12 months with ticagrelor 60 mg BID (PEGASUS) or DAPT-trial-style if low bleeding risk and high ischemic risk (DAPT score ≥ 2). Shorten to 3–6 months (or P2Y12 monotherapy after 1–3 months) if HBR (ARC-HBR: ≥ 1 major or ≥ 2 minor criteria).
- **Chronic coronary syndrome:** 6 months default; 1–3 months if HBR (MASTER DAPT: 1 month in HBR non-inferior); 12+ months if high ischemic risk.
- **DAPT trial:** 30 vs 12 months: $\downarrow$ ST and MI, $\uparrow$ moderate/severe bleeding, $\uparrow$ non-CV death signal. PRECISE-DAPT $\geq 25 \rightarrow$ short DAPT.
- **Triple therapy (AF + PCI):** WOEST, PIONEER AF-PCI, RE-DUAL PCI, AUGUSTUS, ENTRUST-AF PCI: DOAC + clopidogrel with aspirin stopped at discharge (up to 1 week; ≤ 30 days if very high ischemic risk). Clopidogrel is the P2Y12 of choice (avoid prasugrel/ticagrelor in triple). AUGUSTUS 2 $\times$ 2: apixaban < warfarin for bleeding; aspirin $\uparrow$ bleeding without ischemic benefit. After 12 months, OAC alone (AFIRE: rivaroxaban monotherapy).
- **Bridging:** hold clopidogrel 5 d, prasugrel 7 d, ticagrelor 5 d before CABG; cangrelor bridge if recent stent.

Vasoactive and rescue drugs in the lab

Situation	Drug and dose
Coronary spasm / catheter-induced spasm	IC nitroglycerin 100–200 μ g; contraindicated within 24 h of sildenafil/vardenafil, 48 h tadalafil, and in RV infarct/severe AS/HOCM (relative)
No-reflow (microvascular)	IC adenosine 30–100 μ g boluses, IC nicardipine 100–200 μ g, IC verapamil 100–200 μ g (bradycardia; RCA), IC nitroprusside 50–200 μ g, IC epinephrine 50–200 μ g for refractory; deliver distally via microcatheter/aspiration catheter. Nitroglycerin does not help microvascular no-reflow
Radial spasm	Verapamil 2.5–5 mg + nitroglycerin 200 μ g intra-arterial "cocktail"; sedation; warm
Vagal reaction	Atropine 0.5–1 mg IV, fluids, leg raise
Hypotension	Phenylephrine 100–200 μ g bolus; norepinephrine infusion first-line in cardiogenic shock (fewer arrhythmias than dopamine, SOAP II subgroup); vasopressin if refractory
Bradycardia/asystole during rotablation (RCA)	Prophylactic temporary pacer or aminophylline/atropine; caused by adenosine release from microparticles
Anaphylactoid reaction to contrast	Epinephrine 0.3 mg IM (or 0.1 mg IV of 1:10,000 if hypotensive), H1/H2 blockers, hydrocortisone 200 mg IV, airway
Protamine reaction	Stop; fluids, epinephrine, steroids
Reversal of sedation	Naloxone 0.4 mg IV (opioids); flumazenil 0.2 mg IV (benzodiazepines – seizures in chronic users)

Situation	Drug and dose
Tako/stress test-like hyperemia issues	Aminophylline 50–100 mg IV reverses adenosine/regadenoson

Contrast media and contrast nephropathy

- Types by osmolality: high-osmolar ionic (diatrizoate, >1500 mOsm; obsolete), low-osmolar (iohexol, iopamidol, ioversol; ~600–800), iso-osmolar (iodixanol, ~290). Iso- vs low-osmolar: no consistent CIN difference (NEPHRIC vs later trials).
- CI-AKI definition: creatinine $\uparrow \geq 0.3$ mg/dL within 48 h or $\geq 1.5\times$ within 7 days. Peak day 3–5, recovers 1–3 weeks. Risk score: Mehran (hypotension, IABP, HF, age >75, anemia, DM, contrast volume, CKD).
- Prevention that works: **limit contrast volume** (keep volume/CrCl ratio <2–3; volume $\leq 3.7 \times \text{eGFR}$), ****isotonic IV hydration**** (1–1.5 mL/kg/h 6–12 h pre and post; LVEDP-guided in POSEIDON), stop nephrotoxins, radial access, staged procedures, ultra-low contrast techniques (IVUS-guided). Does NOT work: N-acetylcysteine and sodium bicarbonate (PRESERVE), prophylactic dialysis, fenoldopam, statin loading (mixed; still reasonable), forced diuresis without matched hydration (RenalGuard has support). AMACING: no hydration non-inferior in low-risk eGFR >30.
- Metformin: no need to hold if eGFR ≥ 30 and no AKI; hold 48 h after contrast if eGFR <30 or AKI (lactic acidosis risk from accumulation, not nephrotoxicity).
- Contrast allergy prophylaxis: prednisone 50 mg PO at 13, 7, 1 h + diphenhydramine 50 mg 1 h before; emergent: hydrocortisone 200 mg IV q4h until done + diphenhydramine. Shellfish allergy is NOT a risk factor; prior contrast reaction and asthma/atopy are.
- Thyroid: iodine load can precipitate thyrotoxicosis (Jod-Basedow) in autonomous nodules; hypothyroidism (Wolff-Chaikoff) in Hashimoto's. Delays radioiodine therapy 6–8 weeks.
- Gadolinium and CO₂ are not substitutes for coronary angiography (CO₂ only below diaphragm).

Other drug facts that appear on the exam

- Statin loading before PCI (ARMYDA-ACS, NAPLES II) $\downarrow$ periprocedural MI; high-intensity after ACS.
- PCSK9 inhibitors/ezetimibe: LDL <55 mg/dL target for very high risk (ESC).
- Colchicine 0.5 mg daily: COLCOT and LoDoCo2 positive, **CLEAR SYNERGY (2024) negative post-MI**; 2b.
- Lipids: high-intensity statin ($\geq 50\%$ LDL reduction); **2026 ACC/AHA dyslipidemia guideline: LDL <55 mg/dL for very-high-risk ASCVD, <70 for other ASCVD**; add ezetimibe, then PCSK9i/inclisiran/bempedoic acid; measure Lp(a) once.
- Beta-blocker after MI with preserved EF is contested: REDUCE-AMI and REBOOT neutral, BETAMI-DANBLOCK favorable. Transfusion in MI: MINT (2023) restrictive strategy not non-inferior; Hb ≥ 10 may be reasonable (2b).
- Fibrinolysis adjuncts: clopidogrel 300 mg (≤ 75 y) or 75 mg (>75 y); enoxaparin preferred (ExTRACT-TIMI 25; ≥ 75 y no bolus, 0.75 mg/kg q12h); half-dose tenecteplase ≥ 75 y (STREAM); TREAT: switching to ticagrelor after lysis (<75 y) non-inferior for bleeding.
- Contrast maximum dose rule of thumb: 5 mL $\times$ weight (kg) / creatinine, up to 300 mL.
- Nitrates + PDE5 inhibitors: profound hypotension.

- Beta-blockers: avoid in cocaine-associated ACS acutely (unopposed alpha) – phentolamine, benzodiazepines, nitrates.
- Sedation: midazolam 0.5–2 mg + fentanyl 25–50 µg; fentanyl delays P2Y₁₂ absorption.
- Vasopressin and terlipressin not for coronary spasm.
- Warfarin (SCAI 2021): INR ≤1.8 femoral, ≤2.2 radial; no vitamin K. Apixaban/rivaroxaban hold 24–48 h; dabigatran 48 h (72–96 h if CrCl <50); emergencies go radial. DOACs need no bridging; VKA bridging only for mechanical mitral valve or embolism within 3 months (BRIDGE: none for AF).
- DOAC renal clearance: dabigatran ~80%, edoxaban ~50%, rivaroxaban ~35%, apixaban ~27%. Edoxaban 30 mg if CrCl 15–50 (avoid if CrCl >95). VKA still required for mechanical valves, rheumatic MS with AF (INVICTUS) and triple-positive antiphospholipid syndrome.
- HIT duration: HIT with thrombosis 3 months; isolated HIT 4–6 weeks (up to 3 months); DOACs acceptable; warfarin only after platelets ≥150k with ≥5 days overlap.
- Drug-eluting stent drugs: sirolimus/everolimus/zotarolimus (mTOR inhibitors, cytostatic, G1 arrest) vs paclitaxel (microtubule stabilizer, cytotoxic, used on DCB because it is lipophilic and retained). Biolimus (polymer-free, LEADERS FREE in HBR). Polymer types: durable fluoropolymer (Xience), biodegradable PLLA/PLGA (Synergy, Orsiro), polymer-free.
- Brachytherapy (beta, Sr-90) for recurrent ISR; late "edge effect" and late thrombosis with short DAPT historically.

03 – Complications and Vascular Access

Coronary perforation (Ellis classification)

Type	Angiographic finding	Typical cause	Management
I	Extraluminal crater, no extravasation	Oversized balloon/stent	Observe; prolonged balloon; often benign
II	Pericardial or myocardial blush, no jet	Wire exit, oversizing	Prolonged balloon inflation (5–10 min), echo, watch; consider covered stent if persistent
III	Frank extravasation, jet ≥ 1 mm	Oversizing, atherectomy, IVL in eccentric calcium	Immediate balloon occlusion proximal/at site; covered stent (PK Papyrus 5F-compatible; Graftmaster needs ≥ 6 –7F); pericardiocentesis; reverse anticoagulation; surgery if fails
III-CS	Into a cavity (LV, RV, coronary sinus)	Septal collateral wiring	Often benign; coils if large fistula

- Distal/wire perforations (hydrophilic/CTO wires, wire beyond the target): treat with microcatheter-delivered coils, autologous fat, thrombin, gelfoam, or subcutaneous fat/ microspheres; balloon occlusion proximal while preparing.
- Two-guide ("ping-pong") technique keeps a balloon inflated while delivering the covered stent through a second guide, or use a 7–8F guide for both.
- Reverse heparin with protamine only after the wire and balloon are in place and definitive treatment is available (protamine causes stent thrombosis if you still need the wire/stent to be patent; reverse after covered stent deployed). GPI: stop; platelets for abciximab. Bivalirudin: no antidote (short $t_{1/2}$).
- Late tamponade after PCI (hours) is classically from a **wire perforation** noticed only when hypotension develops; keep a low threshold for bedside echo.
- Rotablation, orbital atherectomy, oversized balloons ($>1.2:1$), chronic total occlusions, women, elderly, and calcified tortuous vessels are risk factors.

Dissection (NHLBI classification)

Type	Appearance	Action
A	Minor radiolucency within lumen, no persistent contrast	Usually leave alone
B	Parallel tracts / double lumen, no persistence	Observe if flow normal
C	Extraluminal contrast "cap" with persistence after dye clears	Stent
D	Spiral dissection	Stent (wire true lumen, IVUS)
E	New persistent filling defect	Stent
F	Total occlusion, no flow	Stent / rescue

- Ostial RCA/LM catheter-induced dissection: deep seating of Amplatz/guide, deep dye injection; may propagate retrograde into the aorta (aortocoronary dissection): stent the ostium immediately to seal the

entry, CT aorta; surgery if it extends >40 mm up the aorta.

- Edge dissection on OCT/IVUS: treat if >60° arc, >3 mm long, media involved, or lumen <4 mm².
- Wire in false lumen sign: cannot advance, no pull-back "give", contrast staining, side branches not seen.

No-reflow / slow flow

- Mechanism: distal embolization, microvascular spasm, neutrophil plugging, reperfusion injury. Highest risk: SVG, thrombus-laden STEMI, rotational atherectomy, attenuated plaque on IVUS, long ischemia time.
- Treatment: exclude epicardial cause (dissection, spasm, residual stenosis) first; then IC vasodilators delivered distally (adenosine, nicardipine, verapamil, nitroprusside), GPI, IC epinephrine for refractory; support pressure; IABP if hemodynamic collapse.
- Prevention in SVG: embolic protection device (SAFER trial, PercuSurge distal occlusion ↓MACE 42%; FilterWire non-inferior). In STEMI: routine thrombus aspiration NOT recommended (TASTE, TOTAL: no benefit, ↑stroke); bail-out only. Deferred stenting (DANAMI-3 DEFER) no benefit.

Stent thrombosis

- ARC definitions: definite (angiographic/pathologic), probable (unexplained death ≤30 d or target-vessel MI without confirmation), possible (unexplained death >30 d).
- Timing: acute <24 h, subacute 24 h–30 d (early = 0–30 d), late 30 d–1 y, very late >1 y.
- Causes: early → underexpansion, edge dissection, residual disease, DAPT failure/non-response, bivalirudin without infusion, ACS setting; late/very late → DAPT discontinuation, malapposition, uncovered struts, neoatherosclerosis, hypersensitivity (first-gen DES), polymer inflammation, stent fracture, bioresorbable scaffolds (Absorb).
- Treatment: emergent PCI with intravascular imaging to find the mechanism; potent P2Y12 (ticagrelor/prasugrel); GPI reasonable; avoid adding another stent layer if the problem is underexpansion (post-dilate).

In-stent restenosis

- Mehran classification: I focal (≤10 mm: IA gap, IB margin, IC focal body, ID multifocal), II diffuse intrastent, III proliferative (extends beyond stent), IV total occlusion. Recurrence rises from ~20% (I) to >80% (IV).
- Mechanisms: neointimal hyperplasia (smooth muscle migration, peaks 6–12 months for BMS; later for DES), underexpansion (most common correctable cause—image it), stent fracture, geographic miss, neoatherosclerosis, drug resistance.
- Treatment ladder: intravascular imaging → high-pressure NC balloon/scoring/cutting → DES (different drug reasonable; RIBS IV/V: EES ≥ DCB for BMS-ISR, similar for DES-ISR) or paclitaxel DCB (AGENT IDE: DCB > balloon; FDA-approved 2024; preferred when avoiding a third layer), IVL for calcified neointima/underexpanded stent, ELCA laser, brachytherapy for multiple recurrences, CABG for diffuse multi-layer.

Device loss and retrieval

- Stent loss: attempt to re-advance the wire through it and re-balloon with a small balloon to retrieve or deploy in place; loop snare; twisting two wires; or crush against the vessel wall with another stent if in a non-critical location. Peripheral embolization of an unexpanded stent is usually tolerated. Surgery if in LM/aorta root causing embolization.
- Wire fracture/retained fragment: snare; if small and in a distal branch, stent over it.
- Rotablator burr entrapment ("Kokesi"): avoid by short runs, no decelerations >5000 rpm, pecking; treat by pulling with wire loop and guide extension, second wire/balloon beside burr, or surgery.

Vascular access

Femoral

- Target the common femoral artery: over the middle of the femoral head, below the inguinal ligament (not the crease), above the bifurcation. Use fluoroscopy + ultrasound (FAUST: US ↓ complications, ↑ first-pass). Micropuncture.
- **High stick** (above inguinal ligament/ external iliac): retroperitoneal hemorrhage → hypotension, flank/back pain, ipsilateral leg pain (femoral nerve), vagal symptoms; diagnose with CT; treat with reversal, transfusion, contralateral balloon tamponade/covered stent, surgery rarely. Mortality ~5–10%.
- **Low stick** (SFA/profunda): pseudoaneurysm and AV fistula.
- Pseudoaneurysm: pulsatile mass, bruit; US shows "yin-yang" flow. Treat if >2 cm, painful, expanding, or anticoagulated: US-guided thrombin injection (avoid if short/wide neck → distal embolization; use compression or covered stent/surgery), ultrasound compression (fails with anticoagulation), surgery for infected, rapidly expanding, skin compromise, or neuropathy.
- AV fistula: continuous bruit; most close spontaneously; surgery/covered stent if high-output HF, limb ischemia/steal, or enlarging.
- Closure devices (Angio-Seal collagen/anchor, Perclose suture, MynxGrip sealant): ↓ time to hemostasis and ambulation, but do NOT reduce bleeding or vascular complications in RCTs (device failure raises complications ~3×); infection (Angio-Seal, immunocompromised), device embolization, vessel occlusion in small/diseased arteries (<5 mm, heavily calcified), re-access same site after 90 days for Angio-Seal.
- Femoral artery thrombosis/limb ischemia: larger sheaths (MCS), PAD; check pulses; embolectomy.
- Cholesterol embolization: livedo reticularis, blue toes, eosinophilia, AKI over weeks, ↓ complement; supportive; avoid anticoagulation intensification; statins.

Radial

- Evidence: RIVAL (benefit in STEMI and high-volume radial centers), MATRIX (↓ mortality/ bleeding), SAFARI-STEMI (no difference). Default for PCI in current guidelines (ESC class I).
- Anatomy: radial artery loop (2%), ~~high bifurcation~~ (7%), remnant radial (~1%), tortuosity; arteria lusoria (aberrant right subclavian, ~1%) makes right radial hard.
- Allen/Barbeau testing does not predict ischemic complications (RIVAL); not required but commonly done; Barbeau D (no waveform) is the only abnormal result some avoid.
- Spasm: cocktail (verapamil + nitroglycerin), hydrophilic sheaths, sedation, warm arm, smaller sheaths; if a catheter is trapped: more vasodilators, sedation, wait, nerve block, never yank.

- Radial artery occlusion (RAO 2–10%): prevention = heparin ≥ 5000 U (or 50 U/kg), patent hemostasis (Barbeau), short compression, smallest sheath, distal (snuffbox) radial access; treat early with ulnar compression 1 h. Asymptomatic, but lose the access site.
- Radial perforation: advance the catheter/long sheath over the wire to tamponade and finish the case, then compression; forearm hematoma → compress, BP cuff at 15 mmHg below SBP; compartment syndrome (rare) → fasciotomy.
- Radial for STEMI, with 6F; two-stent bifurcation and 1.75 mm burr need 7F (or 7F sheathless).
- Distal radial (snuffbox): DISCO RADIAL (2022) showed RAO was NOT significantly lower (0.3% vs 0.9%) with higher crossover; it remains a reasonable alternative. RAO treatment: ipsilateral ulnar compression 1 h or ~4 weeks of LMWH; about half recanalize spontaneously.

Other

- Brachial: highest complication (median nerve injury, thrombosis); avoid.
- Axillary/subclavian for large-bore MCS; transcaval; carotid for TAVR.
- Venous: femoral vein medial to artery; internal jugular with US; complications: AV fistula, air embolism.

Systemic complications

- **Stroke** (0.2–0.4% diagnostic; higher with LM/aortic instrumentation, TAVR): immediate neuro exam, stop heparin decision, CT/CTA, thrombectomy in lab or neuro IR; time is brain.
- **Air embolism**: 100% O₂, aspirate through guide, forceful saline injection, support.
- **Contrast reaction**: see pharmacology; hypotension + bradycardia = vagal (atropine), vs hypotension + tachycardia/urticaria/bronchospasm = anaphylactoid (epinephrine).
- **Vagal**: atropine, fluids; common with femoral sheath removal.
- **Cardiac tamponade**: hypotension, equalization, loss of y descent, pulsus paradoxus; echo; pericardiocentesis (subxiphoid or echo-guided apical/parasternal), reverse anticoagulation, autotransfusion, surgery if ongoing.
- **Ventricular arrhythmia**: RCA injection (VF from ostial damping/deep injection, ionic contrast), rotablation, hypokalemia; defibrillate, stop injection, correct.
- **Coronary spasm at catheter tip**: pull back, IC nitro.
- **Aortic dissection** (retrograde from RCA ostium): seal with stent, CT, surgery if extends.
- **Radiation skin injury**: see note 07.
- **Ventricular perforation with pigtail/ temporary wire**: tamponade; pull carefully.
- **Endomyocardial biopsy**: RV perforation ~0.5%, TR from chordal injury, heart block.
- **Post-PCI MI** definitions: 4th UDMI type 4a (troponin $>5\times$ URL with ischemia evidence), SCAI (CK-MB $\geq 10\times$ or troponin $\geq 70\times$ URL); type 4b = stent thrombosis; type 4c = restenosis.

04 – Hemodynamics, Shock, and Mechanical Circulatory Support

Normal pressures (mmHg)

RA 0–8 (mean ~3) · RV 15–30 / 0–8 · PA 15–30 / 4–12 (mean 9–18) · PCWP 4–12 · LV 90–140 / 5–12 · Ao 90–140 / 60–90. Cardiac index 2.5–4.0 L/min/m². PVR <2 WU (normal ≈ 0.3–1.6 WU; ×80 = dyn·s/cm⁵). SVR 800–1200 dyn·s/cm⁵ (10–15 WU).

Waveforms

- RA/LA: a (atrial contraction), c (tricuspid closure), x descent (atrial relaxation), v (venous filling), y descent (rapid ventricular filling).
 - Cannon a waves: AV dissociation; large a: TS, PS, RVH, pulmonary hypertension. Absent a: AF.
 - Large v waves: TR (RA) or MR/VSD/LV failure (PCWP). Note: v wave size depends on LA compliance; a large v wave is not specific for severe MR and its absence doesn't exclude it.
 - Blunted/absent y descent: tamponade (also ↓x? no, x preserved). Steep x and y ("M/W" sign): constriction. Rapid y descent: constriction, restriction, severe TR.
 - Kussmaul sign (RA rises with inspiration): constriction, RV infarct, severe TR, restriction.
- LV: "dip and plateau"/square root sign = constriction or restriction. Spike-and-dome aortic = HOCM. Brockenbrough-Braunwald-Morrow sign: post-PVC beat shows **↑LV–Ao gradient and ↓aortic pulse pressure** (HOCM); in fixed AS, pulse pressure ↑ post-PVC.
- Pulsus paradoxus (>10 mmHg inspiratory SBP fall): tamponade, severe asthma/COPD, massive PE, RV infarct; absent in tamponade with ASD, severe AI, LV dysfunction.
- Pulsus alternans: severe LV failure. Bisferiens: AI ± AS, HOCM. Parvus et tardus: AS.
- Wedge validation: mean PCWP ≤ PA diastolic (except large v waves), PCWP O₂ sat >90%, a/v waves visible, catheter in West zone 3.

Cardiac output

- **Fick**: $CO = VO_2 / (CaO_2 - CvO_2) = VO_2 / [Hgb \times 1.36 \times 10 \times (SaO_2 - SvO_2)]$. Assumed VO_2 125 mL/min/m² (or 3 mL/kg/min; 110 in elderly) — assumed values are the largest error; measured VO_2 preferred. Mixed venous sample from PA. Best in low-output states and TR.
- **Thermodilution**: inaccurate with significant TR (underestimates), intracardiac shunts, very low output, irregular rhythm; average 3 within 10%.
- Hemodynamic derivations: $SVR = (MAP - RAP)/CO \times 80$; $PVR = (mPAP - PCWP)/CO$ (WU); transpulmonary gradient = $mPAP - PCWP$ (>12 abnormal); diastolic pulmonary gradient = $PA_d - PCWP$ (≥ 7 = precapillary component in CpcPH).
- Cardiac power output = $MAP \times CO / 451$ (W); <0.6 W strongest predictor of mortality in cardiogenic shock (SHOCK registry).
- $PAP_i = (PA \text{ systolic} - PA \text{ diastolic})/RAP$; <0.9 predicts RV failure in AMI shock; <1.85 post-LVAD. RA/PCWP >0.63 also suggests RV failure.

Shunts

- Oximetry run: SVC (high and low), IVC, RA (high, mid, low), RV (inflow, body, outflow), PA, and arterial. Significant step-up: $\geq 7\%$ at atrial level, $\geq 5\%$ at ventricular or great-artery level (single-sample criteria; averaged samples $\geq 5\%/\geq 3\%$).
- Mixed venous O₂ (with atrial shunt) = $(3 \times \text{SVC} + 1 \times \text{IVC})/4$ (Flamm). Use SVC alone if no IVC sample; never use RA in the presence of an ASD.
- $Q_p/Q_s = (\text{SaO}_2 - \text{MVO}_2) / (\text{PvO}_2 - \text{PaO}_2)$. Assume $\text{PvO}_2 = 98\%$ if not sampled and no lung disease. $Q_p/Q_s \geq 1.5$ significant (closure indicated with RV enlargement), ≥ 2 large; < 1 = net $R \rightarrow L$.
- Effective vs total flows: $L \rightarrow R$ shunt = $Q_p - Q_{\text{eff}}$; $R \rightarrow L = Q_s - Q_{\text{eff}}$, where Q_{eff} uses MVO_2 and PvO_2 .
- Arterial sat $< 95\%$ on room air without lung disease $\rightarrow$ suspect $R \rightarrow L$ shunt or Eisenmenger.
- Hemodynamic contraindication to shunt closure: $\text{PVR} > 2/3 \text{ SVR}$ or $\text{PAP} > 2/3$ systemic without reversibility (vasodilator test, balloon occlusion) — Eisenmenger physiology.

Valve calculations

- **Gorlin:** $\text{Area} = \text{CO}(\text{mL/min}) / [\text{HR} \times \text{flow period}(\text{s/beat}) \times 44.3 \times \sqrt{(\text{mean gradient})}]$. Aortic: systolic ejection period; mitral: diastolic filling period and constant 37.7 (44.3×0.85). Hakki simplification: $\text{Area} \approx \text{CO}(\text{L/min})/\sqrt{(\text{gradient})}$.
- Gorlin underestimates area in low-flow states (flow-dependent). Use peak-to-peak for screening only; mean gradient from simultaneous LV–Ao tracings (two catheters or dual-lumen; femoral sheath pressure lags and overestimates pulse pressure). Never use pullback in AF or with variable rhythm.
- Severe AS: $\text{AVA} < 1.0 \text{ cm}^2$ ($\text{AVA}_i < 0.6$), mean gradient ≥ 40 , $\text{Vmax} \geq 4 \text{ m/s}$. Low-flow low-gradient (EF $< 50\%$): dobutamine ($5\text{--}20 \mu\text{g/kg/min}$) $\rightarrow$ true severe if AVA stays < 1.0 with gradient ≥ 40 or flow reserve absent (stroke volume $\uparrow < 20\%$) $\rightarrow$ still consider AVR (high risk); paradoxical LFLG with preserved EF $\rightarrow$ CT calcium score ($\geq 2000 \text{ AU}$ men, ≥ 1200 women = severe).
- Mitral stenosis: transseptal LA pressure is the gold standard; PCWP overestimates the gradient (delay + damping; shift the tracing 50–70 ms). Severe MVA $\leq 1.5 \text{ cm}^2$, mean gradient

5–10 at rest; exercise/pacing to unmask.

- Aortic regurgitation: wide pulse pressure, low aortic diastolic, rapid LVEDP rise equilibrating with aortic diastolic in acute severe AI (premature MV closure).
- Prosthetic valves: DVI, effective orifice area; patient–prosthesis mismatch indexed EOA ≤ 0.85 (moderate), ≤ 0.65 (severe).

Constriction vs restriction vs tamponade

Feature	Constriction	Restriction	Tamponade
RA waveform	Prominent x and y (M/W)	Prominent y, $\uparrow a$	Blunted/absent y
LVEDP – RVEDP	$\leq 5 \text{ mmHg}$ (equalized)	$> 5 \text{ mmHg}$	Equalized diastolics (RA $\approx$ PCWP $\approx$ PAd)
RVSP	$< 50\text{--}55$	> 55	Variable

Feature	Constriction	Restriction	Tamponade
RVEDP/RVSP	>1/3	<1/3	—
Dip-and-plateau	Yes	Yes	No
Respiratory LV/RV systolic pressure	Discordant (ventricular interdependence; systolic area index >1.1, most specific)	Concordant	Pulsus paradoxus
Kussmaul	Yes	Possible	Usually absent (no); rises with tamponade only if effusive-constrictive
Other	Pericardial thickening on CT/MRI; BNP low	BNP high; conduction disease; biopsy	Echo effusion, chamber collapse

Effusive-constrictive: RA stays high after pericardiocentesis.

HOCM

- Gradient at rest ≥ 30 (provoked ≥ 50) is severe; provocation: Valsalva, PVC, isoproterenol, amyl nitrate, exercise. Increased by $\downarrow$ preload, $\downarrow$ afterload, $\uparrow$ contractility. Alcohol septal ablation: see structural note. Mavacamten (myosin inhibitor) reduces need for SRT.

Pulmonary hypertension in the lab

- Definition (2022 ESC/ERS): mPAP >20; precapillary PVR >2 WU with PCWP ≤ 15 ; post-capillary PCWP >15; combined if PVR >2 with PCWP >15. Exercise PH: mPAP/CO slope >3.
- Vasoreactivity (inhaled NO 10–20 ppm, epoprostenol, adenosine): positive = mPAP $\downarrow \geq 10$ to <40 with unchanged/ $\uparrow$ CO $\rightarrow$ CCB responder (only idiopathic/heritable/drug-induced PAH).
- Balloon pulmonary angioplasty for inoperable CTEPH; reperfusion lung injury.

Cardiogenic shock

- Definition: SBP <90 (or vasopressors) with CI <2.2 and PCWP >15, hypoperfusion (lactate >2, oliguria, altered mentation). SCAI stages A (at risk) B (beginning: hypotension without hypoperfusion) C (classic: needs drugs/device) D (deteriorating) E (extremis: arrest/ECMO).
- AMI-CS: early revascularization of the culprit (SHOCK: 6-month and long-term mortality benefit); **culprit-only** PCI in the acute setting (CULPRIT-SHOCK: 30-day death/RRT 45.9% vs 55.4% with immediate multivessel). Norepinephrine first vasopressor; dobutamine inotrope; avoid pure vasodilators.
- RV infarct: hypotension, clear lungs, $\uparrow$ JVP, Kussmaul; ST $\uparrow$ V4R; avoid nitrates/diuretics; volume (cautiously), dobutamine, reperfuse RCA, AV synchrony/pacing, RV support.
- Mechanical complications: papillary muscle rupture (posteromedial, single RCA supply, day 2–7, acute MR often with soft murmur $\rightarrow$ surgery, IABP bridge), VSD (anterior MI $\rightarrow$ apical; inferior $\rightarrow$ basal; step-up at RV; surgery, delayed if stable; percutaneous closure possible), free wall rupture (tamponade/PEA; surgery), LV aneurysm vs pseudoaneurysm (narrow neck, posterior/inferior; pseudoaneurysm $\rightarrow$ surgery).

Mechanical circulatory support

Device	Mechanism	Flow	Access	Contraindications / issues
IABP	Counterpulsation: inflates at diastolic notch (diastole) → ↑coronary perfusion; deflates just before systole → ↓afterload. Helium	~0.5–1 L/min (modest; ↓MVO ₂ 15–20%)	7–8F femoral, axillary; tip 1–2 cm below left subclavian	Contra: moderate-severe AI, aortic dissection, severe PAD (relative), uncontrolled sepsis. Timing errors: early inflation → premature AV closure/↑afterload; late inflation → poor augmentation; early deflation → suboptimal unloading/retrograde flow; late deflation → ↑afterload (worst). Balloon rupture (blood in tubing) → remove. IABP-SHOCK II: no mortality benefit in AMI-CS (class III routine). Sizing 40 cc if height 162–182 cm
Impella 2.5 / CP / 5.0 / 5.5	Axial micro-flow pump LV → aorta; unloads LV, ↓LVEDP, ↓wall stress	2.5 / 3.5–4.3 / 5.0 / 5.5+ L/min	12F / 14F (CP, percutaneous femoral) / 21F surgical or axillary (5.0, 5.5)	Contra: mechanical AV, severe AS (AVA <0.6 relative), moderate-severe AI, LV thrombus, VSD (R → L hypoxemia), severe PAD. Issues: hemolysis (plasma-free Hgb >40; reposition/↓P-level), suction alarms (volume, RV failure, tamponade), positioning inlet 3.5 cm below AV (CP), aortic insufficiency, limb ischemia, bleeding, need for ACT 160–180 with heparin purge (bicarbonate purge alternative). DanGer Shock (2024): Impella CP in STEMI-CS ↓180-day death (45.8% vs 58.5%) with more complications. PROTECT II/III: high-risk PCI; not powered for mortality
Impella RP / RP Flex	RA → PA axial pump	~4 L/min	23F femoral venous (RP), 22F IJ (Flex)	RV failure post-MI/LVAD/cardiectomy; contra: RV thrombus, mechanical TV/PV
VA-ECMO	Centrifugal pump + oxygenator; venous drainage → arterial return retrograde	3–6 L/min full support	21–25F venous, 15–19F arterial (+ distal perfusion catheter to prevent limb ischemia)	↑LV afterload → LV distension, pulmonary edema, LV thrombus → vent (IABP, Impella "ECPPELLA", atrial septostomy). Harlequin (north–south/differential hypoxemia): deoxygenated blood from recovering native heart reaches upper body when lungs fail → monitor right radial ABG/right ear sat, ↑vent, convert to VAV. ECLS-SHOCK/ ECMO-CS: no mortality benefit, more bleeding/limb ischemia in AMI-CS. Contra: severe AI, dissection, no exit strategy, uncontrolled bleeding
TandemHeart	LA → femoral artery centrifugal pump via transseptal 21F cannula	3.5–5 L/min	21F venous transseptal, 15–17F arterial	Needs transseptal; cannula migration to RA → R → L shunt; limb ischemia

- Pressure-volume loop teaching: IABP shifts loop slightly left/down (↓afterload); Impella unloads volume and pressure (loop shifts left and narrows, ↓LVEDP); VA-ECMO shifts loop **right** and up (↑afterload, ↑LVEDP) unless vented.
- Hemodynamic support in high-risk PCI: BCIS-1 (elective IABP, no MACCE benefit; late mortality signal), PROTECT II (Impella 2.5 vs IABP: 30-d NS, 90-d per-protocol trend), **CHIP-BCIS3 (ACC 2026): Impella CP in LVEF ≤35% complex PCI gave no benefit and HIGHER CV death (26.7% vs 14.5%); PROTECT**

IV pending. Routine MCS for high-risk PCI is not supported. 2025 ACS guideline: microaxial pump in selected STEMI-CS is 2a; routine IABP or VA-ECMO in shock is Class 3.

- Post-arrest: TTM2 showed 33 °C no better than normothermia; 2025 AHA: temperature control 32–37.5 °C for ≥36 h with fever avoidance.
- Escalation triggers on support: lactate not clearing at 6–12 h, CPO <0.6, PAPI <1, rising vasopressor dose; LV venting if PCWP >18–20 on ECMO, no aortic valve opening, or pulmonary edema.

05 – Structural Heart Intervention

TAVR

Indications (2020 ACC/AHA VHD). Severe symptomatic AS (or asymptomatic with EF <50%, very severe AS, abnormal exercise test, rapid progression, or other cardiac surgery). **Newer evidence:** EARLY TAVR (2024) showed TF-TAVR in asymptomatic severe AS reduced death/stroke/unplanned CV hospitalization vs surveillance; **FDA approved TAVR for asymptomatic severe AS (May 2025).** AVATAR/EVOLVED: early SAVR benefit. **ESC/EACTS 2025: TAVI favored at ≥70 y.** DEDICATE (2024): TAVR non-inferior at low–intermediate risk. NOTION 10-y: similar mortality, less structural valve deterioration with TAVR. **NOTION-3 (2024): PCI of significant CAD (FFR ≤0.80 or ≥90%) before TAVR reduced death/MI/urgent revascularization** (ACTIVATION had been negative); coronary evaluation before TAVR is Class 1. **JenaValve Trilogly: first TAVR approved for pure aortic regurgitation (March 2026).** TMVR: Tendyne (transapical, 2025) and SAPIEN M3 (transseptal, Dec 2025) approved; neo-LVOT <1.7–2.0 cm² predicts LVOT obstruction (LAMPOON or septal ablation to mitigate). Choice: age >80 or life expectancy <10 y → TAVR; <65 or life expectancy >20 y → SAVR (durability); 65–80 shared decision. TAVR favored when: high surgical risk, porcelain aorta, hostile chest, frailty. SAVR favored when: bicuspid with aortopathy needing repair, low coronary height, unfavorable annulus, concomitant CABG/mitral surgery, endocarditis. Pivotal trials: PARTNER 1B (inoperable vs medical: ↓death 20% absolute), 1A (high risk vs SAVR non-inferior), CoreValve US Pivotal (high risk: TAVR superior mortality), PARTNER 2A/SAPIEN 3 and SURTAVI (intermediate risk), PARTNER 3 and Evolut Low Risk (low risk; PARTNER 3 1-y death/stroke/rehosp 8.5% vs 15.1%; 5-y results converged for death). Evolut Low Risk 4-y: TAVR ↓death/disabling stroke. NOTION (all-comers), UK TAVI.

Planning (gated CTA). Annulus area/perimeter (systolic phase), LVOT calcium, sinus of Valsalva diameters, sinotubular junction, coronary heights (LM usually higher than RCA; **<10–12 mm** plus narrow sinuses → coronary occlusion risk), leaflet length, aortic angulation (>48–70° horizontal aorta favors balloon-expandable), iliofemoral size (≥5.5 mm for 14F eSheath; calcification, tortuosity), membranous septum length (short → pacemaker). Sizing: balloon-expandable oversize area 5–15%; self-expanding oversize perimeter 10–25%. Bicuspid: Sievers types (0 no raphe, 1 one raphe most common, 2 two raphe); higher paravalvular leak, annular rupture, pacemaker, aortopathy; use supra-annular/inter-commissural sizing; not in pivotal RCTs.

Devices. Balloon-expandable (SAPIEN 3/Ultra; intra-annular, cobalt-chrome, rapid pacing required, lower pacemaker rate ~5–8%, higher annular rupture risk if LVOT calcium and oversizing) vs self-expanding (Evolut PRO/FX supra-annular nitinol: better hemodynamics in small annuli, recapturable, more pacemakers ~15%, more PVL historically; ACURATE neo2 and Navitor alternatives). SMART trial: self-expanding better hemodynamics/valve dysfunction in small annuli.

Complications and management.

Complication	Predictors	Management
Conduction: new LBBB (10–30%), high-grade AVB needing PPM (~5–15%)	Pre-existing RBBB (strongest), self-expanding valve, deep implant, short membranous septum, LVOT calcium, PR/QRS prolongation post-implant	Temporary pacer; monitoring ≥24 h with new LBBB; ambulatory monitoring; PPM if HAVB, or new LBBB with QRS >150/PR >240 not resolving; implant depth <3–5 mm reduces risk (cusp-overlap technique)
Paravalvular leak	Undersizing, eccentric calcium, bicuspid, malposition	Post-dilation (self-expanding), valve-in-valve, plug closure; moderate+ PVL ↑mortality

Complication	Predictors	Management
Stroke (2–3% at 30 d)	Bicuspid, AF, prior stroke, valve-in-valve, more manipulation	Cerebral embolic protection (Sentinel): PROTECTED TAVR (2022) and BHF PROTECT-TAVI (2025) showed no stroke reduction; not routine
Coronary obstruction (<1%; mortality ~40%)	Low coronary height, narrow SOV, bulky leaflets, valve-in-valve (stented bioprosthesis with externally mounted leaflets, VTC <4 mm)	Protect with wire/undeployed stent ("chimney" snorkel), BASILICA leaflet laceration, ShortCut device; emergent PCI
Annular/root rupture (<1%)	Balloon-expandable, LVOT calcium, >20% oversizing, aggressive post-dilation	Reverse, pericardiocentesis, surgery
Vascular (major 2–5%)	Sheath-to-femoral artery ratio >1.05, calcification, female	Covered stent from contralateral access, balloon tamponade, surgery
Valve embolization/malposition	Undersizing, pacing failure	Second valve, snare into descending aorta
Leaflet thrombosis (HALT ~10–15% subclinical)	Valve-in-valve, no anticoagulation, low flow	Anticoagulate if ↑gradient/symptoms; routine OAC not indicated (GALILEO: rivaroxaban ↑death/bleeding; ATLANTIS)
Hemolysis, endocarditis (~1%/y; Enterococcus, Staph), tamponade (temporary wire/pigtail), AKI		

Antithrombotic after TAVR. Aspirin monotherapy (POPular TAVI cohort A: aspirin alone < aspirin + clopidogrel for bleeding, no ischemic penalty); if OAC indicated, OAC alone (cohort B). Valve-in-valve and prior TAVR: BASILICA; TAV-in-TAV planning (commissural alignment, neoskirt height).

Post-TAVR PCI: coronary access harder with supra-annular valves; commissural alignment during implant; guide catheters via valve cells.

Balloon aortic valvuloplasty

Bridge/palliation only (restenosis ~50% at 6–12 months); retrograde femoral; rapid pacing; sizing 1:1 to annulus. Complications: stroke, severe AI, annular rupture, vascular. Role: hemodynamic instability before TAVR, non-cardiac surgery, to assess symptomatic benefit.

Percutaneous mitral balloon valvuloplasty (PMBV, Inoue)

- Indication: severe rheumatic MS (MVA ≤ 1.5 cm²) with symptoms or PASP >50 (asymptomatic), new AF, and favorable anatomy.
- Wilkins score (leaflet mobility, thickening, calcification, subvalvular thickening; each 1–4): **≤ 8 favorable**; >8 poorer results (but consider in non-surgical candidates); commissural calcification is the strongest negative predictor.
- Contraindications: LA thrombus (TEE mandatory; anticoagulate 2–3 months and repeat), moderate–severe MR ($\geq 2+$), bicommissural calcification, concomitant severe AS/CAD needing surgery.

- Technique: transseptal (fossa ovalis, guided by TEE/ICE), Inoue balloon sized by height (height cm/10 + 10 = max diameter mm), stepwise inflation checking MR by echo.
- Complications: severe MR (2–10%; anterior leaflet tear → surgery), iatrogenic ASD (usually small, closes), tamponade (1%), stroke (1%), death <1%. Restenosis 10–20% at 5 y; repeat PMBV possible.
- Pregnancy: safe second trimester with shielding.

Transcatheter edge-to-edge repair (TEER)

- **Primary (degenerative) MR:** EVEREST II (surgery better for MR reduction, TEER safer); indicated for prohibitive surgical risk with suitable anatomy. PASCAL (CLASP IID) non-inferior to MitraClip.
- **Secondary (functional) MR:** COAPT (US, 614 pts; EF 20–50%, LVESD ≤70 mm, EROA ≥30 mm², GDMT optimized by HF specialist): ↓HF hospitalization (35.8% vs 67.9%) and ↓death (29.1% vs 46.1%) at 2 y. MITRA-FR (France, larger LVs, smaller EROA ≥20 mm², less GDMT optimization): neutral. Concept: "disproportionate" MR relative to LV size benefits. Guideline class 2a for COAPT-like patients; RESHAPE-HF2 (2024) also positive.
- Anatomic criteria (classic EVEREST): central A2/P2 pathology, coaptation length ≥2 mm, coaptation depth <11 mm, flail gap <10 mm, flail width <15 mm, MVA >4.0 cm², mean gradient <4–5 mmHg, no significant leaflet calcification in the grasp zone, no cleft/perforation.
- Transseptal puncture: **posterior and superior**, height 4–5 cm above the mitral annulus (3.5–4 cm for medial/commissural jets); ICE/TEE bicaval + short-axis views.
- Complications: single-leaflet device attachment (1–2%), leaflet tear, iatrogenic mitral stenosis (gradient >5 mmHg; avoid adding clips), chordal entanglement, iatrogenic ASD (close if R → L shunt, RV failure, or Qp/Qs >1.5), tamponade, stroke.
- Tricuspid TEER: TRILUMINATE (↓TR, ↑QoL; no mortality/HF benefit at 1 y); TriClip. Transcatheter TV replacement (EVOQUE, TRISCEND II). Transcatheter MVR (Tendyne; LVOT obstruction risk from neo-LVOT <170 mm²).

PFO and ASD closure

- Platypnea-orthodeoxia (dyspnea and desaturation when UPRIGHT, relieved supine) is a non-stroke indication for PFO closure, as are decompression illness in divers and paradoxical embolism.
- **PFO:** cryptogenic ischemic stroke age 18–60 with high-risk features (large shunt, atrial septal aneurysm), after exclusion of other causes (AF monitoring). RESPECT (long-term), CLOSE, REDUCE, DEFENSE-PFO: closure + antiplatelet superior to antiplatelet alone (RRR ~50%), with ↑new AF (mostly transient, peri-procedural ~5%). RoPE score (higher = more likely PFO-attributable; young, no vascular risk factors, cortical stroke). PASCAL classification combines RoPE with anatomy. Not for migraine (PREMIUM, MIST negative), or decompression illness routinely. Device: Amplatzer PFO occluder, Gore Cardioform. Post: DAPT 1–6 months then aspirin; SBE prophylaxis 6 months.
- **Secundum ASD:** close if RA/RV enlargement, Qp/Qs ≥1.5, or paradoxical embolism, regardless of symptoms; defect ≤38 mm with ≥5 mm rims (deficient retro-aortic rim acceptable but ↑erosion). Contraindicated: PVR >2/3 SVR or PAP >2/3 systemic without reversibility (test occlusion); **2025 ACHD guideline: close if PVR ≤2 WU; consider at 2–<5 WU in expert centers; no closure with high PVR/Eisenmenger;** sinus venosus, primum, coronary sinus ASDs (surgical; sinus venosus covered-stent investigational). Balloon sizing (stop-flow diameter); device ~ stop-flow diameter. Complications: erosion

(0.1–0.3%; deficient aortic rim + oversizing; presents days–years with tamponade/aorto-atrial fistula), device embolization (undersizing, deficient rims), AF/SVT, heart block (large devices near AV node), thrombus, nickel allergy (rare clinical significance), residual shunt. Post: DAPT 1–6 months, aspirin 6 months.

- Older patients: LV diastolic dysfunction may be unmasked (test occlusion with LVEDP; fenestrated device).

Left atrial appendage occlusion

- Indication: non-valvular AF with CHA₂DS₂-VASc ≥ 2 (men)/3 (women) and a good reason not to take long-term OAC (2023 ACC/AHA class 2a; prior 2b). PROTECT AF / PREVAIL (Watchman vs warfarin: non-inferior, ↓hemorrhagic stroke, ↑procedural complications early), PRAGUE-17 (vs DOAC non-inferior), CHAMPION-AF/CATALYST ongoing. Watchman FLX (PINNACLE FLX): 0.5% major complications; Amulet IDE non-inferior to Watchman with more procedural complications, less device-related thrombus at 45 d? (fewer leaks). OPTION: LAAO after AF ablation ↓bleeding.
- Anatomy: TEE/CT at 0/45/90/135°; ostium width and depth; morphologies (chicken wing, windsock, cactus, cauliflower); exclude thrombus (contraindication).
- Transseptal: **inferior and posterior** puncture for LAAO (vs superior for TEER).
- Post: Watchman: warfarin (or DOAC) + aspirin 45 days → TEE/CT → DAPT to 6 months → aspirin; DAPT-only regimen acceptable if OAC-intolerant. Amulet: DAPT. Device-related thrombus 2–4% (↑stroke 3–5×; risk with deep implant >10 mm below the pulmonary ridge, low EF, large LAA; treat with OAC). **Peri-device leak: any leak, even <5 mm, is associated with more ischemic events (NCDR)**; options are plugs, coils, second device. Complications: effusion needing intervention 0.5–1%, embolization <0.5%, stroke.
- Newest trials: **CHAMPION-AF (2026)** (Watchman FLX vs DOAC in OAC-eligible AF): non-inferior for CV death/stroke/SE (5.7% vs 4.8%), less non-procedural bleeding (10.9% vs 19.0%), but ischemic stroke/SE numerically higher (3.2% vs 2.2%). **CLOSURE-AF (2026)** (HBR/OAC-contraindicated vs best medical care): LAAO FAILED non-inferiority. OPTION (2024): after ablation, LAAO ↓ bleeding, non-inferior for stroke. Guideline (2023 AF): 2a with contraindication to long-term OAC, 2b for HBR/preference.

Alcohol septal ablation (ASA)

- Indication: obstructive HCM with NYHA III–IV (or II with severe gradient) despite medical therapy (beta-blocker, verapamil, disopyramide, mavacamten), gradient ≥ 50 mmHg rest/provoked, septal thickness ≥ 15 mm (ideally 16–18; >30 mm less effective), suitable septal perforator, no other surgical indication (e.g., intrinsic MV disease, papillary anomalies favor myectomy). Myectomy preferred in young patients and complex anatomy; ASA for comorbidities/elderly.
- Technique: OTW balloon in first (or target) septal perforator, myocardial contrast echo to confirm the basal septal target without remote enhancement (papillary muscle, RV free wall), 1–3 mL desiccated ethanol slowly, balloon inflated 5–10 min. Temporary pacemaker mandatory.
- Complications: complete heart block ~10–15% needing PPM, mostly within 24–48 h (higher with pre-existing LBBB or first-degree AVB; ASA causes **RBBB** typically, myectomy causes **LBBB**), VT/VF, large infarct with remote territory, coronary dissection, tamponade, death ~1%. Gradient reduction continues for months (remodeling); ~10–20% need repeat/myectomy.
- Medical therapy ladder (2024 AHA/ACC HCM): non-vasodilating beta-blocker, then non-DHP CCB; if still symptomatic, **myosin inhibitor (mavacamten; aficamten FDA Dec 2025), disopyramide, or SRT** (all

Class 1). VALOR-HCM: mavacamten reduced SRT eligibility; MAPLE-HCM: aficamten > metoprolol. Myectomy preferred with intrinsic MV disease, septum >30 mm, or young age.

Paravalvular leak closure

Hemolysis or HF symptoms; Amplatzer vascular plugs; retrograde aortic, transseptal or transapical for mitral; risk of leaflet impingement; rock the device before release.

Coarctation and other adult congenital

- Native or recurrent coarctation with peak-to-peak gradient ≥ 20 mmHg (or < 20 with collaterals/hypertension): covered/bare stents in adults; balloon alone in recoarctation; aneurysm/dissection risk, especially native and > 1.5 balloon/isthmus ratio. Long-term hypertension persists in many.
- Pulmonary valvuloplasty: severe PS (peak > 64 mmHg, mean > 35) or symptomatic moderate–severe domed PS; balloon $1.2\text{--}1.4\times$ annulus; dysplastic valves (Noonan) respond poorly. "Suicide RV" (dynamic infundibular obstruction after relief): fluids + beta-blocker, avoid inotropes.
- Repaired tetralogy: PVR for symptomatic $\geq$ moderate PR; asymptomatic with **RVESVi > 80 mL/m²** (2025 ACHD), RVEDVi ≥ 160 , RV dysfunction, or arrhythmia. Before TPVR (Melody/SAPIEN in conduits; Harmony/ Alterra in native RVOT): coronary compression testing with a balloon in the RVOT; pre-stent conduits.
- Coarctation: 50–75% of coarctation patients have a bicuspid valve (only a minority of BAV have coarctation); Turner, Shone complex, intracranial aneurysm $\sim 10\%$.
- PDA closure (Amplatzer duct occluder; coil if small), VSD closure (perimembranous $\rightarrow$ AV block risk; muscular; post-MI VSD).
- Pulmonary vein stenosis after ablation: angioplasty/stenting.
- Transcatheter pulmonary valve (Melody, SAPIEN in RVOT conduits; Harmony for native RVOT): endocarditis risk with Melody.

Pericardial procedures

- Pericardiocentesis: subxiphoid (toward left shoulder) or echo-guided apical/parasternal (shortest fluid path); confirm with agitated saline; leave drain until < 25 mL/day; complications: RV puncture, coronary laceration, pneumothorax, liver injury.
- Balloon pericardiotomy for malignant recurrent effusions.

06 – Case Selection, ACS Pathways, and Lesion-Subset Technique

STEMI

- Reperfusion goals: FMC-to-device ≤ 90 min (≤ 120 min if transfer); door-in-door-out ≤ 30 min; fibrinolysis within 30 min of arrival if PCI cannot be done ≤ 120 min from FMC (and symptoms < 12 h, ideally < 3 h). Age > 75 : half-dose tenecteplase (STREAM). After lysis: transfer all for angiography within 2–24 h (pharmacoinvasive: TRANSFER-AMI, CARESS-in-AMI); **rescue PCI** if failed lysis ($< 50\%$ ST resolution at 60–90 min, persistent pain, instability). Lysis contraindications: prior ICH, structural cerebrovascular lesion, intracranial neoplasm, ischemic stroke within 3 months, suspected aortic dissection, active bleeding, significant head trauma within 3 months, intracranial/spinal surgery within 2 months, severe uncontrolled HTN, prior streptokinase (for SK).
- Late presenters: PCI for 12–24 h with ongoing ischemia, HF, shock, arrhythmia; asymptomatic stable occlusion 3–28 days: no benefit (OAT) → medical therapy.
- Multivessel disease in STEMI without shock: **complete revascularization** (COMPLETE: $\downarrow$ CV death/MI 7.8% vs 10.5%; either index admission or staged within 45 days; PRAMI, CvLPRIT, DANAMI-3-PRIMULTI, MULTISTARS AMI immediate = staged, BIOVASC). Angiography-guided non-culprit PCI is at least as good as FFR-guided in STEMI (FLOWER-MI, FRAME-AMI favored FFR but small). In shock: culprit-only (CULPRIT-SHOCK).
- Thrombus: no routine aspiration (TASTE, TOTAL); bail-out aspiration for large burden. Deferred stenting not helpful (DANAMI-3-DEFER). Radial preferred. Bivalirudin vs heparin: equal with post-PCI infusion (MATRIX, BRIGHT-4 favored bivalirudin + high-dose infusion).
- Post-MI: DAPT 12 months, high-intensity statin, beta-blocker (REDUCE-AMI/REBOOT neutral if EF preserved; BETAMI-DANBLOCK favorable), ACEi/ARB, MRA if EF $\leq 40\%$ with HF/DM, SGLT2i emerging, cardiac rehab. ICD at ≥ 40 days if EF $\leq 35\%$ (≥ 90 days after revascularization).
- MINOCA (5–6%): CMR to find myocarditis/Takotsubo/true MI; OCT for plaque erosion/SCAD; vasospasm provocation.
- **SCAD**: young women, peripartum, FMD, emotional stress; angiographic types 1 (contrast staining/double lumen), 2 (long smooth narrowing, most common, 2A vs 2B to the terminus), 3 (focal, mimics atherosclerosis; use OCT/IVUS). Conservative management preferred (spontaneous healing $> 90\%$); PCI only for ongoing ischemia/LM/shock (propagation risk; long stents, cutting balloon fenestration); avoid lysis; beta-blockers reduce recurrence; screen for FMD (brain-to-pelvis CTA); avoid pregnancy/OCPs.
- Takotsubo: apical ballooning, LVOT obstruction possible (avoid inotropes).

NSTEMI-ACS

- Risk stratify: GRACE, TIMI, hs-troponin pathways. Timing of invasive strategy: **immediate (< 2 h)** for very high risk (hemodynamic instability/shock, refractory chest pain, life-threatening arrhythmia, mechanical complication, acute HF from ischemia, recurrent dynamic ST changes); ****early (< 24 h)**** for high risk (NSTEMI by troponin, dynamic ST/T changes, GRACE > 140 ; TIMACS, VERDICT); **within 72 h** for

intermediate risk (DM, CKD, EF <40%, early post-MI angina, prior PCI/CABG, GRACE 109–140); ischemia-guided for low risk.

- Complete revascularization is Class 1 in NSTEMI-ACS (2025). SENIOR-RITA (2024): invasive strategy in NSTEMI ≥75 y reduced non-fatal MI without a mortality difference. In cardiac arrest without STEMI, no immediate angiography benefit (COACT, TOMAHAWK).
- Antithrombotic: aspirin + P2Y12 (prasugrel after anatomy or ticagrelor); anticoagulant UFH, enoxaparin, bivalirudin (fondaparinux for conservative). No upstream GPI.

Stable (chronic) coronary syndrome

- COURAGE, BARI 2D, ISCHEMIA (moderate–severe ischemia, no LM: invasive strategy did not ↓ death/MI; ↓ spontaneous MI but ↑ periprocedural MI; better angina control), ISCHEMIA-CKD (no benefit, more dialysis), ORBITA (single-vessel: PCI ≈ sham for exercise time), ORBITA-2 (PCI vs sham without antianginals: PCI improves angina). REVIVED-BCIS2: PCI in ischemic cardiomyopathy EF ≤35% did not ↓ death/HF hospitalization vs OMT. STICH(ES): CABG benefit at 10 y in ischemic cardiomyopathy.
- Revascularize for: symptoms despite OMT (class 1), LM ≥50%, proximal LAD ≥70% in MVD (survival benefit for CABG 2a), MVD with EF ≤35% (CABG 1), diabetes with MVD (CABG over PCI: FREEDOM, BARI 2D), large ischemic burden (debated after ISCHEMIA).
- Left main: SYNTAX score ≤22 → PCI reasonable (class 2a), 23–32 → 2a/2b depending on guideline and heart team, ≥33 → CABG. Trials: SYNTAX (LM subgroup equal), PRECOMBAT, EXCEL (5-y death/stroke/MI 22.0% vs 19.2%, non-significant; all-cause death 13% vs 9.9% controversy; periprocedural MI definition debate), NOBLE (5-y MACCE 28% vs 19%, CABG better), meta-analyses: no mortality difference, more spontaneous MI/revascularization with PCI. 3VD: SYNTAX (CABG better in intermediate/high), FAME 3 (CABG better even with FFR-guided PCI at 1 y; 3-y no difference).
- Use SYNTAX II/heart team; STS score (mortality, morbidity); EuroSCORE II.
- Ischemic testing before PCI: 2021 guidelines require documented ischemia or physiology for intermediate lesions (AUC "rarely appropriate" for PCI of asymptomatic single-vessel without ischemia).
- Diabetes + MVD: CABG (FREEDOM 5-y death/MI/stroke 18.7% vs 26.6%); PCI if low SYNTAX and patient preference.
- Prior CABG: PCI of the native vessel preferred over degenerated graft; redo CABG for diseased LIMA/LAD territory.

Bifurcation lesions

- Medina classification (proximal main, distal main, side branch: 1 = ≥50% stenosis). True bifurcation = 1,1,1 / 1,0,1 / 0,1,1.
- Default: **provisional single-stent** strategy (NORDIC, BBC ONE, CACTUS/ BBK). Steps: wire both branches (jailed SB wire), predilate main (SB predilation only if severely stenotic or calcified — it risks dissection making re-wiring hard), stent main sized to distal reference, **POT** (proximal optimization: short NC balloon at the carina to the proximal reference size), re-wire SB through a distal cell, kissing balloon if SB compromised (TIMI <3, ostial >75%, FFR ≤0.80) or ostial SB dissection, re-POT. Sequential balloon (POT–side–POT) is an alternative to kissing.
- Upfront two-stent when: SB ≥2.5–2.75 mm with ostial disease >5–10 mm (DEFINITION criteria), difficult SB re-access risk, SB supplies large territory, complex distal LM (DKCRUSH-V: DK crush ↓TLF vs

provisional 5.0% vs 10.7% at 1 y; DKCRUSH-II vs culotte; EBC MAIN: provisional non-inferior in LM). Techniques: DK crush (preferred in LM), culotte (wide angle <70°, similar diameters), T/TAP (angle ~90°), mini-crush, reverse crush (bail-out). Always finish with kissing and final POT; 2 stents usually need 7F guide (6F possible for DK crush/culotte with modern stents).

- SB occlusion predictors: plaque shift, ostial SB disease, bifurcation angle <70°, TIMI flow, main branch diameter stenosis (RESOLVE score). Rescue: re-wire, balloon, TAP.
- Imaging (OCT) reduces MACE in bifurcation PCI (OCTOBER).
- LM ostial stenting: extend 1–2 mm into the aorta, disengage guide, use marker/pigtail views. LM: MSA targets from note 01; IVUS strongly recommended.

Chronic total occlusion

- Definition: TIMI 0 for ≥3 months. Indication: refractory angina despite OMT after treating non-CTO lesions is **Class 2b (ACC/AHA 2021)**; no mortality benefit (DECISION-CTO neutral, EuroCTO: QoL benefit; EXPLORE and REVASC: no LVEF benefit). Stop the procedure at air kerma >5 Gy (strongly consider at >7 Gy), contrast >3.7× eGFR, or >3 h; consider an investment procedure. Viability and ischemic burden support the decision.
- J-CTO score: blunt stump, calcification, bending >45°, length ≥20 mm, prior failed attempt (each 1 point; 0 easy, 1 intermediate, 2 difficult, ≥3 very difficult — predicts crossing within 30 min). PROGRESS-CTO: proximal cap ambiguity, moderate/severe tortuosity, absence of interventional collaterals, circumflex CTO (predicts success).
- Dual injection is mandatory for retrograde collateral assessment and distal target definition. Hybrid algorithm: antegrade wire escalation (polymer-jacketed tapered wires like Fielder XT, then Gaia, Confianza/Pilot family; penetration force 0.8–12 g), antegrade dissection re-entry (CrossBoss/Stingray; knuckle wire), retrograde via septal (safer, can be dilated with 1.5 balloon) or epicardial collaterals (never dilate, perforation → tamponade in non-post-CABG; post-CABG tamponade can be loculated), reverse CART (controlled antegrade and retrograde tracking — balloon on antegrade wire, retrograde wire enters), guide extension facilitated. Tip-in, rendezvous techniques. Externalization with RG3/R350 wires.
- Equipment: 8F guides (support), microcatheters (Corsair, Turnpike, Caravel; SuperCross angled), anchor balloon, trapping balloon for wire exchange, knuckle wire (Fielder XT/Gladius).
- Complications: perforation (esp. epicardial collaterals), donor vessel dissection/thrombosis (dangerous: donor vessel supplies both territories — anticoagulate ACT >350 s for retrograde), contrast nephropathy, radiation injury (long cases; air kerma), wire fracture, septal hematoma, equipment entrapment.
- Post-CTO: DES, DAPT 12 months; "subintimal" plaque modification (investment procedure) if failed: re-attempt in 6–8 weeks.

Calcified lesions

- Decide by imaging: IVUS arc >270° or OCT calcium score ≥4 (arc >180°, thickness >0.5 mm, length >5 mm) → modify calcium before stenting. Angiographic clues: calcium both sides of the lumen without contrast; balloon "dog-bone"; non-crossable.

Tool	Mechanism / settings	Pros	Cautions
NC balloon, scoring (AngioSculpt), cutting (Wolverine)	High pressure; controlled incisions	Simple	Perforation; cutting balloon not for tortuous/angulated
Rotational atherectomy (Rotablator/RotaPro)	Diamond burr 1.25–2.5 mm; differential cutting (saves elastic tissue); burr:artery 0.5–0.6; 140–150k rpm (slower for big burrs); runs 15–20 s; no deceleration >5000 rpm; RotaWire floppy (extra support for tortuous)	Non-crossable, non-dilatable, nodular calcium	Slow-flow (verapamil/nitro/heparin in flush), bradycardia/heart block (RCA, dominant LCx → temporary pacer or atropine), perforation, burr entrapment, dissection; avoid in thrombus, SVG, dissection, unprotected LM with severe LV dysfunction (relative); 1.5 mm in 6F, 1.75 in 7F, 2.0 in 8F. ROTAXUS/PREPARE-CALC: better acute gain, no late benefit vs balloon
Orbital atherectomy (Diamondback 360)	1.25 mm eccentric crown; 80k (low) or 120k (high) rpm; bidirectional; ViperWire; ViperSlide lubricant	Single crown for multiple sizes; less entrapment; treats larger vessels at high speed	Perforation (higher with high speed, small vessels), slow flow; ECLIPSE (2024): no benefit vs balloon prep before stenting
Intravascular lithotripsy (Shockwave C ² /C ²⁺)	Sonic pressure waves fracture calcium; balloon 1:1, 10 pulses/cycle (C ² 80 total, C ²⁺ 120), inflate 4 atm, post 6 atm	Deep/eccentric/nodular and concentric calcium, stent underexpansion (off-label), safe for wire, no slow-flow	Ventricular capture ("shocktopics", benign), transient pacing interference; needs crossable lesion (2.5–4.0 mm balloons); DISRUPT CAD III: 92% success
Excimer laser (ELCA)	308 nm UV photoablation; saline flush (contrast → acoustic shock, perforation); 0.9–2.0 mm catheters	Uncrossable, ISR, underexpanded stents, thrombus, SVG	Dissection, perforation; contrast injection during lasing contraindicated

- Stent under-expansion in old calcified stent: IVL, high-pressure NC, ELCA; do not rotatable fresh stents (burr entrapment) though "stentablation" is described for chronic ones.
- Calcium nodules on OCT: protruding, eruptive; highest ISR/ST risk; orbital/rota or IVL.

Saphenous vein grafts

- Graft failure: early (<1 month) technical/thrombosis; 1 month–1 y intimal hyperplasia; >1 y atherosclerosis (diffuse, friable, thrombus). 10-y patency ~50% (vein) vs >90% (LIMA-LAD).
- PCI of SVG: embolic protection device (class 2a; SAFER: distal occlusion !MACE; FIRE: FilterWire equivalent; proximal occlusion Proxis); no-reflow prophylaxis (IC vasodilators, nicardipine); avoid predilation; direct stenting; undersized stents (0.9:1) reduce embolization; DES = BMS (DIVA: no difference; ISAR-CABG DES better at 1 y); covered stents no benefit (SYMBIOT, BARRICADE). Distal anastomotic lesions: balloon alone often OK.
- Prefer native-vessel PCI over degenerated SVG when feasible (class 2a); chronically occluded SVG: do not intervene; prior CABG protects the LM if grafts patent ("protected LM").
- Arterial graft PCI: LIMA spasm (vasodilators), tortuous; avoid oversizing; radial grafts spasm-prone.

- Guides: right JR4/multipurpose for RCA grafts (inferior takeoff), LCB/JR/AL1 for left grafts (superior/anterior takeoff), IM guide for LIMA (left radial access easiest).

Thrombus and other subsets

- Aspiration bail-out; GPI; direct stenting; undersize and postdilate later. OCT for erosion (young, women, smokers) → antithrombotic without stent (EROSION study) in selected cases.
- Ostial lesions: aorto-ostial (RCA/LM) need precise placement, protrude 1 mm; ostial LAD/LCx — consider crossover to LM (bifurcation rules). Szabo technique.
- Small vessels (<2.75 mm): DES thin-strut (BIOSCIENCE), DCB non-inferior (BASKET-SMALL 2).
- Long lesions/diffuse disease: full-metal jacket vs CABG; imaging.
- Tortuosity: extra-support wires, guide extensions (GuideLiner/Telescope 5-in-6, 6-in-7), buddy wire, anchor balloon, deep-seated guide; sheathless guides.
- Aneurysms/ectasia: no stents unless covered; anticoagulation debated.
- Coronary artery bypass graft "string sign" of LIMA: competitive flow — treat the native stenosis, not the LIMA.
- Kounis syndrome: allergic ACS (mast cell mediators); type I vasospasm, II plaque rupture, III stent thrombosis; treat allergy and ACS in parallel (avoid morphine/beta-blockers cautiously).

Guide catheters and wires (fast facts)

- Guide inner diameters: 5F 0.056–0.058", 6F ~0.070–0.071", 7F ~0.078–0.081", 8F ~0.088–0.090". Two stents side by side need $\geq 0.070"$ $\times 2 \approx 7F$ (thin-strut 6F possible); 2.0 burr needs 8F.
- Left system: JL4 femoral (JL3.5 radial), EBU/XB 3.5–4 (extra back-up, ↑dissection risk), AL2 for very superior/anomalous; right: JR4, AL1 for shepherd's crook or SVG, AR mod, MP for inferior RCA/SVG, IM for LIMA; Tiger/TIG single catheter for radial diagnostic.
- Wire tip loads: workhorse 0.5–1 g (BMW, Runthrough, Sion); polymer-jacketed (Fielder FC/XT, Whisper, Pilot 50/150/200: 1.5–4 g — perforation risk); stiff tapered (Gaia 1st–3rd 1.7–4.5 g, Confianza Pro 9/12 g, Hornet 14 g, Miracle 3–12 g); support wires (Grand Slam, Iron Man, Mailman) for device delivery; Rotawire (0.009") for rota; ViperWire for orbital; 0.014" is the standard coronary wire; 0.035" for diagnostics; Wiggle wire for tortuosity.
- Balloons: compliant (grows with pressure), semi-compliant, non-compliant (NC: uniform, high pressure, post-dilation); nominal vs rated burst pressure (RBP; exceed at your own risk); stent nominal, post-dilate to $\geq 1:1$ with NC.
- Stent platforms: cobalt-chrome/platinum-chrome thin struts (60–80 μm); strut thickness ↓ → ↓restenosis/ST; max post-dilation limits (e.g., 2.25–2.5 mm stents to ~3.5 mm; 3.0–4.0 to 5.5–5.75).
- Microcatheters vs OTW balloons: exchange with trapping (in 7F+) or extension wire (300 cm).

07 – Peripheral/Carotid/PE, Radiation Safety, Statistics, and Professionalism

Peripheral arterial disease

- ABI: <0.9 abnormal (0.7–0.9 mild, 0.4–0.7 moderate, <0.4 severe); >1.4 non-compressible → toe-brachial index (<0.7 abnormal). Claudication: supervised exercise (class 1), cilostazol (contraindicated in HF), statins, antiplatelet; rivaroxaban 2.5 BID + aspirin after revascularization (VOYAGER PAD ↓MALE) or in stable PAD (COMPASS).
- Revascularize claudication only if lifestyle-limiting after exercise/OMT; chronic limb-threatening ischemia (rest pain, tissue loss, WIfI staging): revascularize promptly. BEST-CLI: surgical bypass with a good single-segment GSV beat endovascular (MALE/death 42.6% vs 57.4%); without good vein, equal. BASIL-2 (2023): for infrapopliteal disease, vein-bypass-first had MORE amputation/death than endovascular-first. STRIDE (2025): semaglutide increased walking distance in PAD with T2DM. Acute limb ischemia = symptoms <2 weeks; Rutherford I viable (arterial and venous Doppler audible), IIa marginally threatened (minimal sensory loss, arterial often inaudible), IIb immediately threatened (rest pain, motor deficit; emergent revascularization ≤6 h), III irreversible (both signals inaudible; primary amputation). Immediate IV heparin is Class 1. Angiosome-directed revascularization.
- Aortoiliac: stents (covered for TASC C/D, COBEST); iliac stenting excellent patency. Femoropopliteal: DCB (paclitaxel; the 2018 mortality signal was later refuted by FDA/SWEDEPAD/large data), DES (Eluvia, Zilver PTX), nitinol self-expanding stents (avoid the popliteal flexion zone unless Supera interwoven), atherectomy (no RCT advantage; embolic protection), covered stents (Viabahn for long occlusions). Infrapopliteal: balloon, DCB mixed, DES (Esprit BTK: LIFE-BTK positive), avoid bare stents.
- Complications: distal embolization, perforation, dissection (flow-limiting → stent), access; contrast; cholesterol emboli.
- Renal physiology: unilateral RAS is renin-dependent (ACEi lowers BP); bilateral RAS or solitary kidney is volume-dependent and **ACEi raises creatinine (AKI)** with little BP effect. Significance: systolic gradient ≥20, mean ≥10 mmHg, Pd/Pa ≤0.90 or renal FFR ≤0.80.
- Renal artery stenosis: atherosclerotic — stenting NOT beneficial vs OMT (ASTRAL, CORAL); consider for flash pulmonary edema, refractory HTN with bilateral stenosis, progressive CKD with bilateral/solitary kidney stenosis. FMD (young women, "string of beads", medial fibroplasia): PTA alone (no stent), cures HTN in many.
- Mesenteric ischemia: stent celiac/SMA; median arcuate ligament syndrome → surgery.
- Subclavian stenosis: stent for symptomatic steal, arm claudication, before/after LIMA graft ("coronary-subclavian steal"); vertebral protection.
- Aortic aneurysm/EVAR: repair AAA ≥5.5 cm (men), ≥5.0 (women), or growth >1 cm/y. Endoleak types: I attachment site (treat), II retrograde branch flow (most common: lumbar/IMA; treat if sac grows), III component separation/fabric tear (treat), IV graft porosity (self-limited), V endotension. TEVAR for complicated type B dissection (INSTEAD-XL long-term).

Carotid disease

- Symptomatic ≥ 50 –70% → revascularize within 2 weeks; asymptomatic 70–99% selected (life expectancy > 5 y, low periprocedural risk). **CREST-2 (NEJM 2025), asymptomatic ≥ 70 %: CAS + intensive medical management reduced periprocedural stroke/death plus 4-y ipsilateral stroke (2.8% vs 6.0%, NNT ~31); the CEA arm was not significant (3.7% vs 5.3%).** ACST-2: CAS $\approx$ CEA. **CMS NCD (Oct 2023): CAS covered for symptomatic ≥ 50 % and asymptomatic ≥ 70 % regardless of surgical risk**, with shared decision-making and neurologic assessment.
- CEA vs CAS: CREST (4-y composite similar 7.2% vs 6.8%; periprocedural stroke $\uparrow$ with CAS, MI $\uparrow$ with CEA; age > 70 favors CEA; < 70 similar), ACT I (asymptomatic, non-inferior), SPACE/EVA-3S/ICSS (CAS worse in symptomatic). CAS favored when: high surgical risk (contralateral occlusion, prior neck surgery/radiation, tracheostomy, high lesion above C2, restenosis after CEA, recurrent laryngeal palsy). TCAR (flow reversal, transcarotid) has lower stroke than transfemoral CAS in registries.
- Technique: embolic protection mandatory (distal filter or proximal flow reversal — Mo.Ma); predilate small, self-expanding stent (open vs closed cell; closed/mesh for symptomatic), post-dilate minimally; atropine/pacing readiness for carotid sinus bradycardia/hypotension (can persist 24–48 h, treat with fluids/pressors, hold antihypertensives); DAPT ≥ 4 weeks. Contraindications: type III arch/severe tortuosity, heavy circumferential calcium, thrombus, no protection possible, occlusion.
- Complications: stroke (embolic; $\uparrow$ age, symptomatic, long lesions), hyperperfusion syndrome (headache, seizure, ICH after severe stenosis; control BP), bradycardia/hypotension, access. Stent fracture; restenosis (radiation, women).

Venous and pulmonary embolism

- PE risk: massive/high-risk (SBP < 90 or shock) → systemic lysis (alteplase 100 mg over 2 h; contraindication → catheter-directed therapy or surgical embolectomy, ECMO bridge); intermediate-high (RV dysfunction + troponin) → anticoagulation, rescue lysis if deterioration (PEITHO: tenecteplase $\downarrow$ hemodynamic collapse, $\uparrow$ major and intracranial bleeding, no mortality benefit); half-dose lysis (MOPETT) controversial. Catheter-directed: ultrasound-assisted CDT (EkoSonic; ULTIMA, SEATTLE II, OPTALYSE), mechanical thrombectomy (FlowTrieve: FLARE, FLAME registry; Indigo/Penumbra: EXTRACT-PE; PEERLESS: FlowTrieve vs CDT fewer decompensations). HI-PEITHO/PEERLESS II ongoing. PERT teams. IVC filter only if anticoagulation contraindicated; retrieve early (PREPIC 2: no benefit routine filter).
- Iliofemoral DVT: CDT/thrombectomy for limb-threatening (phlegmasia) or selected young patients (ATTRACT: no $\downarrow$ post-thrombotic syndrome overall; benefit in iliofemoral subgroup for severity); May-Thurner: iliac vein stent. Chronic venous: stenting.

Radiation safety

Units. Absorbed dose Gy; equivalent dose Sv (=Gy for X-rays); effective dose mSv (tissue-weighted). Machine metrics: **reference-point air kerma (Ka,r, Gy)** at the interventional reference point 15 cm from isocenter toward the tube — best surrogate for peak skin dose (deterministic risk); **kerma-area product (KAP/DAP, Gy·cm²)** — total energy, best for stochastic risk; **fluoroscopy time** — poorest metric (ignores cine and settings). Peak skin dose is estimated (Ka,r overestimates PSD when views are varied).

Effects. Deterministic (threshold, severity $\propto$ dose): skin — transient erythema 2 Gy (hours–days), main erythema/temporary epilation 3–6 Gy (2–8 weeks), permanent epilation 7 Gy, dry desquamation 10–14 Gy, moist desquamation 15–18 Gy, dermal necrosis/ulcer >18–24 Gy (months–years, delayed); cataract (threshold lowered to 0.5 Gy; posterior subcapsular). Stochastic (no threshold, probability $\propto$ dose): cancer (5%/Sv lifetime), heritable effects. Substantial radiation dose level (SIR/NCRP 168): Ka,r **5 Gy**, KAP 500 Gy·cm², peak skin dose 3 Gy, fluoro time 60 min → document, counsel, follow up skin at 2–4 weeks; Ka,r >15 Gy is a sentinel event (Joint Commission). Typical PCI 1–3 Gy Ka,r; diagnostic 0.5–1 Gy; CTO higher.

Operator limits. US NRC: whole body 50 mSv/y (practical goal <5), **lens 150 mSv/y**, skin/ extremities 500 mSv/y; ICRP: 20 mSv/y effective averaged over 5 y (max 50 in one year) and **lens 20 mSv/y** (know both numbers); lifetime 10 mSv × age (NCRP); **pregnant worker fetus 5 mSv** total, 0.5 mSv/month; public 1 mSv/y. Two badges: collar outside the apron (lens/thyroid estimate) and under the apron at waist; effective dose = 0.5 × under + 0.025 × collar.

Dose reduction rules (patient and operator).

- Distance: inverse square law (double distance = ¼ dose); step back during cine and DSA.
- Time: fewer/shorter cine runs; fluoro save/last-image hold; pulsed fluoro 7.5 fps (vs 15/30).
- Shielding: 0.5 mm lead equivalent apron blocks ~95–98%; thyroid collar; leaded glasses (0.75 mm, side shields; cataract); ceiling shield; table skirts; disposable pads; radial access at the arm exposes the operator more unless shielded.
- Geometry: keep the image receptor **close** to the patient and the table **high** (tube far from skin); avoid steep angulation (LAO cranial/caudal >30° roughly doubles dose and puts the tube on the operator's side in LAO); vary angles to spread skin dose; collimate tightly; use the lowest magnification (each mag step ≈ +50–100% skin dose entrance); avoid "digital zoom" misconceptions; remove the patient's arms from the field; large patients → automatic brightness control raises dose exponentially.
- Scatter: patient is the source; highest on the tube (entrance) side; below-table tube means most scatter is low → table skirts matter; operator dose ~1000× lower than skin dose.
- Fluoroscopy dose modes (low/normal/high), frame rate, copper filtration, larger field of view. Store fluoro instead of cine where image quality allows.
- Pregnancy in patient: not a contraindication when indicated; shield abdomen, minimize; fetal dose <50 mGy negligible risk; deterministic fetal effects >100–150 mGy.

Biostatistics and trial design

- Sensitivity = TP/(TP+FN); specificity = TN/(TN+FP); PPV/NPV depend on prevalence (PPV ↑ with prevalence). LR+ = sens/(1–spec); LR– = (1–sens)/spec. Post-test odds = pre-test odds × LR. ROC AUC 0.5 = useless.
- ARR = control event rate – treatment event rate; RRR = ARR/CER; **NNT = 1/ARR** (NNH for harms); RR, OR (≈RR when outcome rare), HR (instantaneous, time-to-event; Cox); 95% CI excluding 1 (ratios) or 0 (differences) ↔ p < 0.05.
- Type I error (α , false positive), type II (β , false negative), power = 1 – β (usually 80–90%); sample size ↑ with smaller effect, lower α , higher power, more variance. Multiple comparisons inflate type I error (Bonferroni; hierarchical testing).
- Non-inferiority: pre-specified margin δ ; the upper CI bound of the difference must lie below δ ; per-protocol AND ITT analyses both required (ITT biases toward null → toward non-inferiority); superiority can be

tested after non-inferiority is met (closed testing).

- Intention-to-treat (preserves randomization, real-world), per-protocol, as-treated. Composite endpoints: driven by the softest component (revascularization). Surrogate endpoints. Bayesian analysis (posterior probability; used in device trials). Adaptive designs.
- Win ratio: hierarchical composite ranking of patient pairs prioritizing death (used in CHIP-BCIS3). Verification bias: short follow-up misclassifies false negatives as true negatives and inflates apparent NPV.
- Bias: selection (fix: randomization), performance/detection (blinding, sham — ORBITA), lead- time/length-time (screening), recall (case-control), confounding (randomization, multivariable adjustment, propensity scores — cannot fix unmeasured confounders), immortal time bias. Regression to the mean. Hawthorne effect.
- Tests: t-test (2 means), ANOVA (>2 means), chi-square/Fisher (proportions; Fisher for small cells), Mann-Whitney/Wilcoxon (non-parametric), Kaplan-Meier + log-rank (survival), Cox (multivariable time-to-event), logistic regression (binary outcome), kappa (agreement between raters), Bland-Altman (agreement between measurement methods), correlation r (Pearson/Spearman), McNemar (paired proportions).
- Levels of evidence: A (multiple RCTs/meta-analyses), B-R (single RCT), B-NR, C-LD, C-EO. Class 1 (benefit >>> risk), 2a (>>), 2b (≥), 3 no benefit / 3 harm.
- Study designs: RCT > cohort > case-control > case series. Registry (NCDR CathPCI) risk models for mortality, AKI, bleeding; public reporting and risk aversion.

Quality, appropriateness, professionalism

- AUC (2017 stable, 2016 ACS): "appropriate / may be appropriate / rarely appropriate"; rarely appropriate = asymptomatic single-vessel without ischemia, PCI of ≤50% lesions without physiology; requirements for antihypertensive trial in stable CAD.
- Quality metrics: door-to-balloon, radial use, contrast volume, AKI, bleeding, unplanned CABG, mortality (risk-adjusted), same-day discharge (safe for uncomplicated elective PCI), PCI without on-site surgery (acceptable with protocols; CPORT-E, MASS COMM), operator volume ≥50 PCIs/y (institution ≥200).
- Informed consent: capacity, disclosure of risks/alternatives, surrogate decision-makers, emergency exception. Refusing care; DNR in the lab (discuss suspending vs maintaining).
- Conflicts of interest (Sunshine Act), industry relations, disclosure of complications/errors (duty to disclose), impaired colleague (report), peer review protections, sham controls ethics (ORBITA), research consent, HIPAA, patient portal results.
- Radiation and reproductive counseling of staff; pregnant operators may work with badges.

08 – Landmark Trials: One-Liners to Memorize

Format: **Trial** – population – comparison – result – exam takeaway.

Stable CAD / revascularization strategy

- **COURAGE** (2007) – stable CAD – PCI+OMT vs OMT – no ↓death/MI – OMT first.
- **BARI 2D** – diabetes stable CAD – revasc vs OMT – no mortality diff; CABG stratum ↓MI.
- **FREEDOM** – diabetes MVD – CABG vs PCI – CABG ↓death/MI (18.7% vs 26.6% at 5 y), ↑stroke.
- **SYNTAX** – 3VD/LM – CABG vs PES – CABG better overall (5-y MACCE 26.9% vs 37.3%); PCI OK if SYNTAX ≤22.
- **EXCEL** – LM, SYNTAX ≤32 – PCI(EES) vs CABG – 5-y death/stroke/MI 22% vs 19.2% (NS); all-cause death higher with PCI (secondary); controversy about MI definitions.
- **NOBLE** – LM – PCI vs CABG – 5-y MACCE 28% vs 19%: CABG better (non-procedural MI, revasc).
- **PRECOMBAT** – LM – PCI(SES) vs CABG – non-inferior at 1 y, 10-y similar death.
- **ISCHEMIA** – moderate-severe ischemia, no LM – invasive vs conservative – no ↓CV death/MI at 3.2 y median; better angina QoL; ↓spontaneous MI later, ↑procedural MI early.
- **ISCHEMIA-CKD** – eGFR <30/dialysis – invasive no benefit; more stroke/dialysis initiation.
- **ORBITA** – single-vessel, sham – PCI ≈ sham for exercise time (6 weeks).
- **ORBITA-2** – sham, no antianginals – PCI ↓angina symptom score.
- **REVIVED-BCIS2** – EF ≤35% viable myocardium – PCI+OMT vs OMT – no ↓death/HF hosp.
- **STICH/STICHES** – EF ≤35% – CABG vs OMT – 10-y mortality ↓ with CABG.
- **FAME** – MVD – FFR-guided vs angio-guided PCI – ↓MACE, fewer stents.
- **FAME 2** – FFR ≤0.80 – PCI+OMT vs OMT – ↓urgent revasc; no death/MI benefit.
- **FAME 3** – 3VD – FFR-PCI vs CABG – PCI failed non-inferiority at 1 y (10.6% vs 6.9%).
- **DEFER** – FFR ≥0.75 – deferral safe 15 y.
- **DEFINE-FLAIR / iFR-SWEDEHEART** – iFR non-inferior to FFR at 1 y; less adenosine discomfort.
- **FUTURE** – all-comers FFR strategy – stopped early, no benefit, mortality signal.
- **TARGET-FFR** – post-PCI FFR-guided optimization – more patients reach ≥0.90; no MACE diff.

ACS

- **SHOCK** – AMI cardiogenic shock – early revasc vs medical – 6-mo/1-y mortality ↓.
- **CULPRIT-SHOCK** – AMI shock MVD – culprit-only vs immediate multivessel – culprit-only ↓death/RRT (45.9% vs 55.4%).
- **IABP-SHOCK II** – AMI shock – IABP no mortality benefit.
- **DanGer Shock** (2024) – STEMI shock – Impella CP vs standard – ↓180-d death (45.8% vs 58.5%), ↑bleeding/limb ischemia/RRT.
- **ECLS-SHOCK** – AMI shock – VA-ECMO vs standard – no mortality benefit, more bleeding.
- **COMPLETE** – STEMI MVD – complete (staged) vs culprit-only – ↓CV death/MI (7.8% vs 10.5%).

- **PRAMI / CvLPRIT / DANAMI-3-PRIMULTI** – complete revasc ↓MACE (mostly revasc).
- **MULTISTARS AMI / BIOVASC** – immediate = staged complete revasc.
- **FLOWER-MI** – STEMI non-culprit – FFR-guided vs angio-guided – no benefit of FFR.
- **FIRE** – MI ≥75 y – physiology-guided complete revasc ↓death/MI/stroke/revasc.
- **TASTE / TOTAL** – STEMI – routine aspiration no benefit; TOTAL ↑stroke.
- **DANAMI-3-DEFER** – deferred stenting no benefit.
- **OAT** – occluded IRA 3–28 d, stable – PCI no benefit.
- **TRANSFER-AMI / STREAM** – pharmacoinvasive (angio 6–24 h after lysis) ≈ primary PCI; STREAM ↑ICH in >75 y → half-dose TNK.
- **TIMACS / VERDICT** – NSTEMI-ACS – early (<24 h) invasive benefits high-risk (GRACE >140).
- **COACT / TOMAHAWK** – cardiac arrest without STEMI – no benefit of immediate angiography.
- **HORIZONS-AMI** – STEMI – bivalirudin vs UFH+GPI – ↓bleeding, ↓death; ↑acute ST.
- **MATRIX** – ACS – radial ↓mortality/bleeding; bivalirudin ≈ UFH; post-PCI bivalirudin infusion ↓ST.
- **BRIGHT-4** – STEMI – bivalirudin + high-dose infusion vs UFH – ↓death/bleeding.
- **VALIDATE-SWEDEHEART** – bivalirudin ≈ UFH (radial, potent P2Y12).
- **RIVAL** – radial vs femoral – overall NS; benefit in STEMI and high-volume centers.
- **SAFARI-STEMI** – radial vs femoral in STEMI – no difference (stopped early).
- **EROSION** – OCT plaque erosion – antithrombotics without stent feasible.

Antiplatelet/anticoagulant

- **CURE / PCI-CURE** – clopidogrel + ASA in NSTEMI-ACS ↓MACE.
- **CURRENT-OASIS 7** – 600 mg clopidogrel load better in PCI; aspirin dose no difference.
- **TRITON-TIMI 38** – prasugrel vs clopidogrel ACS-PCI – ↓MI/ST, ↑bleeding; harm with prior stroke/TIA, ≥75 y, <60 kg.
- **PLATO** – ticagrelor vs clopidogrel ACS – ↓CV death/MI/stroke, ↓death; ↑non-CABG bleeding, dyspnea; aspirin >100 mg interaction (North American paradox).
- **ISAR-REACT 5** – prasugrel > ticagrelor in ACS-PCI (6.9% vs 9.3%).
- **ACCOAST** – prasugrel pretreatment in NSTEMI – no benefit, ↑bleeding.
- **TRILOGY ACS** – prasugrel in medically managed ACS – no benefit.
- **CHAMPION PHOENIX** – cangrelor vs clopidogrel – ↓periprocedural MI/ST.
- **DAPT** – 30 vs 12 months – ↓ST/MI, ↑bleeding; DAPT score.
- **PEGASUS-TIMI 54** – ticagrelor 60/90 mg 1–3 y post-MI – ↓MACE, ↑bleeding.
- **TWILIGHT** – ticagrelor mono after 3 months DAPT – ↓bleeding, no ↑ischemia.
- **TICO / STOPDAPT-2 / SMART-CHOICE / GLOBAL LEADERS** – P2Y12 monotherapy after 1–3 mo.
- **MASTER DAPT** – HBR, 1-month DAPT non-inferior.
- **HOST-EXAM** – after 12 mo DAPT: clopidogrel mono > aspirin mono.
- **ADAPTABLE** – 81 = 325 mg aspirin.
- **WOEST / PIONEER AF-PCI / RE-DUAL PCI / AUGUSTUS / ENTRUST-AF PCI** – AF + PCI: dual therapy (OAC + clopidogrel) ↓bleeding vs triple; AUGUSTUS 2×2 apixaban better, aspirin harmful.

- **AFIRE** – AF + stable CAD >1 y – rivaroxaban mono ≥ combination.
- **COGENT** – clopidogrel + omeprazole – no ↑CV events, ↓GI bleeding.
- **TAILOR-PCI / POPular Genetics** – genotype-guided: TAILOR negative primary; POPular safe de-escalation in STEMI.
- **COMPASS / VOYAGER PAD** – low-dose rivaroxaban + ASA ↓MACE/MALE, ↑bleeding.
- **COLCOT / LoDoCo2** – colchicine ↓MACE post-MI/CCS.
- **ARMYDA / NAPLES II** – statin loading ↓periprocedural MI.

Devices, imaging, lesion subsets

- **DKCRUSH-V** – distal LM true bifurcation – DK crush ↓TLF vs provisional (5.0% vs 10.7%).
- **EBC MAIN** – LM bifurcation – provisional non-inferior (numerically better) vs 2-stent.
- **NORDIC / BBC ONE / CACTUS** – non-LM bifurcation – provisional ≥ 2-stent.
- **DEFINITION II** – complex bifurcation (DEFINITION criteria) – 2-stent ↓TLF.
- **ULTIMATE** – IVUS-guided DES ↓TVF (1 and 3 y).
- **RENOVATE-COMPLEX-PCI** – IVUS/OCT-guided complex PCI ↓TVF.
- **ILUMIEN IV** – OCT: larger MSA, ↓ST; no TVF difference.
- **OCTOBER** – OCT-guided bifurcation ↓MACE.
- **OCTIVUS** – OCT non-inferior to IVUS.
- **ADAPT-DES** – IVUS use ↓ST/MI (registry).
- **PROSPECT / PROSPECT II** – vulnerable plaque (PB ≥70%, MLA ≤4, TCFA; maxLCBI4mm >324) predicts events.
- **PREVENT (2024)** – preventive PCI of vulnerable plaques ↓composite (driven by revasc).
- **ROTAXUS / PREPARE-CALC** – rota vs balloon before DES: better acute gain, no late benefit.
- **ECLIPSE (2024)** – orbital atherectomy vs balloon prep – no benefit.
- **DISRUPT CAD III** – coronary IVL 92% procedural success, low MACE.
- **DECISION-CTO / EuroCTO** – CTO PCI: no hard-endpoint benefit; EuroCTO ↑QoL.
- **EXPLORE** – CTO PCI after STEMI – no LVEF benefit.
- **SAFER** – SVG PCI – distal occlusion EPD ↓MACE 42%.
- **DIVA** – SVG PCI – DES = BMS.
- **ISAR-CABG** – SVG – DES ↓TLR at 1 y.
- **AGENT IDE** – coronary ISR – paclitaxel DCB > balloon.
- **RIBS IV/IV** – DES-ISR: EES ≥ DCB; BMS-ISR: equal.
- **BASKET-SMALL 2** – small vessels – DCB non-inferior to DES.
- **LEADERS FREE** – HBR – polymer-free DCS + 1 mo DAPT > BMS.
- **ABSORB III/IV** – bioresorbable scaffold – ↑target lesion failure/ST vs EES; withdrawn.
- **XIENCE 28/90, Onyx ONE** – short DAPT after modern DES in HBR.
- **SYMBIOT / BARRICADE** – covered stents in SVG: no benefit/harm.
- **TASTE, TOTAL** – above.
- **FAUST** – US-guided femoral access ↓complications.

- **DISCO RADIAL** – distal radial ↓RAO; equal success.

Structural

- **PARTNER 1B / 1A / 2A / 3** – inoperable / high / intermediate / low risk TAVR.
- **CoreValve US Pivotal / SURTAVI / Evolut Low Risk** – self-expanding across risk.
- **NOTION** – all-comers TAVR vs SAVR 10-y similar.
- **SMART** – small annulus – Evolut > SAPIEN for valve hemodynamics/dysfunction.
- **PROTECTED TAVR** – Sentinel – no ↓overall stroke; ↓disabling stroke (secondary).
- **BHF PROTECT-TAVI** – cerebral protection no benefit.
- **POPular TAVI** – aspirin alone (A) / OAC alone (B) ↓bleeding, no ischemic penalty.
- **GALILEO** – rivaroxaban after TAVR – ↑death/bleeding; ↓leaflet thrombosis (imaging substudy).
- **ATLANTIS** – apixaban after TAVR – not superior.
- **EVEREST II** – MitraClip vs surgery – surgery better MR reduction; TEER safer.
- **COAPT** – secondary MR (disproportionate) – TEER ↓HF hosp/death.
- **MITRA-FR** – secondary MR – neutral.
- **RESHAPE-HF2** – TEER in functional MR – positive.
- **CLASP IID** – PASCAL non-inferior to MitraClip (primary MR).
- **TRILUMINATE** – tricuspid TEER ↓TR, ↑QoL; no mortality/HF hosp benefit.
- **TRISCEND II** – EVOQUE TTVR – ↓TR, ↑QoL.
- **PROTECT AF / PREVAIL / PRAGUE-17 / Amulet IDE / PINNACLE FLX / OPTION** – LAAO.
- **RESPECT / CLOSE / REDUCE / DEFENSE-PFO** – PFO closure ↓recurrent stroke; ↑AF.
- **PREMIUM / MIST** – PFO closure for migraine: negative.
- **SEATTLE II / ULTIMA / OPTALYSE / FLARE / EXTRACT-PE / PEERLESS / PEITHO / MOPETT** – PE.
- **ATTRACT** – DVT CDT – no ↓PTS overall.
- **CREST / ACT I / SPACE / EVA-3S / ICSS / CREST-2 (pending)** – carotid.
- **ASTRAL / CORAL** – renal stenting no benefit.
- **BEST-CLI / BASIL-2** – CLTI: bypass with GSV better; infrapopliteal endovascular ≥ bypass.
- **VOYAGER PAD** – rivaroxaban after revasc ↓MALE.
- **PARTNER 3 / Evolut LR 5-y** – outcomes converge; durability ongoing.

09 – Blueprint Gap Topics

Topics named explicitly in the January 2026 blueprint that the other notes cover thinly. These are finite, factual, and cheap points.

Preoperative cardiac evaluation and revascularization before noncardiac surgery

- Stepwise approach (2024 ACC/AHA perioperative guideline): emergency surgery → proceed with monitoring; active cardiac condition (ACS, decompensated HF, significant arrhythmia, severe valve disease) → treat first; estimate risk (RCRI, NSQIP; elevated risk $\geq 1\%$); functional capacity ≥ 4 METs (climb 2 flights) → proceed without testing; poor/unknown capacity and elevated risk → stress testing only if it will change management. Routine coronary angiography or revascularization to "get through" surgery is not recommended.
- **CARP**: prophylactic revascularization before elective vascular surgery gave no survival benefit. Revascularize only for standard indications (LM, ACS, refractory symptoms).
- **POISE**: starting high-dose metoprolol on the day of surgery reduced MI but increased death and stroke; continue chronic beta-blockers, never start them the day of surgery. Statins: continue; start before vascular surgery. Aspirin: POISE-2 no benefit and more bleeding in patients without recent stents, but continue aspirin in patients with coronary stents when the surgeon allows.
- **Timing of elective noncardiac surgery after PCI**: balloon angioplasty ≥ 14 days; BMS ≥ 30 days; DES for chronic coronary syndrome ≥ 6 months (≥ 3 months if the surgery is time-sensitive and the delay risk exceeds stent-thrombosis risk); DES for ACS ≥ 12 months (≥ 6 months if time-sensitive). Continue aspirin throughout if at all possible; stop the P2Y12 inhibitor for the minimum time (clopidogrel/ticagrelor 5 days, prasugrel 7 days) and resume within 24–48 h; consider a cangrelor bridge for very high-risk recent stents. Never interrupt DAPT within 30 days of DES.
- Perioperative MI is usually type 2 (supply–demand) and often silent; troponin surveillance in high-risk patients (MINS); manage with aspirin, statin, and angiography for type 1 features.

Emergency pacing and electrophysiologic complications of STEMI

- Inferior MI heart block: AV nodal level (narrow escape 40–60/min), often transient, responds to atropine/aminophylline; pace only if symptomatic or hemodynamically unstable.
- Anterior MI with new bifascicular block, Mobitz II, or alternating BBB: infranodal, wide unreliable escape → prophylactic temporary transvenous pacing; high mortality (large infarct).
- Transcutaneous pacing as a bridge; transvenous via right IJ (balloon-tipped) or femoral under fluoroscopy; risks: RV perforation (tamponade), VT, dislodgement, infection.
- VF during cath: defibrillate immediately (200 J biphasic); common triggers are RCA ostial damping/deep injection, conus branch injection, ionic contrast, hypokalemia, long ischemia.
- Antiarrhythmics in the lab: amiodarone 150 mg IV over 10 min then 1 mg/min; lidocaine 1–1.5 mg/kg for VF/VT; procainamide for stable monomorphic VT; magnesium 2 g for torsades; adenosine 6/12 mg for SVT (contraindicated in severe asthma); accelerated idioventricular rhythm after reperfusion is benign, no treatment.

Right heart catheterization and endomyocardial biopsy

- RHC: access IJ (US-guided) or femoral; measure in end-expiration; confirm wedge (a/v waves, sat >90%, mean $\leq$ PA diastolic); thermodilution unreliable with severe TR or shunt; Fick with measured VO₂ preferred in low output.
- Endomyocardial biopsy: RV septum via right IJ (Stanford bioptome) or femoral (long sheath); ≥ 4 –5 samples; indications: transplant rejection surveillance, fulminant/unexplained new HF (giant cell, eosinophilic, lymphocytic myocarditis), suspected infiltrative disease (amyloid, sarcoid), cardiotoxicity monitoring; complications: perforation/tamponade 0.5–1%, TR from chordal injury, arrhythmia, RBBB (dangerous with pre-existing LBBB $\rightarrow$ have pacing ready), pneumothorax, carotid puncture; sampling error in focal disease.

Local anesthetics

- Lidocaine 1% = 10 mg/mL; maximum 4.5 mg/kg (≤ 300 mg) plain, 7 mg/kg (≤ 500 mg) with epinephrine. Onset 1–2 min, duration 30–60 min (2–3 h with epinephrine). Bupivacaine is the most cardiotoxic.
- Toxicity sequence: perioral numbness, metallic taste, tinnitus, agitation, seizures, then bradycardia, hypotension, wide QRS, VT/asystole. Treat with airway/benzodiazepines, and **20% lipid emulsion 1.5 mL/kg bolus then 0.25 mL/kg/min**; avoid vasopressin, use low-dose epinephrine; prolonged CPR/ECMO if needed.
- True allergy is rare (ester agents via PABA; amides rarely; methylparaben preservative); alternative: diphenhydramine 1% infiltration.

Contrast physics and the imaging chain

- Iodine K-edge is 33.2 keV; maximal iodine contrast at tube voltages of ~60–80 kVp; higher kVp penetrates more but reduces iodine contrast. Concentration 320–370 mgI/mL; viscosity rises with concentration and falls with warming (warm iodixanol to 37 °C). Ionic high-osmolar agents (diatrizoate, >1500 mOsm) caused VF, hypotension, and more reactions; non-ionic low-osmolar (iohexol, iopamidol, ioversol; 600–800 mOsm) and iso-osmolar iodixanol (290 mOsm, dimer, most viscous) are standard.
- Imaging chain: generator $\rightarrow$ x-ray tube (tungsten anode; small focal spot for fluoro, large for cine; ~1% of energy becomes x-rays) $\rightarrow$ filtration (inherent aluminum plus added copper removes low-energy photons) $\rightarrow$ collimator $\rightarrow$ patient (source of scatter) $\rightarrow$ anti-scatter grid $\rightarrow$ flat-panel detector (cesium iodide scintillator on amorphous silicon) $\rightarrow$ image processing (edge enhancement, frame averaging) $\rightarrow$ display. Automatic exposure/dose-rate control raises kVp and mA as attenuation rises (large patients, steep angles, magnification).
- Parameters: kVp (energy/penetration), mA (photon number; dose $\propto$ mA), pulse width (motion blur if long), frame rate (dose $\propto$ fps), field of view (smaller FOV/magnification $\uparrow$ dose), source-to-image distance (detector close to patient), table height (patient away from tube).
- Image quality trade-offs: spatial resolution (focal spot, detector pixel, motion), contrast (kVp, iodine, scatter), noise/quantum mottle (mA, frame averaging). Digital subtraction angiography for peripheral work (mask subtraction; motion artifact; less contrast needed).

Basic science bullets the blueprint names

- **Clotting cascade:** intrinsic (XII, XI, IX, VIII; aPTT) and extrinsic (tissue factor, VII; PT/ INR) converge on the common pathway (X, V, II, fibrinogen). UFH potentiates antithrombin against IIa and Xa (1:1); LMWH mostly Xa (3:1); fondaparinux Xa only; bivalirudin/argatroban direct IIa; warfarin II, VII, IX, X and proteins C/S (early prothrombotic window); dabigatran IIa; rivaroxaban, apixaban, edoxaban Xa. Fibrinolytics activate plasminogen → plasmin (alteplase/tenecteplase fibrin-specific; streptokinase not, antigenic).
- **Platelet function:** adhesion (vWF–GPIb at high shear, collagen–GPVI), activation (ADP via P2Y₁₂ and P2Y₁, thromboxane A₂ via COX-1, thrombin via PAR-1 targeted by vorapaxar), aggregation (fibrinogen bridging activated GPIIb/IIIa). Platelet-function testing (VerifyNow PRU >208 = high reactivity; LTA; VASP) and CYP2C19 genotyping are not recommended routinely (GRAVITAS, ARCTIC, TRIGGER-PCI, TAILOR-PCI); useful for individualized decisions.
- **Lipids:** LDL cleared by hepatic LDL receptors; PCSK9 degrades LDL receptors (inhibitors evolocumab/alirocumab; inclisiran siRNA); statins inhibit HMG-CoA reductase; ezetimibe blocks NPC1L1; bempedoic acid inhibits ATP-citrate lyase (liver-specific, less myalgia); icosapent ethyl (REDUCE-IT, ↑AF/bleeding); Lp(a) measure once (genetic, not statin-responsive; pelacarsen trials).
- **Reperfusion injury:** calcium overload, mitochondrial permeability transition pore opening, oxidative burst, neutrophil plugging, microvascular obstruction (CMR MVO predicts remodeling); post-conditioning, cyclosporine (CIRCUS negative), and most cardioprotection trials failed.
- **Restenosis biology:** PTCA restenosis = elastic recoil + negative remodeling + neointima; stent restenosis = neointimal hyperplasia only (smooth-muscle migration/proliferation, peaks 6–12 months for BMS); DES restenosis is focal, later, with neoatherosclerosis. Diabetes: diffuse disease, small vessels, impaired collaterals, endothelial dysfunction, platelet hyperreactivity, more restenosis and progression; CABG advantage.
- **Vascular remodeling:** Glagov compensatory enlargement until ~40% plaque burden; positive remodeling in ACS/vulnerable plaque, negative in stable/diabetic/elderly lesions.
- **Vasoreactivity:** endothelium-dependent (acetylcholine, flow-mediated; abnormal → constriction) vs endothelium-independent (nitroglycerin, adenosine, nitroprusside).

Coronary ectasia and aneurysm

- Ectasia = diffuse dilation $\geq 1.5\times$ the reference (Markis classification I–IV by extent); aneurysm = focal. Causes: atherosclerosis (most), Kawasaki, connective tissue disease, vasculitis, cocaine, post-DES/brachytherapy (mycotic/inflammatory). Associated with abdominal aortic aneurysm (screen).
- Consequences: sluggish flow (TIMI frame count high), thrombus, distal embolization, spasm, ACS despite no stenosis. Management: antiplatelet $\pm$ anticoagulation for thrombotic events (data weak), avoid stenting ectatic segments (sizing, malapposition); covered stents or surgery for large/enlarging aneurysms ($>2\times$ reference or symptomatic); Kawasaki giant aneurysms (≥ 8 mm) get anticoagulation plus aspirin.

Cost-effectiveness, quality of life, and epidemiology

- QALY = years of life $\times$ utility (0 = death, 1 = perfect health). ICER = $\Delta\text{cost} / \Delta\text{QALY}$ between strategies. US willingness-to-pay thresholds commonly \$50,000–\$150,000/QALY ("high value" $< \$50k$, "intermediate"

\$50–150k, "low value" >\$150k per ACC/AHA value statements). A "dominant" strategy is cheaper and more effective; "dominated" is costlier and worse. Cost-effectiveness $\neq$ cost-saving.

- Angina/HF quality-of-life instruments: Seattle Angina Questionnaire (SAQ; frequency, physical limitation, QoL domains), Rose Dyspnea Scale, KCCQ (HF and structural trials), EQ-5D (utilities).
- Epidemiology: incidence (new cases/time) vs prevalence (existing cases at a point; prevalence $\approx$ incidence $\times$ duration); case-fatality; relative vs attributable risk; number needed to screen; lead-time and length-time bias in screening.

Device surveillance, adverse event reporting, and regulation

- FDA device classes: Class I (low risk, general controls), Class II (510(k) "substantial equivalence" to a predicate; most catheters/wires), Class III (life-sustaining/implanted: stents, valves, LAAO, MCS $\rightarrow$ premarket approval (PMA) requiring an investigational device exemption (IDE) trial). Humanitarian device exemption for conditions affecting <8,000 US patients/year (no effectiveness proof required). Post-approval studies and registries (TVT Registry mandated for TAVR/TEER reimbursement).
- Reporting: MedWatch is voluntary for clinicians; under the Safe Medical Devices Act, hospitals must report device-related **deaths to FDA and the manufacturer within 10 working days**, and serious injuries to the manufacturer (or FDA if unknown). Manufacturers report to FDA within 30 days (5 days if remedial action needed). MAUDE is the public database of these reports (passive, under-reported, no denominators). Recalls: Class I (serious harm/death), II, III.
- Drug regulation parallels: Phase I–IV; post-marketing (phase IV) surveillance; black-box warnings.

Consent, documentation, patient safety, professionalism

- Informed consent elements: nature of the procedure, risks (including radiation, contrast, stroke, death), benefits, alternatives (including no treatment), the operator's own outcomes when material, and the patient's capacity (understand, appreciate, reason, express a choice). Emergency exception; surrogate hierarchy; consent for trainee participation; ad hoc PCI should be discussed in advance.
- Procedure documentation (SCAI expert consensus): indication and pre-procedure risk, sedation and ASA/Mallampati, access site and technique, catheters and devices (lot numbers), findings with lesion characteristics and physiology/imaging, interventions and results (TIMI flow, residual stenosis), contrast volume, fluoroscopy time and air kerma/KAP, complications, closure method, estimated blood loss, post-procedure plan and antithrombotic regimen. Time-out before the procedure.
- Patient safety: never events (wrong site, retained foreign body, unintended radiation injury), sentinel event reporting, root-cause analysis (system focus, not blame), disclosure of errors to patients is an ethical duty; impaired or incompetent colleague $\rightarrow$ report to appropriate authority.
- Health equity items the blueprint flags: documented disparities in door-to-balloon times, use of reperfusion and invasive strategies, and referral to cardiac rehab by sex, race, ethnicity, rural residence, and insurance; women present later, more often with atypical symptoms, and have more bleeding/vascular complications, MINOCA and SCAD.
- Conflicts of interest: Sunshine Act disclosure; industry relationships must not influence device selection; research consent must separate the physician-investigator roles.

Aortic disease and chronic aortic dissection (noncardiac vascular)

- AAA screening: one-time ultrasound for men 65–75 who ever smoked. Repair at ≥ 5.5 cm (men), ≥ 5.0 cm (women), growth >1 cm/year, or symptoms. EVAR: lower early mortality, equal long-term, needs lifelong surveillance for endoleak (types I–V, see note 07) and sac growth.
- Type B dissection: uncomplicated $\rightarrow$ medical therapy (HR <60 , SBP <120 , beta-blocker first, then vasodilator); complicated (malperfusion, rupture, refractory pain/hypertension, rapid expansion) $\rightarrow$ TEVAR. Chronic type B: TEVAR for aneurysmal degeneration ≥ 5.5 cm or growth; INSTEAD-XL showed late benefit of pre-emptive TEVAR in uncomplicated dissection (aorta-specific mortality). Type A is surgical; a STEMI mimic when the dissection involves the RCA ostium (inferior ST elevation with chest/back pain, pulse deficit, widened mediastinum) $\rightarrow$ CT before lytics or anticoagulation.